Introduction to Modern Causal Inference

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Introduction

Goals and Approach

Our goal is to get people with any amount of experience in formal mathematics past undergraduate probability to be able to take a vaguely-defined scientific question, translate it into a formal statistical problem, and solve it by constructing a tool that is optimal for the job at hand.

There are many texts and courses that share the same philosophy and have substantial overlap in terms of the pedagogic approach and topics. In particular, we owe a lot to the biostatistics curriculum at UC Berkeley and to more encyclopedic texts like *Targeted Learning.* Nonetheless, we think that the particular selection of topics and tone in this book will make the content accessible enough to new audiences that it's worthwhile to recombine the material in this way.

Philosophy

This book is rooted in the philosophy of modern causal inference. What sets this philosophy apart are the three following tenets:

1. The first is that for most practical purposes, **the point of statistics *is* causal inference**. Ultimately, we humans are concerned with how to make decisions under uncertainty that lead to the best outcome. These are fundamentally causal questions that ask "*what if*" we did A instead of B? The machinery of statistical inference is agnostic to causality, but that doesn't change the fact that our motivation in using it isn't. Therefore there is little point in shying away from causal claims because noncausal claims are not often of any practical utility.
2. That leads to the second, often misunderstood point, which is that **there is no such thing as a method for causal inference**. This isn't a failing of the statistics literature so much as a failing of the naive popularization of new ideas (e.g. blog posts, low-quality papers). The process of statistical inference (for point estimation) only cares about estimating a parameter of a probability distribution and quantifying the sampling distribution of that estimate. There is nothing "causal" about that, and many are surprised to learn that the algorithms used for "causal" inference are actually identical to those used to make noncausal statements. What makes an analysis causal has little if anything

to do with the estimator or algorithm used. It has everything to do with whether or not the researcher can satisfy certain non-testable assumptions about the process that generated the observed data. It's important to keep these things (statistical inference and causal identification) separated in your mind even though they must always work together.
3. The last and perhaps greatest shortcoming of traditional statistics pedagogy is that it generally does not teach you to ask the question that makes sense scientifically and *then* translate it into a statistical formalism. This is partly because, in the past, our analytic tools were quite limited and no progress could be made unless one imposed unrealistic assumptions or changed the question to fit the existing methods. Today, however, **we have powerful tools that can then translate a given statistical problem into an optimal method for estimation** in a wide variety of settings. It's not always totally automatic, but at a minimum it provides a clear way to think about what's better and what's worse. It liberates you from the route if-this-kind-of-data-then-that-method thinking that only makes for bad statistics and bad science.

Taken together, these three points give this movement a clear, unified, and increasingly popular perspective on causal inference. We have no doubt whatsoever that this is the paradigm that will come to dominate common practice in the next decades and century.

Pedagogy

The modern approach to causal inference is already well-established and discussed at length in many high-quality resources and courses. So why have yet another book going over the same material?

Well, there are many different kinds of students with different backgrounds for whom different presentation styles work better! We've chosen an approach for this particular text that focuses on the following principles:

1. **Rigor with fewer prerequisites**. Many students are turned off or intimidated by the amount of mathematical formality that is required to rigorously understand modern causal inference. While you don't actually need a lot of math to use and understand the ideas, you do need *some* math to be able to work independently in the field. Since one of our goals is for you to be able to *construct your own* efficient estimator for a never-before-seen problem, we have to think precisely and not hand-wave away rigor.

   For better or worse, the math you need is built on several layers of prerequisites- real analysis, probability theory, functional analysis, and asymptotic statistics (more on that later). In an ideal world, we would all have taken these courses and be prepared for what's next, but in practice we know many people have gaps in their knowledge.

   In this book we try to address that by providing a lot of foundational background information about these topics as they come up. We do our best to build up intuition for the component pieces of things before diving into implications. We lean heavily on figures and natural english prose to explain these concepts alongside the formal mathematical notation. We emphasize intuition and practical use rather than formalism. At a minimum, we try to be very clear about what the foundations actually *are* so that if you don't understand something you at least know where to look.

2. **Core concepts**. Modern causal inference is a huge field. There are so many exciting applications and theoretical developments that it can be hard for students to find their way at first.

   We've tried to keep thing as short and concise as possible in this book by focusing on core ideas and a limited number of examples. The point of this is to emphasize what is most important and to allow you to learn the fundamentals without getting distracted. That means there are many topics that aren't covered in this book (see section below), but the idea is that this book will give you the tools and mindset you need in order to continue your learning productively.

3. **A welcoming voice.** Modern causal inference is fast-moving and intense. It's easy to feel like you don't belong or aren't good enough to participate. The agnostic, impersonal tone that is best for writing a clear academic paper doesn't always create the warm, emotional connection that is necessary for us to feel secure and ready to learn.

   To deal with that problem, the voice we use throughout this book is informal and decidedly nonacademic. We write in the first person and address you, the reader, directly. Figures are hand-drawn and cartoonish. We provide personal commentary alongside agnostic technical material. All of this is deliberate and meant to help you feel comfortable and guided through this world. You are not alone in learning!

If you find this book isn't for you, that's also great! Here is a list of other resources that cover some of the same ground:

- *Semiparametric Doubly Robust Targeted Double Machine Learning: A Review* - Edward Kennedy
- *Semiparametric Theory and Missing Data* - Anastasios Tsiatis
- *Targeted Learning* - Mark van der Laan, Sherri Rose
- *Unified Methods for Censored Longitudinal Data and Causality* - Mark van der Laan, James Robins
- *Causal Inference: What If?* - Miguel Hernán, James Robins

Rigor with Fewer Prerequisites

Despite our best efforts, you will need *some* background material to get the most out of this book. The "hard" prerequisites listed are non-negotiable (this book will make no sense to you at all if you don't know what a probability distribution is), but the "soft" prerequisites are completely optional. We don't cover these topics here because there are already so many good resources to learn them and there's no need to reinvent the wheel. We provide the curated references below so you know where to go to fill in the holes in your understanding as you go along.

Hard Prerequisites You'll need to know probability theory at the undergraduate level. Random variables, expectation, conditional expectation, probability densities should be familiar to you, and ideally you have at least a vague understanding of what convergence in distribution means.

You also need undergraduate multivariable calculus and linear algebra: integrals, derivatives, vectors, etc. should all be well-worn tools at your disposal.

If you're shaky on these topics, we recommend Khan Academy. We also encourage you to check out the real analysis and measure-theoretic probability resources listed below, which will provide you with a very useful conceptual understanding.

Soft Prerequisites The following topics aren't necessary to grasp the arc of the arguments we'll make, but a fully rigorous understanding isn't possible without them. As we go along, we'll provide parenthetical commentary to elucidate concepts from these subjects that you may not be familiar with. We suggest attempting to read and understand this book all the way through and taking notes on the places where you feel like you're missing something. This will help you prioritize your time if you choose to fill in the gaps. None of this material is beyond you, I promise. It just takes a little time to work through it. I learned it all from these books myself: you may be surprised to hear I don't have any formal training in mathematical statistics!

These books and videos give what I think is the most self-contained treatment of their respective topics that is in line with the didactic approach taken in this book. They tend to have a more conversational tone, provide more background and intuition to the reader, and have plenty of worked examples. As always, however, there are many alternatives that may work better for you depending on your needs. These subjects should be tackled in the order they are listed here:

Field	Topics to Focus On	Resources
1. Intro Real Analysis	sequences, limits, series, continuity, convergence (pointwise, uniform), derivative	*Understanding Analysis* (Ch 1-6) - Abbott

Field	Topics to Focus On	Resources
2. Measure-Theoretic Probability	definition and construction of measure, Lebesgue integral and its properties, Radon-Nikodym theorem, change-of-variables formula, modes of convergence for random variables, independence, central limit theorem, conditional expectation	*Measure Theory Youtube Lectures* - Landim, *Measure, Integral, and Probability* - Capiński & Kopp
3. Functional Analysis in $\mathcal{L}_2$	projection, bounded dual space, Reisz representation	*Introductory Functional Analysis with Applications* (Ch 1-3) - Kreyszig

A few tips for self study: I recommend reading skimming each chapter first and getting an idea of what the main ideas are and how they interconnect to each other and to what you've already done. Take your time to understand what the definitions of the objects are that are used in each theorem (e.g. what exactly is a sequence / random variable / linear bounded functional?). Think of some examples and draw pictures.

Then go through the chapter again and read it slowly, line by line. Reading a line of math should take you as long as it takes to read a paragraph of english because that's often how much information is conveyed therein. Now draw some pictures or think of examples attempting the illustrate the content of the theorems. After you're done, do several of the exercises that the author provides. I've found that it also helps to make a "one-page summary" or "cheatsheet" of all the material in the chapter (definitions, important theorems, and their relationships). This will help you keep all the concepts organized as you go along.

I'll also refer to *Asymptotic Statistics* by A.W. van der Vaart in many places throughout the guide where we need to point to technical material. Relative to the way we write things up here, that book goes over background material much faster, so you may find it more challenging to read through. However, it does contain all the necessary information and details. If you've read the material in the table above and you really want to dot your i's and cross your t's on the material in this book you can try your hand at ch. 1, 2, 6, 7, 8, and 25 of Asymptotic Statistics.

Core Concepts

We've done our best to keep this book lean and targeted on foundational concepts. As a consequence, there are many important and interesting topics that we'll say nothing or little about. For the sake of completeness, we'll list some of them here so you know where to start looking when you're ready to build on your solid foundation.

To keep things simple and consistent, most of the examples in this book focus on inference of the average treatment effect in simple covariate-treatment-outcome observational or randomized studies. However, there are many data structures which are much more elaborate than this. For example, we do not fully discuss inference in longitudinal settings with time-varying covariates and treatments, inference when some outcomes are censored, stochastic interventions, or continuous treatments. Non-IID data (e.g. treatment spillover in a network) is another interesting topic that is not covered. Nonetheless, what is in this book will prepare you to study all of these topics and to understand them as special cases or extensions of the general approach presented here.

While the tools we describe in this book are extremely powerful, they are not *all*-powerful. Some estimands of legitimate interest are not "pathwise differentiable" and therefore cannot be analyzed with these tools. An important example of such an estimand is the conditional average treatment effect, which is the subject of extensive research. There is not yet a unified approach to these problems.

Another interesting and useful setting that we do not discuss is online (or reinforcement) learning. Many topics related to this literature (e.g. off-policy evaluation) are directly addressable with the methods in this book, but here we omit all discussion of scenarios where the treatment policy can be manipulated during data collection as more information comes in. The framework for causal inference we develop in this book does extend to such settings but only with some added complexity that is best understood after learning the basics.

Lastly, the reader should be aware that this book discusses the most popular and widely used framework for causal inference, which is that of potential outcomes in the frequentist population model. Readers may have heard of directed acyclic graphs (DAGs) or nonparametric structural equations models (np-SEM) as alternatives to potential outcomes, but these frameworks are in fact largely equivalent for the vast majority of practical problems so it is really not worth worrying about the differences until your are very deep in the weeds.

Potential outcomes are also used outside the population model in the randomization inference framework, which differs in that only the treatment assignment is assumed to be random (i.e. sampling from a larger population is not a source of variability). Much of what you learn here is also applicable to randomization inference. Potential outcomes are also commonly used in Bayesian approaches to causal inference, which we also don't discuss here.

Topics

This book aims to be a relatively self-contained treatment of the modern approach to causal inference. As we said above, the goal is for you to be able to take a vaguely-defined scientific question, translate it into a formal statistical problem, and solve it using a tool that you optimally construct for the job at hand. Thankfully, there's a well-developed process for doing that called the Statistical Roadmap which we follow closely in the chapters of this book:

1. To begin with, we need to first understand the process of ***statistical inference***. We do that by defining a set of assumptions that represent what we know is true about the real world along with some true quantity of interest. Only then do we propose some kind of method or algorithm that, given data, produces an estimate of this quantity. We should be able to show that, with enough data, our answer gets closer and closer to the truth and that we have some way to quantify the uncertainty due to random sampling. These are the topics we'll work through in chapter 1, closing out with an example that illustrates the danger of blindly applying popular methods without thinking through the question and assumptions.
2. Once we've understood the goal and process of inference, we introduce causality. Causal inference questions are just statistical inference questions but in an imaginary world where we can always observe the result of alternative "what if"s. Of course in the real world we can't see all the "what if"s. We only get to see what happened, not what *could* have happened had we made a different choice, taken a different drug, or implemented a different policy. To link the two worlds together we have to make some assumptions to guarantee that the answer to a statistical inference problem in the real world actually represents the answer to our causal question in the world of "what if"s. Chapter 2 lays out this process (***causal identification***) and gives examples. Chapter 2 also addresses why a well-defined intervention is important to formulating something as a causal question and discusses broadly applicable methods that quantify the impact of potentially erroneous identifying assumptions.
3. Once we have an identification result in hand, we are completely done with causality and can focus on solving the real-world inference problem. But for better or for worse there are thousands of inference methods, each of which work under different assumptions and target different parameters. Choosing among these is daunting and often puts the cart before the horse because we're forced to define our question such that an existing method can solve it. Chapter 3 lays the foundation to address this problem by showing that for a wide range of assumptions and questions, there is generally a way to quantify what the best possible estimator is in terms of the ***efficient influence function***. The efficient influence function completely describes the behavior of the optimal estimator in large samples. This optimal estimator always converges to the right answer with more data and has the lowest legitimate uncertainty possible under the given assumptions. We work through examples of deriving efficient influence functions for the average treatment effect in standard observational and randomized experiments.
4. Finally, chapter 4 does the hard work of putting the theory from chapter 3 into practice. Here we show how we can actually build an ***efficient estimator***- one which has the efficient influence function. In other words, given a statistical problem, we can construct an optimal method to solve it. We outline the three main strategies for doing this, which are called bias-correction, estimating equations, and targeted maximum likelihood. We show the different ways in which these strategies ensure that the constructed estimator

has the right behavior. All of them ultimately attain the same goal and are theoretically equivalent in large-enough samples, but the differences are relevant for how the constructed estimators perform in practice.

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Part I

Efficiency Theory

In the previous chapters we learned how to pose a good causal question and examined what kinds of causal assumptions we might have to make in order to translate the causal estimand to an observable estimand. But once an observable estimand is established, we're done with causality for the remainder of the analysis. At this point in a study we are left with a purely statistical problem: we want to estimate some aspect of a distribution (the estimand) given some finite samples from that distribution (the data).

The fact that identification and estimation can be cleanly separated is a wonderful feature of the modern approach to causal inference that was not widely appreciated or articulated for a long time. The causal problem is that we don't observe the right "columns" (the potential outcomes) in the "population dataset". The statistical problem is that we don't observe all of the "rows". These are fundamentally different issues that require different approaches, though there is some interplay if you get into the weeds (Aronow et al. 2025). In this part of the book we'll dedicate ourselves to the latter, purely statistical problem.

We will begin by examining how to define whether an estimator is in some sense "optimal" for a given estimand. If you've already done some applied causal inference, you probably already know that there are many different practical approaches for estimating causal effects (matching, regression, inverse probability weighting, etc.). How can we judge these against each other when in reality we never observe the "truth" that all of them are trying to estimate? What reason might we have to choose one over another? Once we've developed a characterization of optimality, we will see how to use that knowledge to construct optimal estimators for many kinds of estimands.

As a heads up, this chapter is technical, so you will see a lot of pointers to the appendices, which explain foundational concepts in mathematical statistics. As a reminder, these are targeted at readers who have taken a course in basic probability and statistics, but perhaps not real analysis or measure-theoretic probability. The appendices by no means cover all of what there is to know about their respective topics. While I strive to be as rigorous as possible, the primary goal is not rigor but instead to give the reader a sense of why and how the concepts are most often used in asymptotic statistics so that they can begin to distinguish between the main ideas and the mathematical details in the literature.

1 Asymptotic Linearity

1.1 What do we want from estimators?

Let's forget about what we know about estimators for a second and start from scratch. What is it that we *want* from an estimator in the first place? Can we set some "deal-breaker" conditions or some "nice-to-haves"?

Recall that an estimator $\hat{\Psi}$ is a function that takes a dataset (which we represent as an empirical distribution) $\mathbb{P}_n$ and returns a number. We use $\hat{\psi} = \hat{\Psi}(\mathbb{P}_n)$ to denote the *estimate*, which is a random variable. In practice, however, the term "estimator" is often used to refer to both the function $\hat{\Psi}$ and the random variable $\hat{\psi}$ interchangeably.

Any estimator $\hat{\Psi}$ can be characterized by the distribution of $\hat{\psi}$ at each sample size n from each distribution $\mathbb{P} \in \mathcal{P}$. What would we want these sampling distributions to look like relative to the true value $\psi = \Psi(\mathbb{P})$?

Why not demand that the estimate is always *exactly* equal to the estimand, i.e. $\hat{\Psi}(\mathbb{P}_n) = \Psi(\mathbb{P})$ for all distributions $\mathbb{P}$ in the model $\mathcal{P}$? That might seem completely absurd but it's worth taking a second to ask why that is really such a crazy thing to demand. It is less extreme to ask that our estimate is correct *on average*, but even this is usually not possible to do. So let's settle for asking that our estimator is *asymptotically* unbiased: the estimate is more or less right on average given a large-enough sample size. Informally: $\mathbb{E}[\hat{\psi}] \approx \psi$ for n large enough.

If I have two estimators that are both unbiased, which is better? Since the total estimation error $\mathbb{E}[(\hat{\psi} - \psi)^2]$ can be decomposed to $(E[\hat{\psi}] - \psi)^2 + \mathbb{V}[\hat{\psi}]$ (by adding and subtracting $\mathbb{E}[\hat{\psi}]$ inside the square and expanding), the unbiased estimator with the smallest overall error will be the one with the smallest variance. Therefore we should desire estimators that have small (asymptotic) sampling variance, subject to being (asymptotically) unbiased.

Let's throw in a practical consideration. When we estimate a quantity we usually report confidence intervals or p-values alongside the point estimate. Those inferential statistics depend on having some estimate of the sampling variance of our estimator, which we usually construct based on a theoretical understanding of what that sampling distribution should be. Many commonly used estimators are known to be asymptotically normal, which greatly facilitates inference. Therefore another desired criterion might be that our estimator should have a theoretically-understood sampling distribution.

Therefore a sensible list of desiderata for an estimator might be:

1. asymptotically unbiased
2. small asymptotic sampling variance
3. well-understood sampling distribution

1.2 Asymptotic Linearity

In light of these criteria, we will simplify the problem of finding an optimal estimator to finding an optimal estimator among those that are *asymptotically linear*. I'll lay out what that means, argue for why it's a reasonable restriction given our criteria above, and then explain why this restriction is helpful in characterizing optimality.

Asymptotic Linearity

Definition 1.1. As before, assume that Z_i are drawn IID from some distribution $\mathbb{P}$. An *asymptotically linear* estimator $\hat{\psi}$ is one that can be written like

$$\hat{\psi} = \psi + \frac{1}{n}\sum_i \phi(Z_i) + o_{\mathbb{P}}(n^{-1/2})$$

for some mean-zero function ϕ with mean zero $\mathbb{E}[\phi(Z)] = 0$ and finite variance $\mathbb{V}[\phi(Z)] < \infty$.

Warning

If you don't really understand what $o_{\mathbb{P}}(n^{-1/2})$ means in the definition above or why it implies the sample mean term "dominates" asymptotically, you should work through all of Appendix A before returning here.

The function ϕ is called the *influence function* of the estimator $\hat{\psi}$. A better name for it could be the "averaging transformation" of the data for a given estimator because once the data have been transformed by ϕ, the estimator can be closely approximated by the truth plus a sample average of those transformed variables (which have mean zero). In large samples, sample averages of mean-zero variables behave like Gaussian random noise. It's somewhat remarkable that a huge number of very complicated estimators actually behave this way, as we will see shortly. By definition, every asymptotically linear estimator has an influence function that satisfies the above criteria: having one is what makes the estimator asymptotically linear.

Example 1.1 (Sample Mean)**.** The sample mean estimator is one example of an asymptotically linear estimator. To see that, write $\hat{\psi} = \frac{1}{n}\sum Z_i$ as

$$\hat{\psi} - \psi = \frac{1}{n}\sum \underbrace{(Z_i - \psi)}_{\phi(Z_i)} + 0.$$

For this estimator, the remainder term is exactly 0, which clearly satisfies the requirement of being $o_{\mathbb{P}}(n^{-1/2})$.

The influence function for an estimator also depends on the distribution of the data. For example, in the case of the sample mean we saw that $\phi(z) = z - \psi$ which we should really write as $\phi_{\mathbb{P}}(z) = z - \Psi(\mathbb{P})$ to emphasize that this particular influence function depends on the mean of Z *under the distribution* $\mathbb{P}$. Often we write something like $\phi_{\mathbb{P}}(z)$ or even $\phi(\mathbb{P})(z)$ to mean the influence function of a given estimator *at a particular distribution* $\mathbb{P}$. But more often than not we leave the $\mathbb{P}$ part implicit to minimize notational clutter.

In fact, it's more appropriate to say that an estimator is asymptotically linear *for some set of distributions*, i.e. within some model. But usually the restriction on the set of distributions is not particularly meaningful so we just say the estimator is "asymptotically linear (under regularity conditions)" and leave it at that.

Exercise 1.1. Let's presume that Z are drawn from a distribution $\mathbb{P}$ such that $\sqrt{n}\left(\frac{1}{n}\sum Z_i\right)$ diverges as a random variable (see Section A.3.2 for a definition of divergence of random variables). A "textbook example" of such a distribution is Pareto$(1, 1.5)$, but this happens for many "fat-tailed" distributions. You can ask your favorite large language model if you want to see a proof.
Why does this tell you that the sample mean cannot be asymptotically linear at $\mathbb{P}$?
The point of this example isn't to convince you that the sample mean isn't asymptotically linear (it is for essentially all distributions we care about), it's just to cement the idea that asymptotic linearity of an estimator technically depends on what distributions you're talking about and to better understand the definition by considering a counterexample.

It's totally possible for two different estimators to have the same influence function (at each distribution). This is because two different algorithms can turn out to behave very similarly as the sample size increases. For example, we could construct a very silly alternative estimator for the mean by adding $1/n$ to our sample mean estimate. This extra term is certainly converging to zero at better than a $\sqrt{n}$ rate so it is negligible and the influence function for our alternative estimator is the same as for the sample mean. We'll also see the idea of two different estimators sharing an influence function come up later when we discuss different strategies for building optimal estimators: the estimators we get from using each strategy may be different, but they all have the same influence function and therefore all behave similarly in large-enough samples.

At this point you may be worried about how you can take your favorite estimator and derive its influence function. It's often possible to prove an estimator is aymptotically linear, but in this book we actually won't need to do that. We're going to go the other direction: starting from the estimand, we're going to come up with a way to determine the "best" possible influence function and then *construct* an estimator around it.

1.3 Benefits of AL Estimators

Given our desiderata, asymptotically linear (AL) estimators are a very sensible class of estimators to restrict ourselves to. For one, AL estimators are guaranteed to be consistent: that is, they converge to the estimand in probability and are thus asymptotically unbiased. This is guaranteed by the fact that any influence function ϕ is mean-zero so the associated sample mean converges to 0 by the law of large numbers.

> **Exercise 1.2.** Consider the silly estimator $\hat{\psi} = 0$ when attempting to estimate the population mean $\mathbb{E}[Z]$ in a model containing all possible finite-mean distributions of Z. Is this estimator asymptotically linear across the entire model? Why or why not?

Another related benefit of asymptotically linear estimators is that they satisfy our practical requirement of understanding the sampling distribution. By the central limit theorem, $\psi + \mathbb{P}_n\phi$ looks more and more like $\psi + \mathsf{N}(0, \mathbb{V}[\phi(Z)]/n)$ as the sample size increases. The remaining term $o_{\mathbb{P}}(n^{-1/2})$ is negligible in comparison (by definition). More often you will see this written formally as

$$\sqrt{n}(\hat{\psi} - \psi) \rightsquigarrow \mathsf{N}(0, \sigma_*^2)$$

where $\sigma_*^2 = \mathbb{V}[\phi(Z)] = \mathbb{P}\phi^2$ (since ϕ has mean zero).

We call σ_*^2 the *asymptotic variance* of our estimator. You should think of $\hat{\psi} - \psi$ as a normal distribution that is shrinking towards a spike at 0: its variance is decreasing at a rate of $n^{-1/2}$ by the central limit theorem (see Example A.6 in Section A.3.2). We therefore "stabilize" the shrinking distribution of $\hat{\psi} - \psi$ by "blowing it up" by a factor of $\sqrt{n}$. The asymptotic variance σ_*^2 is the variance of this stable normal distribution (compare rows 1 and 3 of the figure in Figure A.2). It is *not* the limit of the sampling variance $\mathbb{V}[\hat{\psi}]$ as a function of n because that limit is always zero on account of the fact that the estimate is converging to the estimand.

At any finite sample, the standard error of the estimate is roughly given by $\sigma = \frac{\sigma_*}{\sqrt{n}}$. Since the asymptotic variance is given by the variance of the influence function, understanding the influence function is all we need to understand the sampling distribution.

Constructing asymptotically valid confidence intervals and p-values for any asymptotically linear estimator is therefore as simple as estimating $\sigma_*^2 = \mathbb{P}\phi^2$. If we can construct an estimate $\hat{\sigma}_*$ that is close to σ_* for large-enough n then we can estimate the standard error of our estimate with $\hat{\sigma} = \frac{\sigma_*}{\sqrt{n}}$ we will get asymptotic 95% coverage of ψ with an interval of the form $[\hat{\psi} - 2\hat{\sigma}/\sqrt{n}, \hat{\psi} + 2\hat{\sigma}/\sqrt{n}]$ (this is a "Wald-type" confidence interval).

> **Exercise 1.3** (Wald Interval)**.**
>
> a. Assuming that $\sqrt{n}(\hat{\psi} - \psi) \sim \mathsf{N}(0, \sigma^2)$, prove that $\hat{\mathbf{C}} = [\hat{\psi} \pm 2\sigma/\sqrt{n}]$ is a 95% confidence interval, i.e. $\mathbb{P}\{\psi \in \hat{\mathbf{C}}\} > 0.95$.
> b. Assuming asymptotic linearity, how does our actual knowledge of the sampling

distribution of $\hat{\psi}$ compare to what we assumed in part (a)?

c. In the real world our confidence interval is $[\hat{\psi} \pm 2\hat{\sigma}/\sqrt{n}]$. How does this compare to what we used in part (a)?
d. Given (b) and (c), do you think the real-world confidence interval will generally have better-than 95% coverage for all sample sizes?

Example 1.2 (Variance of the sample mean)**.** In the case of the sample mean estimator $\hat{\psi} = \frac{1}{n}\sum Z_i$ we previously saw the influence function was $\phi(z) = z - \psi$. A plug-in estiamate of $\mathbb{P}\phi^2$ is therefore given by $\frac{1}{n}\sum \hat{\phi}(Z_i)^2 = \frac{1}{n}\sum(Z_i - \hat{\phi})^2$

Example 1.3 (Variance of ordinary least squares coefficients)**.** Let's say we're trying to estimate the regression coefficients in a linear model. Let $(Y, X) \sim \mathbb{P}$ with $X \in \mathbb{R}^d$. Let $\beta_0 = \text{argmin}_\beta \mathbb{E}\left[(Y - X\beta)^2\right]$ be the coefficients that minimize the population mean squared error. This will be our estimand. We haven't assumed the Y vs. X relationship is truly linear, we're just estimating the best linear approximation. As an estimator we will use the ordinary least squares (OLS) estimate $\hat{\beta} = \text{argmin}_\beta \frac{1}{n}\sum\left[(Y_i - X_i\beta)^2\right]$.

It is a well-known result that the OLS estimator is asymptotically linear under mild regularity conditions. For one proof of this see chapter 5 of Asymptotic Statistics (but it's not important at the moment). The influence function is

$$\phi(x, y) = \mathbb{E}[XX^\top]^{-1}x(y - x^\top\beta)$$

You may recognize this because it's where the "sandwich" estimator (also known as "Huber-White" or "robust" estimator) of the standard error comes from. The (asymptotic) sandwich standard error is a plug-in estimate for the quantity $\mathbb{E}[XX^\top]^{-1}\mathbb{E}[X(Y - X^\top\beta)^2X^T]\mathbb{E}[XX^\top]^{-1}$, which is nothing other than $\mathbb{P}\phi\phi^\top$, i.e. the equivalent of $\mathbb{P}\phi^2$ when ϕ is a vector-valued function (as is the case when the estimand is a vector).

In this example the dependence on $\mathbb{P}$ is through the expectation $\mathbb{E}[XX^\top] = \mathbb{E}_{\mathbb{P}}[XX^\top]$ and the true best regression coefficients $\beta = \beta_{\mathbb{P}}$. As usual, I've left off the explicit dependence on $\mathbb{P}$ in the notation.

The interesting thing about this example is that computing the estimate involves a fairly complex algorithm: either gradient descent or a series of matrix multiplications and inversions. It's not a "simple" estimator like the sample mean. Nonetheless, in large enough samples, the estimator behaves as if if were just a sample mean of a transformation of the data.

OLS is a special case of a huge class of estimators called M-estimators. Others include sample quantiles, robust regressions, and many others. All of these are asymptotically linear under mild regularity conditions.

In our list of desired characteristics we also wrote that we wanted estimators to have as small a variance as possible. Asymptotic linearity by itself doesn't guarantee small variance, but it does give us a very easy way to compare two estimators. If we have estimators $\hat{\psi}_1$ and $\hat{\psi}_2$ which are both asymptotically linear, then all we need to know to decide which one is better (in large-enough samples) is to compare their asymptotic variances which are given by $\mathbb{P}\phi_1^2$ and $\mathbb{P}\phi_2^2$.

You may also recognize $\mathbb{P}\phi^2$ as the squared $\mathcal{L}_2(\mathbb{P})$ norm of ϕ. Therefore we are comparing $\|\phi_1\|$ to $\|\phi_2\|$. In other words, the better estimator is whichever has the "smaller" influence function! In the following sections this is the key insight we will use to construct optimal estimators.

Warning

If you are unfamiliar with $\mathcal{L}_2$ spaces or don't understand why $\mathbb{P}\phi^2$ is a norm, you should work through through Section B.1 and Section B.2 now.

The theory of optimality we are building is only valid for asymptotically linear estimators. If there were many practically useful estimators that were *not* AL, this could be a problem because we would not able to usefully compare them. We'll discuss this more in a later section but the short answer is that AL estimators are an extremely broad class and we're not leaving much of anything on the table.

2 Regularity

While restricting ourselves to asymptotically linear estimators is a good start, unfortunately it is not quite enough to develop a sensible theory of optimality. In this section we'll first argue for why a further restriction is not only sensible but useful. Then we will come up with and refine a definition that suits our needs.

2.1 Hodges' Estimator

Let's presume we're interested in estimating the mean $\psi = \Psi(\mathbb{P}) = \mathbb{E}_{\mathbb{P}}[Z]$ from n IID draws of a scalar random variable Z that we know is normally distributed with standard deviation $\sigma = 1$. Our statistical model is therefore $\mathcal{P} = \{\mathsf{N}(\psi, 1) : \psi \in \mathbb{R}\}$, a one-dimensional parametric model. Obviously this is a contrived example where we know a lot about Z, but the point is to show that even in this extremely favorable setting we can be led astray by mathematical arguments that put our attention in the wrong place.

We'll consider two estimators:

$$\hat{\psi} = \frac{1}{n} \sum Z_i$$

$$\tilde{\psi} = \begin{cases} \hat{\psi} & |\hat{\psi}| > n^{-1/4} \\ 0 & |\hat{\psi}| \leq n^{-1/4} \end{cases}$$

The first estimator is just the usual sample mean. The second is what is called *Hodges' estimator*. When the sample is large, Hodges' estimator almost always returns the sample mean. But when the sample is small, Hodges' estimator often ignores the sample mean and returns zero.

Let's compare the asymptotic sampling distributions of the two. Since Z are normal random variables with unit variance and mean ψ, we know that the sample mean has the exact distribution $\mathsf{N}(\psi, 1/n)$. Thus it follows that

$$\sqrt{n}(\hat{\psi} - \psi) \rightsquigarrow \mathsf{N}(0, 1)$$

Warning

If you are not familiar with convergence in distribution you should work through Appendix A now.

Hodges' estimator, however, has the following asymptotic distribution:

$$\sqrt{n}(\hat{\psi} - \psi) \rightsquigarrow \begin{cases} \mathsf{N}(0,1) & \psi \neq 0 \\ \mathsf{N}(0,0) & \psi = 0 \end{cases}$$

Derivation of the asymptotic distribution of Hodges' estimator

Let's prove that $\sqrt{n}(\tilde{\psi} - \psi) \rightsquigarrow \mathsf{N}(0,0)$ when $Z \sim \mathsf{N}(0,1)$. This will require some understanding of what is meant formally by convergence in probability and a common probability tool called Chebyshev's inequality. It's not important to understand this proof if you're comfortable with the conclusion, I just provide it for completeness and as a refresher for those who are familiar with this kind of argument.

Since convergence in distribution and in probability are equivalent when the limit is a point mass (see Appendix A), the statement is the same as saying that $\sqrt{n}(\tilde{\psi} - 0) \xrightarrow{\mathbb{P}} 0$. That's what we will prove.

We can write Hodges' estimator as $\tilde{\psi} = 1(|\hat{\psi}| > n^{-1/4})\hat{\psi}$. Then, for any $\epsilon > 0$,

$$\begin{aligned} \mathbb{P}\left\{|\sqrt{n}\tilde{\psi}| > \epsilon\right\} &= \mathbb{P}\left\{1(|\hat{\psi}| > n^{-1/4})|\hat{\psi}| > \epsilon/\sqrt{n}\right\} \\ &= \mathbb{P}\left\{|\hat{\psi}| > \epsilon\sqrt{n} \,\middle|\, |\hat{\psi}| > n^{-1/4}\right\} \mathbb{P}\left\{|\hat{\psi}| > n^{-1/4}\right\} \\ &\leq \mathbb{P}\{|\hat{\psi}| > n^{-1/4}\} \end{aligned}$$

by definitions and total probability. We can bound the last term by applying Chebyshev's inequality to the distribution of $\hat{\psi}$ and using the probability threshold $t = n^{-1/2}$:

$$\mathbb{P}\{|\hat{\psi}| > n^{-1/4}\} = \mathbb{P}\left\{|\hat{\psi}| - \mathbb{E}[\hat{\psi}] > n^{-1/4}\right\} \overset{\text{Cheb.}}{\leq} \frac{\mathbb{V}[\hat{\psi}]}{n^{-1/2}} = \frac{1}{\sqrt{n}} \to 0$$

where we used the fact that $\hat{\psi} = \frac{1}{n}\sum Z \sim \mathsf{N}(0, 1/n)$ when $Z \sim \mathsf{N}(0,1)$.

This proves the desired convergence. Perhaps surprisingly, the proof still works if we are interested in proving that $r_n(\tilde{\psi} - \psi)$ converges in probability to 0 when the mean of Z is zero for an arbitrary rate r_n. In other words, this convergence happens arbitrarily quickly as far as the asymptotics are concerned.

Nearly identical arguments are used to show that Hodges' estimator has the same asymptotic distribution as the sampling mean when the true mean is not zero. This is a fine exercise to try on your own to practice your probability theory.

Like the sample mean, Hodges' estimator is definitely asymptotically linear: the influence function is

$$\phi_{\mathbb{P}}(z) = \begin{cases} z - \mathbb{E}[Z] & \mathbb{E}[Z] \neq 0 \\ 0 & \mathbb{E}[Z] = 0 \end{cases}$$

which is certainly a $\mathbb{P}$-mean-zero $\mathcal{L}_2(\mathbb{P})$ function for all $\mathbb{P}$ with finite mean of Z. Therefore if we were defining optimality among AL estimators, we would have to consider Hodges' estimator as a candidate.

Exercise 2.1. Judging by the asymptotic distributions, do you think the sample mean or Hodges' estimator is better?

The sample mean and Hodges' estimator are asymptotically the same across most distributions. But in any case where $\mathbb{E}[Z] = 0$, Hodges' is clearly better: its variance goes to zero even if blown up by the square root of n whereas the sample mean's variance does not. Therefore by these arguments Hodges' is the better estimator overall.

But if this is true, why do we use the sample mean instead of Hodges' estimator? And moreover why couldn't we reason the same way about a Hodges'-style estimator that has a special preference for some other value instead of zero? Wouldn't that estimator also be better than the sample mean for that special value and otherwise the same?

The reason we don't use Hodges' estimator is that the asymptotic arguments hide that this estimator actually performs much worse than the sample mean for most values of n and most possible distributions of Z. You can see this in a simulation (or work it out analytically):

The y-axis of Figure 2.1 shows the relative efficiency of Hodges' estimator, defined as the ratio of MSEs between Hodges' estimator and the sample mean: $\mathbb{E}[(\tilde{\psi} - \psi)^2]/\mathbb{E}[(\hat{\psi} - \psi)^2]$. If this is greater than one, then Hodges' estimator is better than the sample mean, if less than one, it is worse.

As you can see, regardless of what n is, Hodges' estimator is worse (and sometimes *much* worse) than the sample mean for the majority of the plotted values of ψ. So then why do the asymptotic arguments suggest that Hodges' estimator is actually better than the sample mean?

The problem is that the asymptotics are only concerned with the behavior of each estimator at any particular distribution of Z. In our contrived model, the distributions of Z are all normal with unit variance so a single distribution corresponds to a value of ψ. If you look at a single nonzero point on the x-axis (say, $\psi = 1/2$) and compare the relative efficiency as n increases you'll see that it does converge to 1: both estimators are asymptotically equivalent. At the point $\psi = 0$, however, the relative efficiency converges to 0 meaning that Hodge's estimator is infinitely more efficient at that point asymptotically. We can repeat our simulations but plot

```
`summarise()` has grouped output by 'n'. You can override using the `.groups`
argument.
```

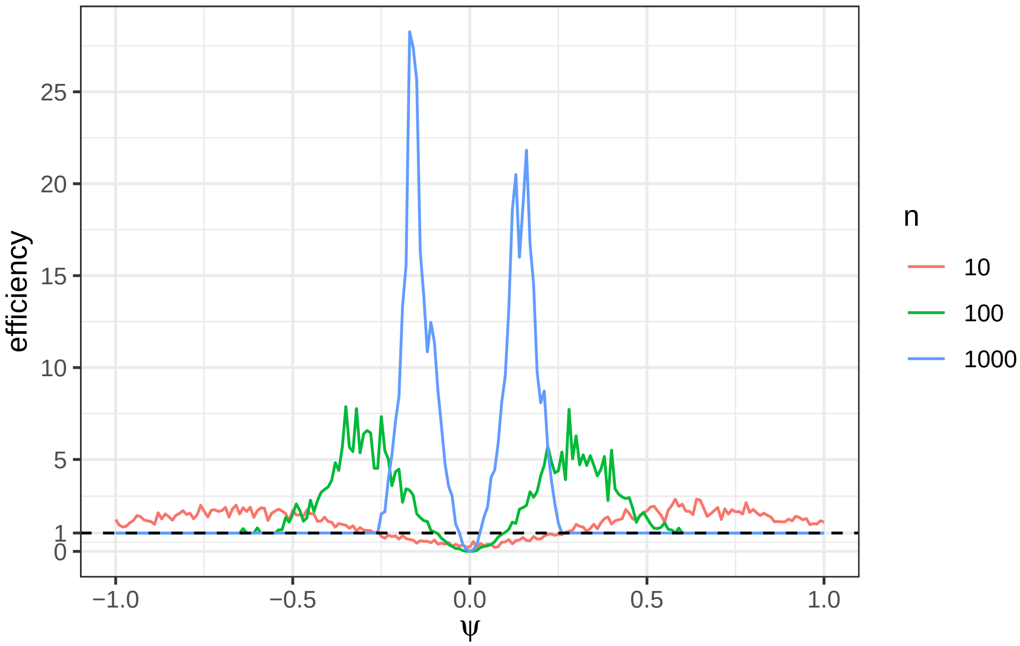

Figure 2.1: Relative efficiency of Hodges' estimator relative to the sample mean across ψ for a few values of n.

them a little differently in order to see this. Instead of plotting efficiency against ψ and slicing by n, we'll instead plot efficiency against n and slice by ψ (you could imagine a 3D plot with efficiency on the z-axis and n and ψ on the x and y).

```
`summarise()` has grouped output by 'n'. You can override using the `.groups`
argument.
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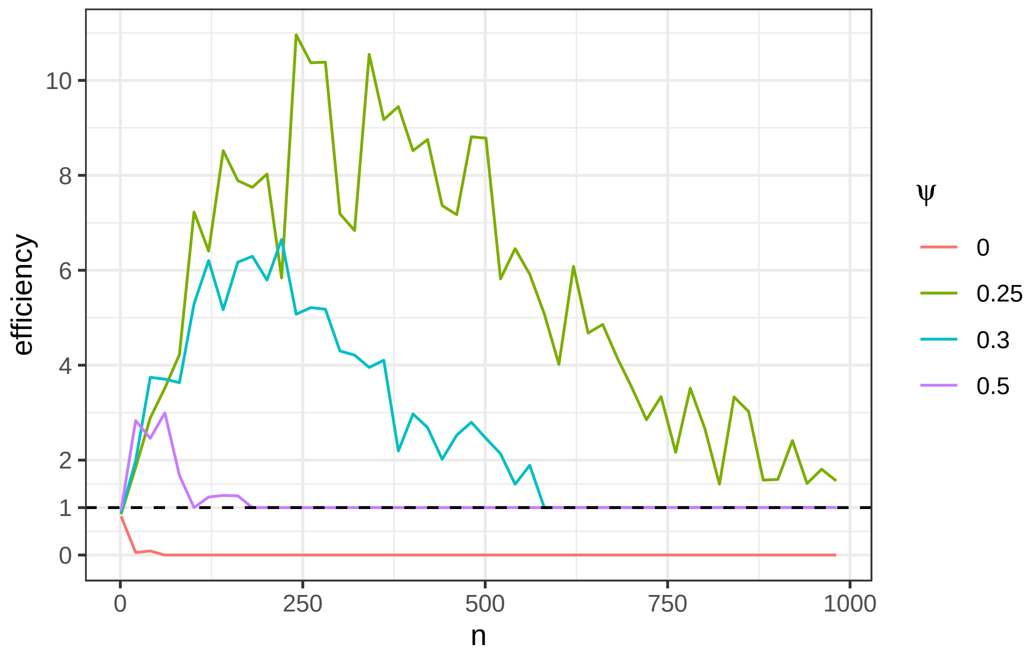

Figure 2.2: Relative efficiency of Hodges' estimator relative to the sample mean across n for a few values of ψ.

In Figure 2.2 you'll see that the bad behavior of Hodges' estimator is temporary for each value of $\psi \neq 0$. As n increases, the estimator eventually behaves like the sample mean. Since the asymptotics don't care about any initial behavior of the sequence, they completely ignore the fact that Hodges' estimator is usually much worse than the sample mean because those problems are "temporary" as far as the behavior at any fixed distribution is concerned.

The issue with that is that is that the problem doesn't actually disappear, it just goes to another distribution! As you can see in our first plot, the region of bad behavior simply moves as we increase n. If you're standing still at a given $\psi \neq 0$ you'd think everything is fine, but if you look across all values of ψ you notice that we're just moving the problem around. That is precisely what the asymptotic arguments ignore: they are incapable of looking at the behavior of each estimator *at multiple distributions simultaneously*. They only consider a single distribution at a time. We call this *pointwise* asymptotic behavior because we're looking at each "point" (distribution) $\mathbb{P}$ in the model $\mathcal{P}$ one-at-a-time. Once we look across all values of

ψ at the same time, it's clear that Hodges' estimator is much worse than the sample mean.

Asymptotic linearity is only concerned with pointwise convergence, but to prevent the bad behavior of Hodges' estimator we need some kind of *uniform* convergence: we'd like the sampling distributions of the estimates at each $\mathbb{P}$ to "arrive" or "approach" their limits at the same time. Physically you can imagine flattening a marshmallow: if you push down at a point on the marshmallow (e.g. by squeezing it between your thumb and index finger), the mass will can just go elsewhere. But if you sandwich it between your palms, you ensure that it gets squeezed down everywhere at the same time. That's "uniform" convergence.

> Warning
>
> See Section A.2.1 for the definition of uniform convergence of functions. This is a common type of convergence that we often want to generalize for other kinds of objects. In what follows we will present a notion of "uniform" convergence in distribution for random variables.

Estimators like Hodges' estimator are called *superefficient* for reasons that will be explained in a few sections' time. In general, superefficient estimators perform better than other estimators on some very small set of distributions at the cost of arbitrarily bad performance that moves around over the other distributions as n increases. We want to rule out that sort of behavior so that we can compare estimators asymptotically without getting burned by their finite-sample behavior.

2.2 Regularity in a Parametric Model

We've identified a problem with looking only at AL estimators: there are always AL estimators that look good in terms of their pointwise asymptotics but are actually bad because they just move the problem around in a way that makes it invisible to the asymptotics.

To rule this out, we'll impose an addition condition called *regularity*. Informally, regularity means that the estimator approaches its limit distribution in a (locally) uniform way across the model so that asymptotics can't hide bad performance that moves around the model.

We will start with a toy definition of regularity that we will build up to the formal definition in the next few sections. That will require us to introduce and explain a few abstractions but these will also come in handy later so it's worth doing. For now, though, here's a first draft:

> Regularity in the Normal Location Model
>
> **Definition 2.1.** An estimator $\hat{\Psi}$ is *regular* at the point $\mathbb{P} = \mathsf{N}(0, 1)$ in the model $\mathcal{P} =$

$\{\mathsf{N}(\psi, 1) : \psi \in \mathbb{R}\}$ if there is a single, fixed distribution $\mathbb{D}$ such that

$$\sqrt{n}\left(\hat{\Psi}(\mathbb{P}_n^{(n)}) - \Psi(\mathbb{P}^{(n)})\right) \rightsquigarrow \mathbb{D}$$

for any sequence of distributions $\mathbb{P}^{(n)} = \mathsf{N}\left(\psi_n, 1\right)$ such that $\sqrt{n}\psi_n$ is bounded.

First, let's clarify the notation. Since $\mathbb{P}_n$ is used to denote the (random) empirical distribution of n draws from a distribution $\mathbb{P}$, I'm using superscripts to denote a sequence of (fixed) distributions $\mathbb{P}^{(1)}, \mathbb{P}^{(2)} \ldots \mathbb{P}^{(n)} \ldots$. In this convention, $\mathbb{P}_n^{(m)}$ would be the empirical distribution of n draws from distribution $\mathbb{P}^{(m)}$.

Before understanding the definition above, let's break down the following standard asymptotic statement which says that $\hat{\psi}_n = \hat{\Psi}(\mathbb{P}^{(n)})$ is a $\sqrt{n}$-asymptotically normal estimate of $\psi = \Psi(\mathbb{P})$:

$$\sqrt{n}\left(\hat{\Psi}(\mathbb{P}_n) - \Psi(\mathbb{P})\right) \rightsquigarrow \mathsf{N}(0, 1)$$

For each sample size n, the left-hand side here is a random variable corresponding to the scaled and centered sampling distribution of the estimator, drawing the n data points from the distribution $\mathbb{P}$. You can imagine standing still at the distribution $\mathbb{P} \in \mathcal{P}$: first drawing many datasets of size $n = 1$ and passing them each through $\hat{\Psi}$ to generate the sampling distribution of $\hat{\psi}_1 - \psi$, then drawing many datasets of size $n = 2$ to generate the distribution of $\hat{\psi}_2 - \psi$, etc. We're just (repeatedly) drawing larger and larger datasets from the same distribution to feed through the estimator and comparing the results to the estimand. The right-hand side says that those (centered and scaled) sampling distributions look more and more like the standard normal the larger n gets.

Now compare that to our definition of regularity. The only difference is that instead of $\hat{\Psi}(\mathbb{P}_n)$ we have $\hat{\Psi}(\mathbb{P}_n^{(n)})$ and instead of $\Psi(\mathbb{P})$ we have $\Psi(\mathbb{P}^{(n)})$. What that means is that we are no longer necessarily "standing still" at a single distribution $\mathbb{P} \in \mathcal{P}$ while drawing larger datasets. Instead, we are typically *moving closer* to $\mathbb{P}$ as we increase the size of our draws.

Concretely, we start at $\mathbb{P}^{(1)} = \mathsf{N}(\psi_1, 1)$. We draw many datasets of size $n = 1$ from that distribution and pass them each through $\hat{\Psi}$, comparing the results to the current ground truth ψ_1\$. Now we move to $\mathbb{P}^{(2)} = \mathsf{N}(\psi_2, 1)$. We draw many datasets of size $n = 2$ from $\mathbb{P}^{(2)}$ and pass them each through $\hat{\Psi}$, comparing the results to the *new* ground truth ψ_2. We continue like this: at each n, we arrive at the distribution $\mathbb{P}^{(n)} = \mathsf{N}(\psi_n, 1)$, draw many datasets of size n, and generate the sampling distribution of the estimation error relative to the current ground truth. An estimator is said to be regular in the normal location model at the standard normal if that error distribution, scaled by $\sqrt{n}$, approaches a fixed distribution no matter how we choose the sequence of "truths" ψ_n as long as $\sqrt{n}\psi_n$ stays bounded.

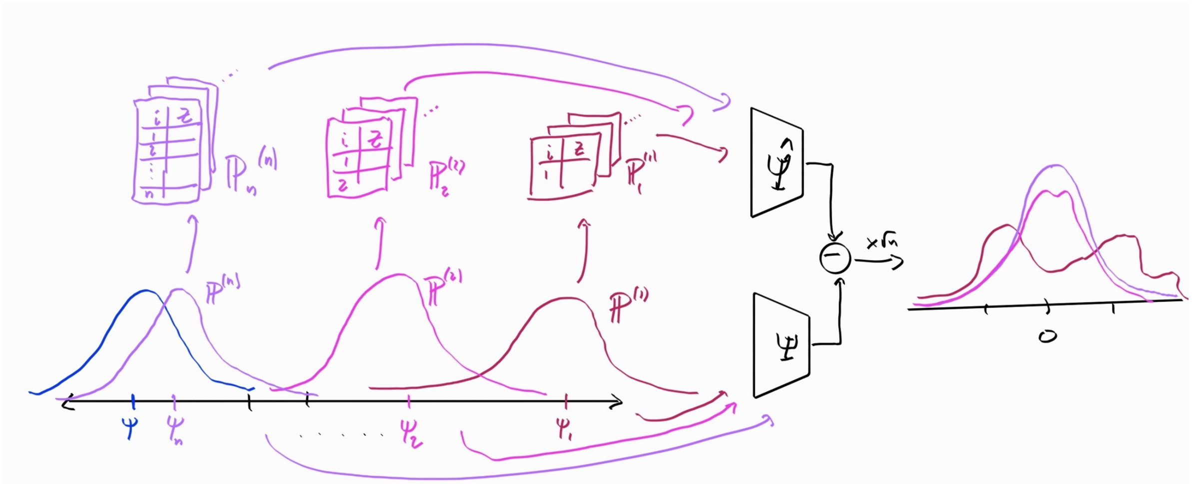

Figure 2.3: The asymptotics in the definition of regularity.

Draw a figure based on Figure 2.3 the depicts the standard asymptotic statement

$$\sqrt{n}\left(\hat{\Psi}(\mathbb{P}_n) - \Psi(\mathbb{P})\right) \rightsquigarrow \mathsf{N}(0, 1).$$

What parts of Figure 2.3 do you have to modify to accomplish this?

To summarize: the process in the definition of regularity is the same process as in the standard $\sqrt{n}$ asymptotic convergence statement. The only difference is that we're now allowed to shift the data generating distribution under the estimator and estimand rather than leaving it fixed. In the definition of regularity, however, the right-hand side must always be the same fixed distribution $\mathbb{D}$. In other words, no matter how we move the data-generating distribution around in a shrinking neighborhood of $\mathbb{P}$ while we increase the sample size, the estimator behaves more and more like it would if we stood still at $\mathbb{P}$. The fact that the limit is the same regardless of whether we're moving the ground truth distribution around (as long as it "converges quickly" to the true distribution) is the key to regularity. Satisfying that property is what forces our estimator to have (locally) uniformly good convergence across distributions and prevents the bad behavior of Hodges' estimator.

In our definition we only consider sequences ψ_n such that $\sqrt{n}\psi_n$ stays bounded. This means that $\psi^n\$$ is converging to the reference value $\psi = 0$ relatively quickly because even if we "blow up" ψ_n by an increasing factor of $\sqrt{n}$, the result *still* converges to zero. The fact that regularity is only concerned with *quickly*-converging sequences makes it more *lenient* than if we required the same kind of asymptotic convergence in distribution along *all* sequences $\psi_n \to \psi$. We want to a-priori rule out estimators whose bad performance will escape asymptotic detection, but we want to do this in the most lenient way possible so that we don't accidentally rule out some good estimators too. That's why we say that regularity enforces a kind of *local* uniform convergence in distribution: all the sequences of distributions we want to check still converge (quickly) to the distribution of interest. Another way to think of this is that the sequences we care about are all perturbations with in a "$\sqrt{n}$-sized" neighborhood of the distribution of interest.

Exercise 2.2. Prove that the sample mean is a regular estimator at any distribution $\mathsf{N}(\mu, 1)$ according to Definition 2.1:

a) If $\mathbb{P}$ is such that $Z_i \overset{\text{IID}}{\sim} \mathsf{N}(\mu, \sigma^2)$ for any $\mu \in \mathbb{R}, \sigma > 0$, what is the exact distribution of $\hat{\Psi}(\mathbb{P}_n) = \frac{1}{n} \sum Z_i$?
b) Using your answer from part (a), what is the exact distribution of $\sqrt{n}\left(\hat{\Psi}(\mathbb{P}_n^{(n)}) - \Psi(\mathbb{P}^{(n)})\right)$ at any given n, assuming $\mathbb{P}^{(n)} = \mathsf{N}(\psi_n, 1)$? What is the CDF $F_n(z)$ of this distribution at each n?
c) Recall that pointwise convergence of the CDFs is the definition of convergence in distribution (Definition A.6). Let $\sqrt{n}(\psi_n - \psi) \to 0$ (as examples you can try $\psi_n = \psi + \frac{1}{n}$ or $\psi_n = \psi$). Prove that under any such sequence, the sequence of CDFs $F_n(z)$ from part (b) converge at each point to the same limit $F_{\mathsf{N}(0,1)}(z)$, the CDF of the standard normal.

Exercise 2.3. In a derivation above we proved that Hodges' estimator has excellent asymptotic properties if we stand still at $\mathbb{P}^{(n)} = \mathsf{N}(0, 1)$:

$$\sqrt{n}\left(\tilde{\Psi}(\mathbb{P}_n^{(n)}) - \Psi(\mathbb{P}^{(n)})\right) \rightsquigarrow \mathsf{N}(0, 0).$$

Now we'll prove that Hodges' estimator is *not* a regular estimator according to this definition by showing that

$$\sqrt{n}\left(\tilde{\Psi}(\mathbb{P}_n^{(n)}) - \Psi(\mathbb{P}^{(n)})\right) \rightsquigarrow \mathsf{N}(0, 1)$$

if we move toward the true distribution like $\mathbb{P}^{(n)} = \mathsf{N}(\frac{1}{n}, 1)$.

a) Presume that $\hat{\psi} \sim \mathsf{N}(\mu, \sigma^2)$. Given that $\tilde{\psi} = 1(|\hat{\psi}| > n^{-1/4})\hat{\psi}$, what is the CDF of Hodges' estimate $\tilde{\psi}$ for any value of n? It may help you to draw a picture of the "density function" for Hodges' estimator for some fixed n (relative to the normal it has a spike at 0 comprised of the mass that was in the tails $|z| > n^{-1/4}$).
b) Using your answer from part (a), what is the exact distribution of $\sqrt{n}\left(\tilde{\Psi}(\mathbb{P}_n^{(n)}) - \Psi(\mathbb{P}^{(n)})\right)$ at any given n, assuming $\mathbb{P}^{(n)} = \mathsf{N}(\frac{1}{n}, 1)$? What is the CDF $F_n(z)$ of this distribution at each n?
c) Argue that this CDF $F_n(z)$ converges pointwise to the limit $F_{\mathsf{N}(0,1)}(z)$, the CDF of the standard normal.

Since the asymptotic distribution of Hodges' estimator is different depending on how we approach $\mathsf{N}(0, 1)$ with $\sqrt{n}\psi_n \to 0$ (standing still at $\psi_n = 0$ or with $\psi_n = \frac{1}{n}$), we conclude that Hodges' estimator is not regular at this distribution.

The definition we're working with at the moment is specific to the normal model $\mathcal{P} = \{\mathsf{N}(\psi, 1) : \psi \in \mathbb{R}\}$. We built a very specific sequence of distributions $\mathbb{P}^{(n)} = \mathsf{N}(\psi_n, 1)$ that get closer and closer to the true distribution. Moreover we required the "distance" to the true distribution to close at the $n^{-1/2}$ rate or faster. Our challenge in the proceeding sections will be figuring out a sensible way to generalize our definition to nonparametric models. But to do that we'll first have to establish some notions of "geometry" in that setting.

2.3 Paths and Scores

Our goal at our moment is to generalize our definition of regularity by establishing some "geometry" in our model so we can talk about a sequence of distributions converging in some required sense. In our current definition, a model $\mathcal{P}$ is just a set of distributions: there is no sense of direction or distance that is inherent to it. I think of a naked model like that as a big soup of distributions: everything is free to jumble around. We need a little more structure to make sense of regularity in a nonparametric model.

For the remainder of this section we'll work within models where every distribution $\mathbb{P}$ has a density function p. Formally, we will assume that all our distributions are absolutely continuous with respect to Lebesgue measure (don't worry if you don't know what that means). The results we will discuss actually do not depend on that assumption: the theory works exactly the same way for general models. However, the intuition is harder to get across in the general case, especially for readers who are new to measure-theoretic probability. I'll abuse notation where convenient and use $\mathcal{P}$ to refer to the set of densities p as well as the set of distributions (measures) $\mathbb{P}$. If you know some measure theory, you can for now imagine that all distributions in the model have a density with respect to an arbitrary dominating measure $\bar{\mathbb{P}}$ and replace references to $\mathcal{L}_2$ below with $\mathcal{L}_2(\bar{\mathbb{P}})$. Throughout the rest of the book I will also write integrals like $\int f(z)\ p(z)\,\mathrm{d}z$ with the standard differential $\mathrm{d}z$ so that they are comprehensible without measure theory. If you know measure theory, you can replace these with Lebesgue integrals with respect to the relevant dominating measure.

The first step to imposing a structure that will be useful to us is to think in terms of density functions rather than distributions (measures). If all the elements of the model have densities, we can think of the model as a collection of densities. Since densities are functions, we might be tempted to say they are elements of $\mathcal{L}_2$ because then everything is great: we have lengths, we have angles, we have all sorts of useful identities and relationships. But unfortunately that's not immediately possible. Recall Definition B.2 that $\mathcal{L}_2$ only includes functions that are square-integrable (have finite norm). However, there are some density functions for which $\int p(z)\,\mathrm{d}z$ may blow up.

Example 2.1. The function $p(z) = \frac{1(0<z<1)}{2\sqrt{z}}$ is a valid density because it is positive and integrates to one:

$$\frac{1}{2}\int_0^1 \frac{\mathrm{d}z}{\sqrt{z}} = 1.$$

But it is not square-integrable: the integral of its square is

$$\frac{1}{4}\int_0^1 \frac{\mathrm{d}z}{z} = \infty.$$

Warning

We will be drawing on properties of $\mathcal{L}_2$ substantially from here on out so this is a good time to work through Section B.2 if you haven't already.

Thankfully, this is an easy problem to fix! Instead of working with the density functions directly, we can work with the *root densities*. Since $\int p \, \mathrm{d}z = 1$, it's clear that $\int (\sqrt{p})^2 \, \mathrm{d}z = 1$. In other words, the square root of any density function is square-integrable and thus a function in $\mathcal{L}_2$. Moreover, all of these root densities have unit norm in $\mathcal{L}_2$! I'll use the symbol $\sqrt{\mathcal{P}} = \{\sqrt{p} : p \in \mathcal{P}\}$ to denote the collection of square roots of densities in the model. Because $\sqrt{\mathcal{P}}$ contains only positively valued functions with unit norm, we can visualize it as some subset of the unit ball in the "positive orthant" of $\mathcal{L}_2$. In describing "orthants" I am drawing on the intuition described in Section B.1.1 where I imagine the "axes" of $\mathcal{L}_2$ corresponding to the possible arguments of the function.

In addition to the model, in Figure 2.4 you can also see what we're going to call a *differentiable path* through the model. A path (also called a one-dimensional submodel) is intuitively what it sounds like: if you were standing at a point p (or $\mathbb{P}$) in the model and you started walking in some direction, you'd be tracing out an example of a path. Formally, a path is a one-dimensional set of distributions in the model, parameterized by some $\epsilon \in \mathbb{R}$. By "parameterized" I mean that every distribution in this set can be assigned a unique number ϵ. If you consider p_0 (or $\mathbb{P}_0$) to be the "start" of the path, you can think of the parameter ϵ as encoding "how far" you are along the path (and perhaps in which direction). The density (or distribution) indexed by the particular number ϵ is often written p_ϵ (or $\mathbb{P}_\epsilon$). Paths are going to be very useful tools for us to define what we mean by distributions getting close to each other in a sense that is required for a general definition of regularity. Paths will also come up in many other places so it is worth taking some time to understand them here.

Path (1D Submodel)

Definition 2.2. A *path* is any set of distributions $\mathcal{P}_\epsilon \subseteq \mathcal{P}$ that can be put into 1-to-1 correspondence with any interval of the reals containing 0.

Example 2.2. Presume the model is the set of all continuous distributions. Pick any two points $p_0, p_1 \in \mathcal{P}$. Then the set of densities $\left\{\left(\frac{1}{2} - \epsilon\right) p_0 + \left(\frac{1}{2} + \epsilon\right) p_1 : \epsilon \in [0,1]\right\}$. This is an

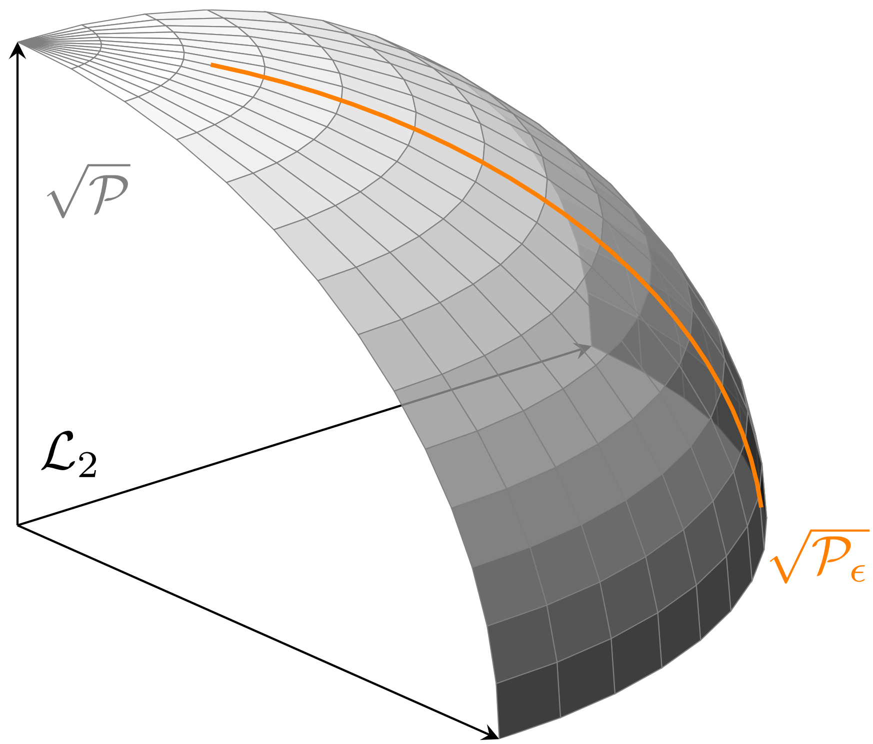

Figure 2.4: The model $\sqrt{\mathcal{P}}$ embedded in $\mathcal{L}_2$. In the drawing $\mathcal{L}_2$ is represented as 3D space but this is not meant to be literal- this is a 3D cartoon representing infinite-dimensional space. The model is represented as the visible quarter section of the unit sphere. That part of the unit sphere represents all densities. If the model were restricted in some way (e.g. all zero-mean distributions), the model could be visualized as some subset of the shown surface.

example of a *convex* path. You can think of this as the straight-line path starting at p_0 and ending at p_1 (at $\epsilon = 1/2$). Some algebra shows that another way to write the elements of this path is $p_\epsilon = p_0 + \epsilon(p_1 - p_0)$.

Example 2.3. Presume the model includes all normal distributions. Then the set of distributions corresponding to $\mathsf{N}(\epsilon, 1)$ is a path parameterized by $\epsilon \in \mathbb{R}$.

Exercise 2.4. Presume the model is the set of all mean-zero random variables. Why is $\{\mathsf{N}(\epsilon, 1) : \epsilon \in \mathbb{R}\}$ not a path here?

Example 2.4. Presume the model is the set of all densities. For $\epsilon \leq 0$ let p_ϵ be the density of $\mathsf{N}(\epsilon, 1)$ and for $0 < \epsilon$ let p_ϵ be the density $\mathsf{Unif}(0, 1 + \epsilon)$.

It's important that every element of the path is an element of the model. A path, by definition, stays within the model. Later we will see that when we have a smaller (more restricted) model we can sometimes construct estimators that have smaller asymptotic variance. This should make sense to you: the more you already know and can encode as a modeling assumption, the less you have to rely on the data and the more confident you can be in your result. Mathematically, the way that we show this is by arguing that there are "fewer" paths passing through a given point when the model is more restricted. If paths were allowed to "leave" the model, this wouldn't be the case.

Our definition of path is very general. Too general, in fact, to correspond nicely to the mental model we have from Figure 2.4. As Example 2.4 shows, we can easily construct paths that don't feel "continuous". To make a definition that reflects our picture of a continuous, smooth path through the model we need to impose more conditions.

For example, we might impose that paths be continuous in the sense that we would want the $\mathcal{L}_2$ distance between two (root) distributions $\sqrt{p_0}$ and $\sqrt{p_\epsilon}$ along the path to go to zero if $\epsilon \to \epsilon_0$. That would rule out the path in Example 2.4. But it turns out that what we need in order to generate a useful theory of efficiency is not just continuity, but also differentiability. That is, we are interested in continuous paths, but specifically those that do not have any "kinks". By that we mean that we're interested in *differentiable* paths. Since we're talking about a path through infinite-dimensional space, let's spend a little time developing our intuition about derivatives before we define a differentiable path.

Given a function $f : \mathbb{R} \to \mathbb{R}$, recall that its derivative at the point $x = 0$ (if it exists) is defined as $\dot{f}_0 = \lim_{x \to 0} \frac{f(0) - f(x)}{x}$. Another way to write this is that there exists a number $\dot{f}_0$ such that

$$\left| \frac{f(0) - f(x)}{x} - \dot{f}_0 \right| \overset{x \to 0}{\to} 0.$$

By $x \to 0$ we mean that the statement holds when replaced by *any* sequence x_n which converges to zero.

The derivative is usually thought of the "slope" of the function f at the point in question. You can think of it also as the "direction" that f is going at that point: in the simplest interpretation, a positive derivative means that f is increasing (going up), a negative derivative means that it's decreasing (going down). The magnitude of the derivative gives the rate of increase or decrease.

Let's generalize this a bit. You should recognize that the value of the derivative $\dot{f}_0$ is a number specifically because the function f outputs real numbers. If f returned finite-dimensional vectors, for example, then the numerator in the "slope" fraction $\frac{f(0)-f(x)}{x}$ would also be a vector and the derivative $\dot{f}_0$ would also have to be a vector in order for the difference of the two to be defined. Now that we're working with the difference of two vectors instead of the difference of two numbers, the appropriate measure of distance is a norm on the vector space rather than the absolute value (which is the usual norm on $\mathbb{R}$). To formalize this, let $\mathcal{V}$ be a vector space with a norm $\|\cdot\|_{\mathcal{V}}$. We'll say that $f : \mathbb{R} \to \mathcal{V}$ is differentiable at zero if there is some element $\dot{f}_0 \in \mathcal{V}$ such that

$$\left\| \frac{f(0) - f(x)}{x} - \dot{f}_0 \right\|_{\mathcal{V}} \overset{x \to 0}{\to} 0.$$

Paths, as we have defined them (Definition 2.2), can be thought of as functions mapping a value $\epsilon \in \mathbb{R}$ to a (root) density $\sqrt{p_\epsilon} \in \mathcal{L}_2$. We can therefore apply the above reasoning, letting ϵ play the role of x, letting the mapping $\epsilon \mapsto \sqrt{p_\epsilon}$ play the role of f, and letting $\mathcal{L}_2$ play the role of the generic vector space $\mathcal{V}$. Since in this case f maps numbers to $\mathcal{L}_2$, the derivative of f will also be an element of $\mathcal{L}_2$, i.e. a square-integrable function.

Differentiable in Quadratic Mean (DQM), Score Function

Definition 2.3. A path $\mathcal{P}_\epsilon$ is *differentiable in quadratic mean* (or just *differentiable* for short) at $\epsilon = 0$ if there is a function $\dot{\sqrt{p_\epsilon}} \in \mathcal{L}_2$ such that

$$\left\| \frac{\sqrt{p_\epsilon} - \sqrt{p_0}}{\epsilon} - \dot{\sqrt{p_\epsilon}} \right\|_{\mathcal{L}_2} \overset{\epsilon \to 0}{\to} 0.$$

We define $s = 2\frac{\dot{\sqrt{p_\epsilon}}}{\sqrt{p_0}}$ to be the *score function* of the path $\mathcal{P}_\epsilon$ at the point $\epsilon = 0$. Often the score can be written as the derivative of the log-likelihood evaluated at zero:

$$s(z) = \left.\frac{\mathrm{d}}{\mathrm{d}\epsilon}\right|_{\epsilon=0} \log p_\epsilon(z).$$

The score function for a path is always in $\mathcal{L}_2(\mathbb{P}_0)$ and always mean-zero at $\mathbb{P}_0$ in the sense that $\mathbb{P}_0 s = \int s(z)\ p_0(z)\,\mathrm{d}z = 0$.

Sometimes you may also see the DQM condition written out more explicitly and with the score function, e.g.

$$\int \left[\frac{\sqrt{p_\epsilon(z)} - \sqrt{p_0(z)}}{\epsilon} - \frac{1}{2} s(z) \sqrt{p_0(z)} \right]^2 \mathrm{d}z \overset{\epsilon \to 0}{\to} 0.$$

This is equivalent but in my opinion makes it opaque that $\frac{1}{2}s(z)\sqrt{p_0(z)}$ just represents the $\mathbb{R} \to \mathcal{L}_2$ derivative of $\epsilon \mapsto \sqrt{p_\epsilon}$.

Proof of Claims about Scores

Proof that $s \in \mathcal{L}_2(\mathbb{P}_0)$: By definition, $\dot{\sqrt{p_\epsilon}} \in \mathcal{L}_2$ and $\dot{\sqrt{p_\epsilon}} = \frac{1}{2} s \sqrt{p_0}$. Then

$$\|s\|^2_{\mathcal{L}_2(\mathbb{P}_0)} = \int s^2(z) p_0(z) \, \mathrm{d}z = \int \left(\dot{\sqrt{p_\epsilon}}(z) \right)^2 \mathrm{d}z = \left\| \dot{\sqrt{p_\epsilon}} \right\|^2_{\mathcal{L}_2} < \infty.$$

Proof that $\mathbb{P}_0 s = 0$: For this proof to make sense you'll need to have worked through Section B.3.

The definition of DQM is the same as saying that $\epsilon^{-1}(\sqrt{p_\epsilon} - \sqrt{p_0}) \overset{\mathcal{L}_2}{\to} \frac{1}{2} s \sqrt{p_0}$ as $\epsilon \to 0$ in the sense of Definition A.3. Since ϵ^{-1} diverges, there is no way for that series to converge unless $\sqrt{p_\epsilon} \overset{\epsilon \to 0}{\to} \sqrt{p_0}$. This is a rather formal way of saying that if the path $\epsilon \mapsto \sqrt{p_\epsilon}$ is differentiable, then it certainly must be continuous since that is a weaker condition.
As a conseqeunce of that continuity, $\sqrt{p_\epsilon} + \sqrt{p_0} \to 2\sqrt{p_0}$. Now we write

$$\mathbb{P}_0 s = \int s(z) \; p_0(z) \, \mathrm{d}z = \int \left(\frac{1}{2} s(z) \sqrt{p_0(z)} \right) \left(2\sqrt{p_0(z)} \right) \mathrm{d}z,$$

i.e. the $\mathcal{L}_2$ inner product between $\frac{1}{2}s\sqrt{p_0}$ and $2\sqrt{p_0}$. Notice that these two arguments are the limits of $\epsilon^{-1}(\sqrt{p_\epsilon} - \sqrt{p_0})$ and $\sqrt{p_\epsilon} + \sqrt{p_0}$ respectively because of what we showed above. Since the inner product is continuous (see Example B.7), the inner product at the limits is the same as the limit of the inner products at the sequences, i.e.

$$\begin{aligned} \left\langle \frac{1}{2} s \sqrt{p_0}, 2\sqrt{p_0} \right\rangle &= \lim_{\epsilon \to 0} \epsilon^{-1} \left\langle \sqrt{p_\epsilon} - \sqrt{p_0}, \sqrt{p_\epsilon} + \sqrt{p_0} \right\rangle \\ &= \lim_{\epsilon \to 0} \epsilon^{-1} \left(\|\sqrt{p_\epsilon}\| - \|\sqrt{p_0}\| \right) \end{aligned}$$

For any density p, we already showed that $\|\sqrt{p}\| = 1$, i.e. the root-densities are on the unit ball. Thus the difference of norms on the right-hand side cancels and we get 0 as desired.

For those of you who know measure theory, you can check that this definition does not actually require a global dominating measure for the entire path which is used to define the densities. For each sequence $\epsilon_n \to 0$ needed to check the DQM criterion you may construct a bespoke dominating measure using the convex combination of the countable number of distributions $\mathbb{P}_{\epsilon_n}$.

Instead of demanding that $\epsilon \mapsto \sqrt{p_\epsilon}$ be differentiable in the mean-squared sense at $\epsilon = 0$, we might instead consider asking for the existence of the derivative $\frac{\mathrm{d}}{\mathrm{d}\epsilon}\sqrt{p_\epsilon(z)}$ at every single point z. When both the z-pointwise derivative $\frac{\mathrm{d}}{\mathrm{d}\epsilon}\sqrt{p_\epsilon(z)}$ and the mean-square derivative $\dot{\sqrt{p_\epsilon}}$ exist they certainly agree with each other, but sometimes one exists and not the other. There are paths that are not z-pointwise differentiable but are still DQM: since the square-mean derivative integrates over the squared difference of the finite-ϵ "slope" term and the limit term in z, it does not care if that slope term blows up for some points z: they simply fall out of the integral. Similarly there are z-pointwise differentiable paths that are not DQM: refer to the results and examples in Section 7.2 of Van der Vaart (2000) if you are curious to learn more. The reason we care about DQM is because it will turn out later that this is precisely the condition required to establish something called local asymptotic normality which we will use to prove a very important result.

Even though DQM is the condition we need, it's still informative to consider the pointwise derivative because it explains why we write the derivative as $\dot{\sqrt{p_\epsilon}} = \frac{1}{2}s\sqrt{p_0}$ using the score function s. Starting with the pointwise derivative and using the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon}\sqrt{p_\epsilon(z)} = \frac{1}{2}(p_\epsilon(z))^{-1/2}\frac{\mathrm{d}}{\mathrm{d}\epsilon}p_\epsilon(z) = \frac{1}{2}\sqrt{p_\epsilon(z)}\frac{\mathrm{d}}{\mathrm{d}\epsilon}\log p_\epsilon(z)$$

where in the last line we used $\frac{\mathrm{d}}{\mathrm{d}\epsilon}\log p_\epsilon(z) = \frac{1}{p_\epsilon(z)}\frac{\mathrm{d}}{\mathrm{d}\epsilon}p_\epsilon$. Therefore, when the pointwise derivative exists, the score function s is equal to the rate of change of the log-likelihood at $\epsilon = 0$. For almost all practical purposes you can take this to be the definition of the score for a path at p_0, i.e. $s(z) = \frac{\mathrm{d}}{\mathrm{d}\epsilon}\log p_\epsilon(z)|_{\epsilon=0}$. From a pointwise perspective, the score is telling you how much to increase or decrease the value of the (log) density at each value z in order to move from one density in the path to another. This is shown in Figure 2.5.

I find it most helpful to think of the score function at a point p_0 in the path as the "direction" that the path is moving when it passes through that point. You can see this is the case geometrically in Figure 2.6. In the figure I've drawn $s\sqrt{p_0}$ as the *tangent vector* to the path at the point $\sqrt{p_0}$. This is justified by the fact that $\frac{1}{2}s\sqrt{p_0}$ is the derivative of $\epsilon \mapsto \sqrt{p_\epsilon}$ and therefore encodes the "direction" that function (path) is going. Another way of seeing it is that $\frac{1}{2}s\sqrt{p_0}$ is the "slope" of the best linear approximation to $\epsilon \mapsto \sqrt{p_\epsilon}$ at the point $\sqrt{p_0}$. So, roughly speaking,

$$\sqrt{p_\epsilon} \approx \sqrt{p_0} + \frac{1}{2}\epsilon(s\sqrt{p_0}).$$

This is exactly what I've drawn in Figure 2.6: the actual path of densities $\sqrt{p}_\epsilon$ (the left-hand side of the above display) and what would happen if you followed the current direction $s\sqrt{p}$ starting at $\sqrt{p}$ (the right-hand side).

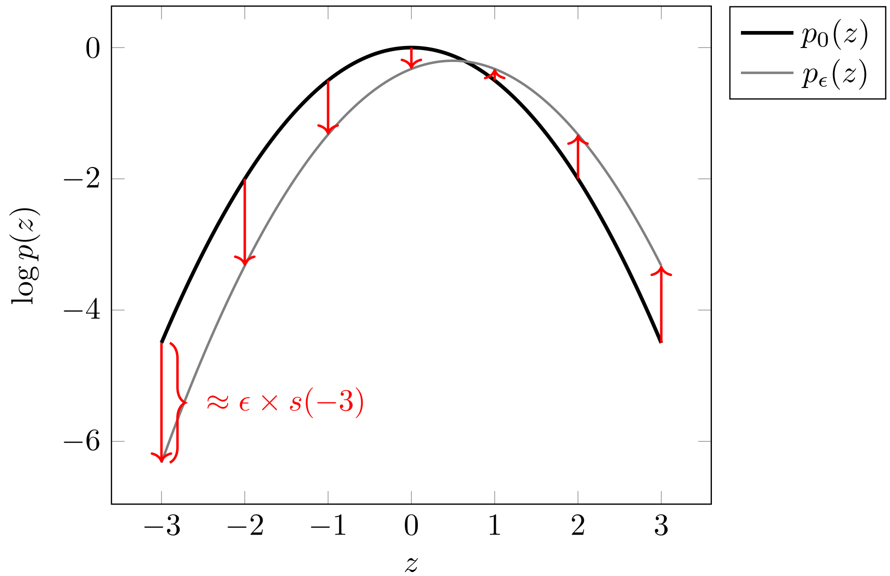

Figure 2.5: The (log-)density at ϵ for any value z can be approximated by adding the score (times ϵ) to the starting (log-)density.

To make an analogy, let's say you were following a path from New York to San Francisco. A "path" would be equivalent to a line on the map indicating the route you take. Your "score" at some point along the route would be the direction that the route tells you to go in at that point (e.g. in New York your "score" might be "West").

Technically it's not the scores themselves that are tangent to paths in the model, but the scores times the current root-density that are tangent to the root-path. Nonetheless we often speak informally and talk about the scores themselves being tangent, etc, ignoring the fact that we had to take roots of the densities in our model to get everything embedded into $\mathcal{L}_2$. In line with that oversimplification, I sometimes find it helpful to think in terms of or refer to cartoons like Figure 2.7.

Example 2.5. Fix an initial point p_- and final point p_+ in a model and consider the convex path $\{p_0 + \epsilon(p_1 - p_0) : \epsilon \in [0, 1]\}$ discussed in Example 2.2. For this path, the (pointwise

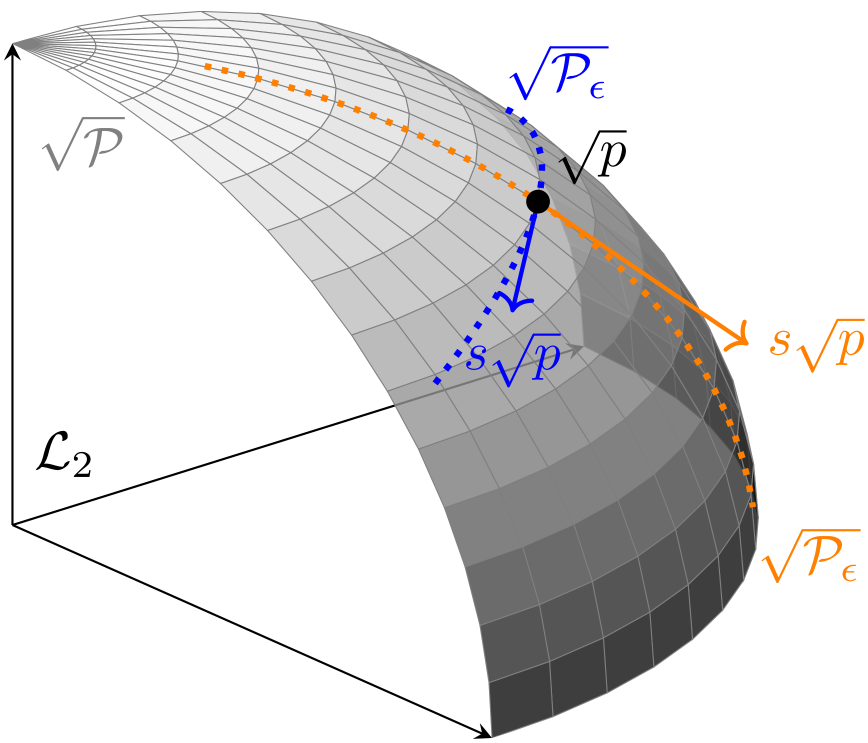

Figure 2.6: The score s (times $\sqrt{p}$) is tangent to the path $\sqrt{\mathcal{P}_\epsilon}$ at $\sqrt{p}_0$. In the figure I've shown two different paths passing through the same point $\sqrt{p}$ (in two different colors) and their respective scores.

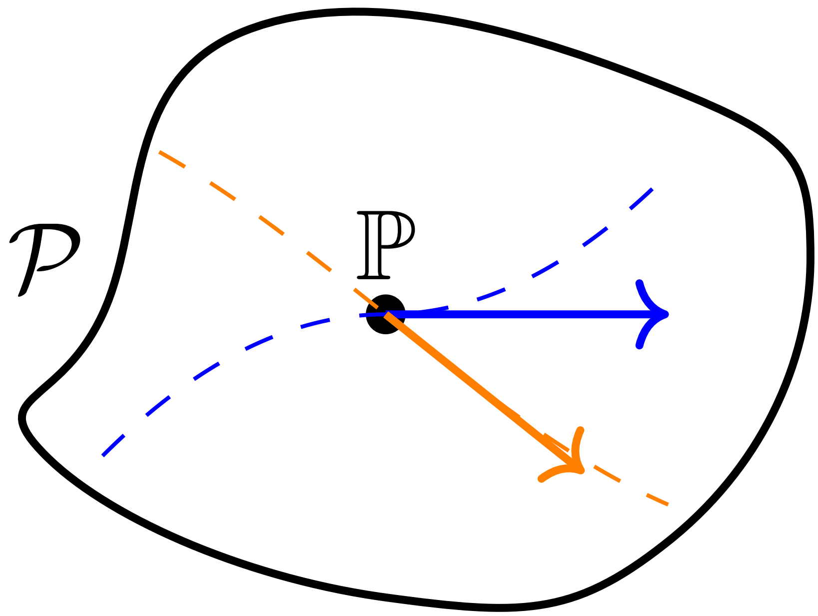

Figure 2.7: A cartoon of the model and a distribution $\mathbb{P}$ within it. Two differentiable paths are depicted with dotted lines (in different colors) and the scores for those two paths at $\mathbb{P}$ are shown as tangent vectors in the corresponding colors.

derivative version of the) score at $\epsilon = 0$ is defined and is equal to

$$\begin{aligned}
s &= \frac{\mathrm{d}}{\mathrm{d}\epsilon} \log\left(p_0 + \epsilon(p_1 - p_0)\right)\Big|_{\epsilon=0} \\
&= \frac{1}{p_0 + \epsilon(p_1 - p_0)}(p_1 - p_0)\Big|_{\epsilon=0} \\
&= \frac{p_1 - p_0}{p_0} \\
&= \left(\frac{p_1}{p_0} - 1\right)
\end{aligned}$$

A little algebra shows that this path can also be written as $p_\epsilon = (1 + \epsilon s)\, p_0$.

Exercise 2.5. While it's most helpful to think of scores as encoding "direction" within a model, scores also encode the "speed" at which a path goes.
Presume that the score for a path $\epsilon \mapsto p_\epsilon$ is s. In terms of s, what is the score for the reparametrized path $\epsilon \mapsto p_{c\epsilon}$ for some $c \in \mathbb{R}$? To solve this problem you can use the pointwise derivative version of the score for convenience. For intuition, note that for $c > 1$, the reparametrized path is the same as the original, just traversed "faster", whereas for $c \in (0, 1)$ it is the same path traversed more "slowly" (and for $c < 0$, in reverse).

Paths and scores are extremely useful objects in nonparametric asymptotics and we will use them many times in the rest of the book.

2.4 Regularity

Equppied with these tools we can craft a definition of regularity that is more general than Definition 2.1. In this section I'll return using superscripts to denote a sequence of distributions so that the subscript n can indicate the empirical distribution. For example, in the context of a path, I'll use $\mathbb{P}^{(\epsilon)}$ rather than $\mathbb{P}_\epsilon$ (and I'll use distributions $\mathbb{P}$ rather than densities p since we can now be more general).

First let's generalize the model. In our previous definition we had the normal model $\mathcal{P} = \{\mathsf{N}(\psi, 1) : \psi \in \mathbb{R}\}$. This is actually one example of a differentiable path within a larger, possibly nonparametric model (e.g. all continuous distributions). So let's replace the normal model with a generic differentiable path that includes the true distribution (at $\mathbb{P}_0$ so $\epsilon = 0$ is the true value, for convenience). Next, we can replace the condition that $\sqrt{n}\psi_n$ stays bounded with an equivalent condition on the parameter in the path. That yields our general pathwise definition of regularity:

Pathwise Regularity

Definition 2.4. An estimator $\hat{\Psi}$ is *regular* at $\mathbb{P}_0$ in the differentiable path $\mathcal{P}_\epsilon$ if there is a single, fixed distribution $\mathbb{D}$ such that

$$\sqrt{n}\left(\hat{\Psi}(\mathbb{P}_n^{(n)}) - \Psi(\mathbb{P}^{(n)})\right) \rightsquigarrow \mathbb{D}$$

for *any* sequence of distributions $\mathbb{P}^{(n)} = \mathbb{P}^{(\epsilon_n)}$ such that $\sqrt{n}\epsilon_n$ remains bounded.

In other words, an estimator is regular in a given differentiable path if the limiting distribution doesn't depend on what distributions you draw from asymptotically as long as they move to true distribution quickly enough. Recall that $\sqrt{n}\epsilon_n$ staying bounded (not diverging) means that $\epsilon_n \to 0$ faster than the $\sqrt{n}$ factor blows up; another way to write this condition is $\epsilon_n = O(n^{-1/2})$. The "uniform" behavior of regular estimators across "local" distributions in the path is what prevents the bad behavior of Hodges' estimator. The defects of irregularity escape detection in pointwise asymptotic arguments where the distribution is fixed.

To generalize our definition to a larger, possibly nonparametric model, we simply require that the limiting distribution $\mathbb{D}$ is also the same no matter which differentiable path through $\mathbb{P}$ you choose. In other words, we require the "uniformity" of convergence in distribution across all sequences of distributions converging quickly to the truth along differentiable paths.

Regularity

Definition 2.5. An estimator $\hat{\Psi}$ is *regular* at a point $\mathbb{P}$ in the model $\mathcal{P}$ if is is regular with the same limiting distribution $\mathbb{D}$ along all paths $\mathcal{P}_\epsilon \subseteq \mathcal{P}$ that are differentiable at $\mathbb{P}_0 = \mathbb{P}$. An estimator is regular in a model $\mathbb{P}$ if it is regular at every point $\mathbb{P} \in \mathcal{P}$.

We do not care about the behavior of the estimator along non-differentiable paths because these paths don't go smoothly towards the true distribution. It would not be sensible to assume that any estimator should behave in a predictable or consistent way if we moved along a non-differentiable path. Moreover we do not care about sequences of distributions along paths that do not converge quickly enough (at a $n^{-1/2}$ rate) to zero: we have no need to impose uniformity outside of "neighborhoods" of distributions that are roughly of "radius" $n^{-1/2}$ in order to get a nice, general theory of optimality.

Proving that an estimator is not regular is only as difficult as finding a single differentiable path for which it is not regular. We already did that for Hodges' estimator in Exercise 2.3. To directly prove an estimator *is* regular you have to prove regularity for all differentiable paths. There are ways to do this but more often it is easier to prove regularity *indirectly*. In the next chapter we will prove a powerful result that lets us characterize the set of all regular, asymptotically linear estimators. We can then show an estimator is regular by proving it is asymptotically linear and also meets the conditions required for membership in that set. More importantly, the characterization we develop will let us precisely identify the *most efficient* regular and asymptotically linear estimator.

2.5 Is Regularity Too Strict?

By restricting our attention to *regular* asymptotically linear estimators we are trying rule out the kind of bad performance exhibited by Hodges' estimator. But Hodges' estimator is just one example. It's rational to ask whether there are any *good* estimators that regularity rules out. In practice, the short answer is no, but in theory the long answer is that it depends on your perspective. I won't give a detailed account of all the arguments but I will give a brief summary that should at a minimum hold you over until you can dig deeper. Nonetheless, if you are interested in these questions I recommend revisiting them *after* you have digested the practical implications of efficiency theory that we'll discuss in the next chapters.

The first argument that usually gets rolled out in favor of enforcing regularity is that irregular estimators can only perform asymptotically better than regular estimators on an "infinitesimally small" fraction of all possible distributions in any parametric model. We saw this for Hodges' estimator: it has the same asymptotic performance as the sample mean in the normal model everywhere except for the *single distribution* where $\psi = 0$. That's the only point at which Hodges' estimator can beat the sample mean in large-enough samples. The theoretical justification for this claim is something called the almost-everywhere (or Hajek-Le Cam) convolution theorem. You can see section 8.6 of Van der Vaart (2000) for further details. The idea is that if irregular estimators can only beat regular estimators in very rare cases, it's no big deal to exclude them. As a bonus, the convolution theorem also shows that the most efficient regular estimator is asymptotically linear. This justifies restricting our search to asymptotically linear estimators once we're already comfortable with regularity since we lose nothing (asymptotically) by the further restriction.

A second argument for regularity follows from the local asymptotic minimax theorem (Theorem 8.11 and Lemma 8.13 in Van der Vaart (2000)). Given an estimate $\hat{\psi}$, we can imagine calculating a scaled error $E_{n,\mathbb{P},\hat{\Psi}} = \sqrt{n}(\hat{\psi} - \psi)$. This error is indexed by the sample size n, the distribution $\mathbb{P}$ we're drawing data from and calculating the estimand ψ from, and of course the estimator $\hat{\Psi}$. Since $\hat{\psi}$ is random, $E_{n,\mathbb{P},\hat{\Psi}}$ is also a random variable: we get a different error with a different sampled dataset. We can also define a loss function $l : \mathbb{R} \to \mathbb{R}$ which takes the error and returns its "cost". We'll require that the loss be "bowl-shaped": symmetric around 0 and either increasing or flat moving away from zero. For example, we might use the squared loss function $l(e) = e^2$ if we want to strongly penalize large errors but be relatively more lenient towards small errors, or we could use $l(e) = 1(|e| > c)$ for a constant penalty outside a region and no penalty within it.

The local asymptotic minimax theorem is concerned with the expected loss $\mathbb{E}_{\mathbb{P}}[l(E_{n,\mathbb{P},\hat{\Psi}})]$. Clearly we would like to find an estimator that makes the expected loss as small as possible. But the theorem tells us that there is a limit to how low the expected loss can get if we want the loss to be *uniformly* low in some sense around a distribution $\mathbb{P}$. Let $\mathbb{P}^{(n)}$ denote any sequence $\mathbb{P}^{(\epsilon_n)}$ in a differentiable path with $\mathbb{P}_0 = \mathbb{P}$ such that $\sqrt{n}\epsilon \to 0$. Informally, the theorem establishes that in most models, if we consider a large enough set of possible differentiable paths and sequences and take n to be large enough, then there is *no* estimator for which the worst-case value (over paths) of $\mathbb{E}_{\mathbb{P}^{(n)}}[l(E_{n,\mathbb{P}^{(n)},\hat{\Psi}})]$ can be meaningfully better than $\mathbb{E}_{Z\sim\mathsf{N}(0,\sigma^2_{\mathbb{P}})}[l(Z)]$. The quantity $\mathbb{E}_{Z\sim\mathsf{N}(0,\sigma^2_{\mathbb{P}})}[l(Z)]$ is a sort of universal "speed limit" for good locally-uniform estimation of the estimand at $\mathbb{P}$ as measured by the loss l. In the next few chapters we'll say a lot more about the mysterious quantity $\sigma^2_{\mathbb{P}}$ and how to derive it.

It's natural to ask whether there are estimators that can meet this bound, i.e. whether there are estimators that are *locally asymptotically minimax* for a loss l. If such an estimator were available, there would be very good reason to prefer it over alternatives. The second part of the local asymptotic minimax theorem establishes that, roughly speaking, only *regular* estimators can be locally asymptotically minimax for scalar estimands. One implication of this theorem is that the undesirable behavior exhibited by Hodges' estimator is characteristic of *all* superefficient estimators. If there is no *irregular* locally asymptotically minimax estimator, that means that irregular estimators *all* have regions of bad performance with in the model that shift towards the point of superefficiency as n increases.

Together, the convolution and local asymptotic minimax theorem provide strong reasons to prefer regular estimators, but ultimately both are asymptotic statements and say nothing about how irregular estimators might perform in finite samples. Moreover, parts of the local asymptotic minimax theorem don't hold if the estimand is a vector rather than a scalar. For example, the James-Stein estimator is an *irregular* estimator of a vector-valued mean that *also* locally asymptotically minimax (in a sense defined for vector estimands) for the squared error loss (see Section 8.8 of Van der Vaart (2000) or Chapter 2 of Ibragimov and Has' Minskii (2013)). But even this is a qualified improvement: while James-Stein improves on the sample mean for squared error loss, there are other losses for which it can be much worse.

Ultimately, the strongest argument in favor of assuming regularity is a practical one: if you assume regularity, you get a very clear theory of optimality that works in immense generality for many statistical models and estimands. It's possible to cook up small-sample scenarios where irregular estimators can win, but given a general setup and without very specific assumptions it's not usually clear how to *generate* a good irregular estimator from scratch. Moreover it can be difficult to characterize the sampling distribution of irregular estimators, making it hard to construct confidence intervals and p-values. In contrast, our efficiency theory for regular estimators will allow us to build great estimators with easy ways to do inference for essentially any situation.

3 The Efficient Influence Function

In Chapter 1 we defined *asymptotic linearity* (AL) and discussed why AL estimators are nice: they are asymptotically unbiased, are easy to do inference for, and can be (asymptotically) compared by looking at their influence functions. In Chapter 2 we argued that we should further restrict the class of desirable estimators to those that are *regular* so that bad finite-sample behavior can't be hidden in the asymptotic comparisons. In this chapter we will establish the central result of efficiency theory: in any model where the estimand varies smoothly enough over the different distributions, there is a best-possible regular asymptotically linear (RAL) estimator.

Really what we'll prove is that there is a best-possible *influence function*: multiple estimators may have that influence function but by virtue of that fact they will behave similarly in large-enough samples. We call the best-possible influence function the *efficient* influence function (EIF). The EIF sets a limit on how good any RAL estimator can be at estimating some estimand in a some model. If you have a RAL estimator with that influence function, you can rest assured that it is the best you can do (asymptotically).

In this chapter we will prove our central result and discuss some important implications. In the next chapter we'll show how to derive the EIF in the most common scenarios. After that, in the last part of the book, we will demonstrate how to use knowledge of the EIF to *construct* efficient estimators.

3.1 Characterizing Influence Functions

Not every possible function corresponds to a valid RAL estimator of an estimand in a particular statistical model. Clearly it would be wonderful if we could find a RAL estimator with $\phi(z) = 0$ because this estimator would have no variance at all in large samples. But if that were possible, statistics would be easy. So $\phi(z) = 0$ must not be an influence function for any sensible estimand in any sensible model. So, for a given estimand and statistical model, what functions are influence functions of RAL estimators, and what functions aren't?

We already know that any regular estimator by definition behaves in a particular way asymptotically along a quickly-converging sequence of distributions. Remarkably, it turns out that asymptotic linearity also implies specific asymptotic behavior along a subset of these sequences.

Asymptotic linearity along differentiable paths

Theorem 3.1. *Let $\mathcal{P}_\epsilon \subseteq \mathcal{P}$ be any differentiable path with score function s and let $\mathbb{P}^{(\epsilon_n)}$ be a sequence of distributions along this path with $\epsilon_n = c_n n^{-1/2}$ for any converging sequence of real numbers $c_n \to c$. Let $\mathbb{P} = \mathbb{P}^{(0)}$ denote the true data-generating distribution.*
If an estimator $\hat{\Psi}$ is asymptotically linear with influence function ϕ, then

$$\sqrt{n}\left(\hat{\Psi}(\mathbb{P}_n^{(\epsilon_n)}) - \Psi(\mathbb{P})\right) \rightsquigarrow \mathsf{N}(\mathbb{P}[s\phi], \mathbb{P}\phi^2).$$

Proof Outline

A thorough proof of this result is too technical for this book so I will only give an outline here. While I do encourage you to dig into the theory if you are curious (since it has other applications as well), I think this is best done *after* understanding the implications of the result in our present context.
Let p and p_{ϵ_n} denote the densities of $\mathbb{P} = \mathbb{P}^{(0)}$ and $\mathbb{P}^{(\epsilon_n)}$, respectively (you can either assume we're working in a model with densities or recall that we can always construct densities along sequences of a differentiable path).
The first step is to show that any differentiable path is in fact a *locally asymptotically normal* (LAN) model. This is proved as Theorem 7.2 in Van der Vaart (2000) and while the arguments are accessible with basic asymptotic statistics, I don't think they add too much insight. In LAN models we only consider sequences of the type $\epsilon_n = c_n n^{-1/2}$ where $c_n \to 1$. This kind of sequence is a special case of the sequences where $\sqrt{n}\epsilon_n$ stays bounded that we considered in the definition of regularity. The reason we need to consider this special case is that these are the sequences for which $\sqrt{n}(\sqrt{p_{\epsilon_n}} - \sqrt{p})$ converges in $\mathcal{L}_2$ to a fixed limit $\frac{1}{2}s\sqrt{p}$ (by Definition 2.3). The $\sqrt{n}$ convergence of the root densities is required to show that

$$\sum_i^n \log \frac{p_{\epsilon_n}(Z_i)}{p(Z_i)} = \frac{1}{\sqrt{n}}\sum_i^n s(Z_i) - \frac{1}{2}\mathbb{E}[s(Z)^2] + o_{\mathbb{P}}(1)$$

along such sequences. When the above holds for a model along such sequences, we say the model is LAN.
The LAN property has nothing to do with any estimator, it is purely a statement about what the densities of nearby distributions p_{ϵ_n} look like relative to p as we sample more data from them and move closer to p. The log density ratio shown above is actually exactly of the form you get if you instead look at the log density ratio of two normal distributions with the same variance but different means. That's why the property is called local asymptotic normality. So the result says that as you "zoom in" to a point on a differentiable path while sampling more data, the model looks more and more like a set of normal distributions with different means. This happens for any path where the densities vary smoothly (differentiably).

Incredibly, local asymptotic normality tells us enough about what the densities of nearby data-generating distributions look like to infer how any asymptotically linear estimator behaves along these sequences of distributions. The implied pathwise behavior is

$$\sqrt{n}\left(\hat{\Psi}(\mathbb{P}_n^{(\epsilon_n)}) - \Psi(\mathbb{P})\right) \rightsquigarrow \mathsf{N}(\mathbb{P}[s\phi], \mathbb{P}\phi^2)$$

as desired. The technical mechanism for inferring this is called Le Cam's 3rd Lemma (see Example 6.7 in Van der Vaart (2000)). To prove the result in generality requires contiguity arguments (Chapter 6 of Van der Vaart (2000)), but if you assume that every distribution in the model has a density and that all the densities have the same support (set of values z for which the density is nonzero), then you can prove an equivalent result using only "basic" arguments (continuous mapping and a characteristic function argument). Either way, I don't think the proof gives much intuition that's relevant to our purposes so I've omitted it.

This is remarkable because asymptotic linearity is a pointwise property: it describes only what happens if the estimator is evaluated with more and more data sampled from the *same* distribution. By itself it says nothing about what would happen if we also *changed* that distribution while letting n grow. But if we take into account differentiability along the path suddenly we can understand that changing pathwise behavior. Moreover the pathwise behavior depends crucially on what the score s is for the path in question: notice that the mean of the limiting distribution is $\mathbb{P}s\phi$.

The benefit of knowing this is that now we can compare to the definition of regularity (Definition 2.4), which *also* tells us how the estimator behaves along sequences including the kind we considered in Theorem 3.1, but this time in a way that does *not* depend on what the score is for the path:

$$\sqrt{n}\left(\hat{\Psi}(\mathbb{P}_n^{(\epsilon_n)}) - \Psi(\mathbb{P}^{(\epsilon_n)})\right) \rightsquigarrow \mathbb{D}.$$

Recall that $\mathbb{D}$ is some fixed distribution that by definition is not allowed to depend on the sequence or path. If we let $\epsilon_n = 0$ (a valid choice), the resulting sequence is $\sqrt{n}\left(\hat{\Psi}(\mathbb{P}_n) - \Psi(\mathbb{P})\right)$ which converges to $\mathsf{N}(0, \mathbb{P}\phi^2)$ when the estimator is asymptotically normal (Definition 1.1). Since the limiting distribution $\mathbb{D}$ must be the same across all paths for the estimator to be regular at $\mathbb{P}$, it must be that $\mathbb{D} = \mathsf{N}(0, \mathbb{P}\phi^2)$.

Therefore, combining this and Theorem 3.1, for any estimator that is both regular *and* asymptotically linear,

$$\sqrt{n}\left(\hat{\Psi}(\mathbb{P}_n^{(\epsilon_n)}) - \Psi(\mathbb{P})\right) \rightsquigarrow \mathsf{N}(\mathbb{P}[s\phi], \mathbb{P}\phi^2)$$
$$\sqrt{n}\left(\hat{\Psi}(\mathbb{P}_n^{(\epsilon_n)}) - \Psi(\mathbb{P}^{(\epsilon_n)})\right) \rightsquigarrow \mathsf{N}(0, \mathbb{P}\phi^2)$$

for any sequences $\mathbb{P}^{(\epsilon_n)}$ along a differentiable path with $\epsilon_n = c_n n^{-1/2}$ and $c_n \to 1$. Comparing the left- and right-hand sides above, the only way that both asymptotic statements can be true simultaneously is if

$$\sqrt{n}\left(\Psi(\mathbb{P}^{(\epsilon_n)}) - \Psi(\mathbb{P})\right) \to \mathbb{P}[s\phi].$$

Because $\epsilon_n \sim 1/\sqrt{n}$ and the choice of differentiable path was arbitrary, this implies the following result.

Influence Functions of RAL Estimators

Theorem 3.2. *If ϕ is an influence function of an estimator that is RAL at $\mathbb{P} \in \mathcal{P}$, then*

$$\lim_{\epsilon\to 0} \frac{\Psi(\mathbb{P}^{(\epsilon)}) - \Psi(\mathbb{P}^{(0)})}{\epsilon} = \mathbb{P}s\phi$$

for any differentiable path $\mathcal{P}_\epsilon = \{\mathbb{P}^{(\epsilon)} : \epsilon \in \mathbb{R}\}$ with score function s.
We often abbreviate the left-hand side of the above with the notation $\dot{\Psi}(s)$ to indicate that it is a derivative of Ψ at $\mathbb{P}$ in the "direction" s in which case the identity is written cleanly as

$$\dot{\Psi}_{\mathbb{P}}(s) = \mathbb{P}s\phi.$$

Any asymptotically linear estimator with influence function ϕ that satisfies this identity for all differentiable paths is regular at $\mathbb{P} \in \mathcal{P}$.

Proof

At first it may seem like there is nothing to prove since substituting $\sqrt{n} \approx \epsilon_n^{-1}$ appears to do the trick, but just because

$$\lim_{n\to\infty} \frac{f(\epsilon_n) - f(0)}{\epsilon_n} = \dot{f}$$

for all sequences $\epsilon_n = c_n n^{-1/2}$ with $c_n \to 1$ does not obviously imply that the same is true for *any* sequence $\epsilon_n \to 0$ (here I am abbreviating $f(\epsilon) = \Psi(\mathbb{P}^{(\epsilon)})$ and $\dot{f} = \mathbb{P}s\phi$). The latter convergence along any sequence $\epsilon_n \to 0$ is what we need for the left-hand side above to define a derivative.
To do this rigorously requires a little more effort. The rigorous proof is a bit tedious and not at all important to understand if you are happy to believe the result. I include it here only because it relies only on standard tools from analysis and I have not found a thorough version elsewhere in the literature.
In terms of the abbreviations with f, the display above Theorem 3.2 says that

$$\sqrt{n}[f(\epsilon_n) - f(0)] \to \dot{f}.$$

Since $\epsilon_n = c_n n^{-1/2}$, the above is equivalent to

$$c_n \epsilon_n^{-1}[f(\epsilon_n) - f(0)] \to \dot{f}.$$

Since $c_n \to 1$ this happens only if $\epsilon_n^{-1}[f(\epsilon_n) - f(0)] \to \dot{f}$. Therefore $\epsilon_n^{-1}[f(\epsilon_n) - f(0)] \to \dot{f}$ for all $\epsilon_n - c_n n^{-1/2}$ with $c_n \to 1$. However to prove the theorem we need this to hold for all $\epsilon_n \to 0$. To show this, we start by picking any sequence $\eta_m \to 0$. Our goal is be to show that this can be written in a way similar to $c_n n^{-1/2}$ with $c_n \to 1$ so that the difference quotient $\eta_n^{-1}[f(\eta_n) - f(0)]$ converges for this sequence by what we've already shown.

Let $n_m = \lfloor 1/\eta_m^2 \rfloor$ be the value of $1/\eta_m^2 \geq 0$ rounded down to the nearest integer for each m. Since $\eta_m \to 0$ by assumption, n_m is a sequence of positive integers that is eventually diverging. If we define $c_m = \eta_m n_m = \eta_m \sqrt{\lfloor 1/\eta_m^2 \rfloor}$ then $|c_m| \to 1$ because the function $\sqrt{\lfloor 1/\eta^2 \rfloor}$ gets closer and closer to $\sqrt{1/\eta^2}$ as η gets smaller.

If you care to see that rigorously, let $\zeta = 1/\eta^2$, which must diverge. The desired convergence

$$\left|\sqrt{1/\eta^2} - \sqrt{\lfloor 1/\eta^2 \rfloor}\right| \overset{\eta \to 0}{\to} 0$$

is true if

$$\left|\sqrt{\zeta} - \sqrt{\lfloor \zeta \rfloor}\right| \overset{\zeta \to \infty}{\to} 0.$$

but

$$\left|\sqrt{\zeta} - \sqrt{\lfloor \zeta \rfloor}\right| = \sqrt{\zeta} - \sqrt{\lfloor \zeta \rfloor} \leq \sqrt{\lfloor \zeta \rfloor + 1} - \sqrt{\lfloor \zeta \rfloor} = \frac{1}{\sqrt{\lfloor \zeta \rfloor + 1} + \sqrt{\lfloor \zeta \rfloor}}$$

The final expression on the right converges to zero for any $\zeta \to \infty$, which establishes $|c_m| \to 1$.

We have therefore established that any sequence $\eta_m \to 0$ can be represented as a sequence $c_m/\sqrt{n_m}$ where $|c_m| \to 1$ and n_m are integers diverging to infinity. This is almost of the form we need except for the fact that 1) n_m is an arbitrary diverging sequence of integers rather than the counting sequence $\{1, 2, 3 \ldots\}$ and 2) we have $|c_m| \to 1$ instead of $c_m \to 1$.

To resolve this we will first assume $c_m \to 1$ and show that the convergence of the difference quotient along sequences $c_n/\sqrt{n}$ actually implies convergence along sequences $c_m/\sqrt{n_m}$ first with n_m strictly increasing and then with n_m an arbitrary diverging sequence of integers. After that is done we will address the issue of the absolute value on c_m.

For any increasing sequence n_m and sequence $c_m \to 1$, build the "supersequence" $\tilde{c}_n = c_m$ for all $n_m \leq n < n_{m+1}$ so that $c_m = \tilde{c}_{n_m}$ by construction and $\tilde{c}_n \to 1$ since $c_m \to 1$. Again by construction, $\tilde{\eta}_m = \tilde{c}_{n_m}/\sqrt{n_m}$ is a subsequence of $\epsilon_n = c_n/\sqrt{n}$ which is of the desired form. Because of that, $\tilde{\eta}_m^{-1}[f(\tilde{\eta}_m) - f(0)]$ is a subsequence of $\epsilon_n^{-1}[f(\epsilon_n) - f(0)]$ and thus must converge to the same limit $\dot{f}$.

Now let n_m be an arbitrary, diverging, positive sequence of integers and $c_m \to 1$. Since n_m is diverging, any subsequence n_{m_k} is also diverging. From any diverging subsequence we can take a strictly increasing subsequence $n_{m_{k_j}} < n_{m_{k_{j+1}}}$. Along this sub-subsequence,

$c_{m_{k_j}} \to 1$. Therefore, letting $\bar{\eta}_j = c_{m_{k_j}}/\sqrt{n_{m_{k_j}}}$, we know that $\bar{\eta}_j^{-1}[f(\bar{\eta}_j) - f(0)] \to \dot{f}$ because $\bar{\eta}_j$ satisfies the conditions required from the paragraph above. Since every subsequence $\left(\frac{c_{m_k}}{\sqrt{n_{m_k}}}\right)^{-1}\left[f\left(\frac{c_{m_k}}{\sqrt{n_{m_k}}}\right) - f(0)\right]$ has a further subsequence that converges to a common limit $\dot{f}$, that means the original sequence $\left(\frac{c_m}{\sqrt{n_m}}\right)^{-1}\left[f\left(\frac{c_m}{\sqrt{n_m}}\right) - f(0)\right]$ also converges to that common limit. This is a consequence of the fact that a sequence converges if and only if all of its subsequences converge to the same limit.

Now we deal with the issue of the absolute value on the sequence c_m. To do this we consider a sequence along a path $\mathcal{P}_\epsilon$ indexed by $\epsilon_n = c_n n^{-1/2}$ with $c_n \to -1$ instead of $c_n \to 1$. This is exactly the same as a sequence with $c_n \to 1$ but along the "reversed" path $\mathcal{P}_{-\epsilon}$ which has score $-s$ (see Exercise 2.5). Therefore along such sequences $c_n\epsilon_n^{-1}[f(\epsilon_n) - f(0)] \to -\dot{f}$ by our starting proposition. Multiplying both sides by -1 we obtain that $(-c_n)\epsilon_n^{-1}[f(\epsilon_n) - f(0)] \to \dot{f}$ for any sequence $c_n \to -1$ and $\epsilon_n = c_n n^{-1/2}$. Since $-c_n \to 1$, by the argument at the beginning of this proof we know that $\epsilon_n^{-1}[f(\epsilon_n) - f(0)] \to \dot{f}$ for any sequence $\epsilon_n = c_n n^{-1/2}$ now for either $c_n \to 1$ *or* $c_n \to -1$.

We may now return to our original sequence $c_m = \eta_m n_m$ whose absolute value converges to 1. In a mild abuse of notation let c_m^+ be the subsequence of the non-negative entries of c_m and let c_m^- be the negative entries (with corresponding subsequences n_m^+ and n_m^- which are both diverging sequences of integers). If c_m^+ has a finite number of elements then $c_m \to -1$ and otherwise if c_m^- has a finite number of elements then $c_m \to 1$. Either way $\eta_m^{-1}[f(\eta_m) - f(0)] \to \dot{f}$ by the argument of the paragraph above. In the case that both c_m^+ and c_m^- are infinite sequences, $|c_m| \to 1$ implies that $c_m^+ \to 1$ and $c_m^- \to -1$. Therefore we may divide eta_m into the two corresponding sequences η_m^+ and η_m^- and since $(\eta_m^\pm)^{-1}[f(\eta_m^\pm) - f(0)] \to \dot{f}$ for either part of the sequence it must be that $\eta_m^{-1}[f(\eta_m) - f(0)] \to \dot{f}$ for the entire sequence.

Lastly, the claim that any asymptotically linear estimator with influence function ϕ that satisfies $\dot{\Psi}_{\mathbb{P}}(s) = \mathbb{P}s\phi$ for all differentiable paths is automatically regular at $\mathbb{P} \in \mathcal{P}$ comes from running the argument above in reverse: if the conclusion holds and asymptotic linearity holds, then regularity must hold or else a contradiction arises (see also Proposition 1 in Chapter 3 or Bickel et al. (1993)).

This is an astounding result because it connects two seemingly unrelated things: the left-hand side is the rate of change in the estimand as we move the underlying distribution along a path. That has nothing to do with any estimator. The right-hand side is some measure of the "angle" between the influence function of an estimator and the direction of the path. By assuming regularity and asymptotic normality we have managed to connect these two seemingly unrelated quantities. What this means about the set of possible influence functions may still be unclear, but for the moment I hope you can appreciate the result as a very interesting connection nonetheless!

3.2 Tangent Sets

Theorem 3.2 holds simultaneously for *all* differentiable paths in the model through $\mathbb{P}$ (and all their induced scores). To unlock the important insights from the theorem we therefore need to be able to refer to that set of scores.

Tangent Set

Definition 3.1. The *tangent set* $\dot{\mathcal{P}}_{\mathbb{P}}$ at a point $\mathbb{P} \in \mathcal{P}$ is the set of all scores of all differentiable paths through that point. When the point of interest $\mathbb{P}$ is clear from context we leave the subscript off and just write $\dot{\mathcal{P}}$ for the tangent set.
$\dot{\mathcal{P}}_{\mathbb{P}} \subseteq \{f \in \mathcal{L}_2(\mathbb{P}) : \mathbb{P}f = 0\} \equiv \mathcal{L}_2^0(\mathbb{P})$ since any score s of a differentiable path through $\mathbb{P}$ is in $\mathcal{L}_2(\mathbb{P})$ and has mean zero by Definition 2.3.

Very often the tangent set is a closed vector space (i.e. it is closed under sums, scaling, and limits; see Definition B.1). When this is the case, some authors refer to the tangent set as the tangent *space*. If the tangent set is not already a closed vector space, the term “tangent space” usually refers to the linear closure of the tangent set (all linear combinations of scores and their $\mathcal{L}_2$-limits).

Looking at Figure 3.1, it’s clear why we call this the the “tangent” set: the plane $\dot{\mathcal{P}}_{\mathbb{P}}\sqrt{p}$ is tangent to the surface of the root-model $\sqrt{\mathcal{P}}$ at the density $\sqrt{p}$. That is also why we use the evocative dot notation for the tangent set: $\dot{\mathcal{P}}_{\mathbb{P}}\sqrt{p}$ contains all of the derivatives $\dot{\sqrt{p_\epsilon}}$. Although the picture in fact shows the root-model and tangent set times the root density, we often say informally that the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$ itself is “tangent” to the model $\mathcal{P}$ at the distribution $\mathbb{P}$.

Since it’s useful to think of scores as directions we can move away from $\mathbb{P}$ in the model, the tangent set is simply the set of all such directions. If a model is “large” in a neighborhood of $\mathbb{P}$ in the sense that there are many directions you can go in and stay in the model, then the tangent set will be large. If a model is “small” around $\mathbb{P}$ then the tangent set will be small.

Example 3.1. Consider the model $\mathcal{P} = \{\mathsf{N}(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}$ with densities $p_{\mu,\sigma^2}(z)$ and the point $\mathbb{P} = \mathsf{N}(0, 1)$ with density $p(z) = (2\pi)^{-1/2}e^{-z^2/2}$ of the standard normal. Since this is a two-dimensional parametric model, the tangent set at p is all linear combinations of

$$
\begin{aligned}
s_\mu(z) &= \left.\frac{\partial p_{\mu,\sigma^2}(z)}{\partial \mu}\right|_{\mu=0,\sigma^2=1} = z \\
s_{\sigma^2}(z) &= \left.\frac{\partial p_{\mu,\sigma^2}(z)}{\partial \sigma^2}\right|_{\mu=0,\sigma^2=1} = \frac{z^2-1}{2}
\end{aligned}
$$

Thus in this case $\dot{\mathcal{P}}$ is a two-dimensional set of functions.

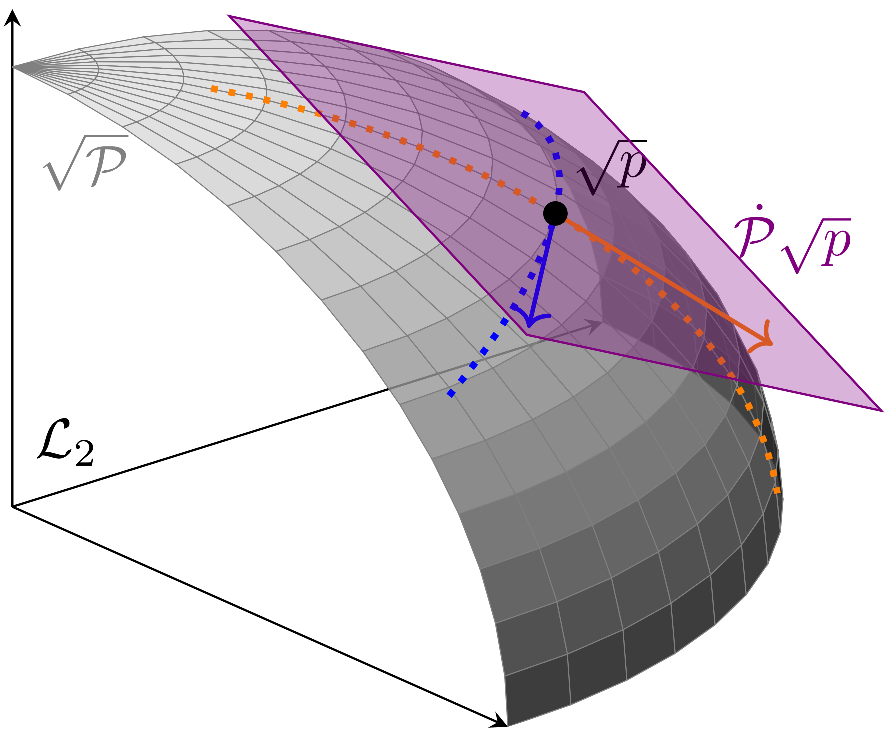

Figure 3.1: A visual representation of the tangent set $\dot{\mathcal{P}}$ along with the model $\sqrt{\mathcal{P}}$, two differentiable paths, and two scores (times $\sqrt{p}$). The set $\dot{\mathcal{P}}\sqrt{p}$ denotes the set of all scores times the root density: $\{s\sqrt{p} : s \in \dot{\mathcal{P}}\}$.

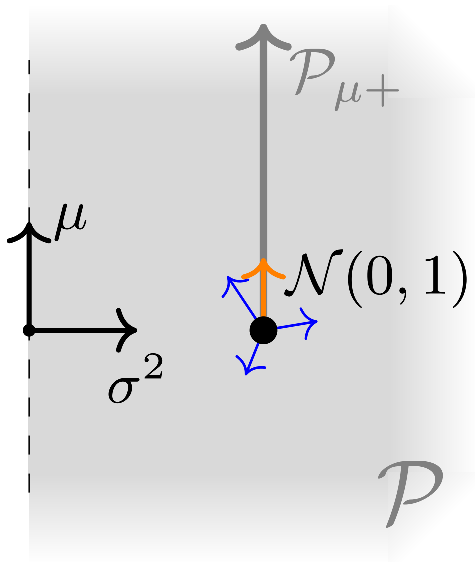

Figure 3.2: A cartoon representation of the models $\mathcal{P}$ and $\mathcal{P}_{\mu+}$ discussed in this example. The arrows emanating from $\mathsf{N}(0,1)$ indicate scores of paths in $\mathcal{P}$, but only the orange arrow is a score of a path in $\mathcal{P}_{\mu+}$.

If we instead restricted the model to be $\mathcal{P}_\mu = \{\mathsf{N}(\mu, 1) : \mu \in \mathbb{R}\}$ by fixing $\sigma^2 = 1$ (presume we knew this a-priori), then the tangent set would be all linear combinations of s_μ alone, i,e, $\{cs_\mu : c \in \mathbb{R}\}$, a one-dimensional subset of the tangent set for the larger model.

Lastly, if we considered the even smaller model $\mathcal{P}_{\mu+} = \{\mathsf{N}(\mu, 1) : \mu \geq 0\}$, then the point $\mathbb{P} = \mathsf{N}(0,1)$ would be on the "edge" of this model. Consequently there is no differentiable path that goes in the "negative μ" direction from $\mathbb{P}$ and therefore the tangent set would be restricted to $\{cs_\mu : c \geq 0\}$, again a subset of the tangent space in the larger parent model. On the other hand, if we considered the point $\mathbb{P} = \mathsf{N}(1,1)$ in this same model, the tangent space set would be $\{cs_\mu : c \in \mathbb{R}\}$, the same as in the larger model where μ is allowed to be negative. This is because $\mathsf{N}(1,1)$ is in the "interior" of the restricted model, so the mean can be decreased without any problem.

This example demonstrates why the tangent set is specific to a particular point in a particular model. The notation $\dot{\mathcal{P}}_\mathbb{P}$ makes the model dependence very explicit and that is why I prefer it to something more cumbersome like $\mathcal{S}_{\mathcal{P},\mathbb{P}}$. The tangent set depends crucially on where the point is within the model because that affects what directions it is legal to move in while remaining in the model (recall that paths are subsets of the model). The shape and size of the tangent set is determined entirely by the *local* character of the model at $\mathbb{P}$. We saw in the example above that the tangent set at $\mathsf{N}(0,1)$ became smaller if we restricted the mean in the normal model to be non-negative, but at $\mathsf{N}(1,1)$ the tangent set was unaffected by that restriction. There are many more examples of specific tangent sets in Chapters 3 and 4 of

Bickel et al. (1993) but for now we can proceed with the intuition we have without getting bogged down in details.

In Definition 3.1 we noted that $\dot{\mathcal{P}}_{\mathbb{P}} \subseteq \mathcal{L}_2^0(\mathbb{P})$ (recall we defined this as the set of $\mathcal{L}_2(\mathbb{P})$ functions with $\mathbb{P}$-mean zero). Some models are so "large", however, that the inclusion is actually an equality. If the model is big enough (locally), you can move in any direction.

Example 3.2. Let p be the set of all distributions of continuous random variables (i.e. distributions with standard densities) and pick some point $\mathbb{P} \in \mathcal{P}$. For any bounded function $s \in \mathcal{L}_2^0(\mathbb{P})$ we can construct the convex path $p_\epsilon = (1 + \epsilon s)p_0$ (bounding s is required so that p_ϵ remains a density and the mean-square and pointwise scores coincide). As we saw in Example 2.5, s is the score for such a path at $\mathbb{P}$ and thus the tangent set at that point includes at least all bounded functions $s \in \mathcal{L}_2^0(\mathbb{P})$.

This is good enough for intuition, but to remove the boundedness condition on the scores and formally extend to all $\mathcal{L}_2^0(\mathbb{P})$ we can instead use the paths $p_\epsilon = \frac{e^{\lambda(\epsilon s)}}{\int e^{\lambda(\epsilon s(z))} p_0(z)\,\mathrm{d}z} p_0$ with $\lambda(t) = 2(1 + e^{-2t})^{-1}$. These paths also have scores s and generate valid densities for any $s \in \mathcal{L}_2^0(\mathbb{P})$. For details see Example 1 in Chapter 3 of Bickel et al. (1993).

We give this kind of model its own name.

Locally Nonparametric Model

Definition 3.2. A model $\mathcal{P}$ is *locally nonparametric* at $\mathbb{P} \in \mathcal{P}$ if $\dot{\mathcal{P}}_{\mathbb{P}} = \mathcal{L}_2^0(\mathbb{P})$.

This is the kind of model we will work with most often for the remainder of this book. Locally nonparametric models are popular in practice because they enforce essentially no assumptions on the data-generating distribution. In fields like medicine, economics, or business, we often cannot say much about the world with great confidence (the distribution of variables, the functional form of their relationships, etc.). We will see shortly that the larger the model is, usually the more uncertainty we get in the final estimate. This makes sense: the more assumptions you're willing to make, the more confident you can be of an estimate. But if you use a model that is too small (i.e. assume things that are not actually true) then the estimate may be biased and confidence intervals will typically underestimate the true sampling uncertainty. Working with a locally nonparametric model is therefore a maximally robust approach, or at worst conservative.

3.2.1 Factoring

There are, of course, many practical situations where we *do* know something. For example, in our familiar covariate-treatment-outcome (X, A, Y) setup, if the treatment is allocated randomly by coin flip, then we know for a fact that $\mathbb{P}(A = 1|X) = 1/2$. This is an important restriction on the model. Due to the ability to experimentally control treatment assignments, there are many cases in causal inference where one or more factors of a probability distribution are known exactly, but nothing is certain about the others. Thankfully, it's not hard to figure out the tangent set in such models.

To simplify things, we'll work in a generic model defined over two variables (Y, Z) (you can think of $Z = (A, X)$ if you please). We'll also presume again that all our distributions have densities and that the pointwise definition of the scores coincides with the mean-square derivative definition to facilitate the explanation. All of the results below also hold in the mean-square sense of scores in models without densities but everything is more cumbersome to prove without adding any intuition. I've put sketches of those proofs into boxed-off sections that you can safely ignore.

Since every distribution $\mathbb{P}(Y, Z)$ can be factored into $\mathbb{P}(Y, Z) = \mathbb{P}(Y|Z)\mathbb{P}(Z)$, every differentiable path of distributions $\mathbb{P}^{(\epsilon)}(Y, Z)$ induces a path of marginal distributions $\mathbb{P}^{(\epsilon)}(Z)$ and a path of conditional distributions $\mathbb{P}^{(\epsilon)}(Y|Z)$. Let's pick some arbitrary path with score s. We can take derivatives of the log-likelihoods along the induced conditional and marginal paths to obtain scores (in terms of the pointwise definition):

$$
\begin{aligned}
s_Z(z) &= \partial_0 \log p_\epsilon(z) \\
s_{Y|Z}(y, z) &= \partial_0 \log p_\epsilon(y|z)
\end{aligned}
$$

where I'm abbreviating the operator $\frac{\mathrm{d}}{\mathrm{d}\epsilon}\Big|_{\epsilon=0}(\cdot)$ with $\partial_0(\cdot)$. When the meaning is clear from context I'll abuse notation a bit and write $s(z)$ for $s_Z(z)$ and similarly $s(y|z)$ for $s_{Y|Z}(y, z)$, the same way as is usually done for marginal and conditional density functions. If we iterate this over all differentiable paths and collect the resulting scores we generate what I'll call the marginal and conditional tangent sets $\dot{\mathcal{P}}_{\mathbb{P}}(Z)$ and $\dot{\mathcal{P}}_{\mathbb{P}}(Y|Z)$.

Let's see how these marginal and conditional scores are related to the joint score for a given path. First off, because

$$
\log p_\epsilon(z, y) = \log(p_\epsilon(y|z)p_\epsilon(z)) = \log p_\epsilon(y|z) + \log p_\epsilon(z),
$$

when we apply ∂_0 to both sides we get

$$
s(y, z) = s(y|z) + s(z)
$$

So the joint score is simply the sum of the marginal and conditional scores. But we can further characterize these scores in terms of the joint score. By the chain rule, $\partial \log p_\epsilon = \frac{\partial p_\epsilon}{p_\epsilon}$, so evaluating at $\epsilon = 0$ and shifting terms around gives

$$
\partial_0 p_\epsilon = p_0 \partial_0 p_\epsilon = p_0 s \tag{3.1}
$$

This is true for the joint, conditional, and marginal densities/scores. If we start with this equation in terms of the joint density/score and integrate with respect to y on both sides we have

$$\int \partial_0 p_\epsilon(y,z)\,\mathrm{d}y = \int p_0(y,z)s(y,z)\,\mathrm{d}y$$

On the left-hand-side, if we are allowed to exchange integration w.r.t z and differentiation w.r.t. ϵ (formally this is possible under mild restrictions on the model), then we get $\partial_0 \int p_\epsilon(y,z)\,\mathrm{d}y = \partial_0 p_\epsilon(z) = p_0(z)s(z)$ by applying Equation 3.1 to the marginal $p_\epsilon(z)$. Dividing both sides of the above by $p_0(z)$ gives

$$s(z) = \int \frac{p_0(y,z)}{p_0(z)} s(y,z)\,\mathrm{d}y = \int p(y|z)s(y,z)\,\mathrm{d}y = \mathbb{E}[s(Y,Z)|Z=z]$$

and therefore $s(y|z) = s(y,z) - \mathbb{E}[s(Y,Z)|Z=z]$ since the conditional and marginal scores sum to the joint.

We now have explicit formulations of the marginal and conditional scores in terms of the joint score. Besides their relation to the joint score, these scores also have some interesting properties. You may recall from Definition 2.3 that $\mathbb{P}s = 0$, i.e. $\mathbb{E}[s(Y,Z)] = 0$. The conditional and joint score obey similar conditions:

$$\begin{aligned}\mathbb{E}[s(Z)] &= 0\\ \mathbb{E}[s(Y|Z)|Z] &= 0.\end{aligned}$$

The first identity comes from the explicit definition of $s(z)$ because $\mathbb{E}_Z[s(Z)] = \mathbb{E}_Z[\mathbb{E}_Y[s(Y,Z)|Z]] = \mathbb{E}[s(Y,Z)] = 0$ by total expectation and since the joint score is mean zero. The second comes from

$$\begin{aligned}\mathbb{E}_Y[s(Y|Z)|Z] &= \mathbb{E}_Y[s(Y,Z) - \mathbb{E}_Y[s(Y,Z)|Z]|Z]\\ &= \mathbb{E}_Y[s(Y,Z)|Z] - \mathbb{E}_Y[s(Y,Z)|Z]\end{aligned}$$

We can also show that the marginal and conditional score functions are orthogonal to each other in $\mathcal{L}_2(\mathbb{P})$. Recall that two functions are orthogonal if their inner product is zero and that the $\mathcal{L}_2(\mathbb{P})$ inner product of two functions is the expectation of their product (see Appendix B). The expected value of the product of the conditional and marginal scores is

$$\mathbb{E}[s(Z)s(Y|Z)] = \mathbb{E}[\mathbb{E}[s(Y,Z)|Z](s(Y,Z) - \mathbb{E}[s(Y,Z)|Z])].$$

Abbreviating $S = s(Y,Z)$, by total expectation, $\mathbb{E}[\mathbb{E}_S[S|Z]S] = \mathbb{E}_Z[\mathbb{E}_S[\mathbb{E}_S[S|Z]S|Z]] = \mathbb{E}_Z[\mathbb{E}_S[S|Z]\mathbb{E}_S[S|Z]]$ so the above is zero.

This is actually a general fact about conditional expectation that is used in many places throughout probability theory: a random variable minus its conditional expectation is orthogonal to that conditional expectation. In fact, the conditional expectation $\mathbb{E}[f(Y,Z)|Z]$ of a random variable $f(Y,Z)$ is precisely its orthogonal projection onto the subspace spanned by all $\mathcal{L}_2(\mathbb{P})$ functions of Z alone: the best possible $\mathcal{L}_2(\mathbb{P})$ approximation of $f(y,z)$ that can be attained with functions of z. This is illustrated in Figure 3.3.

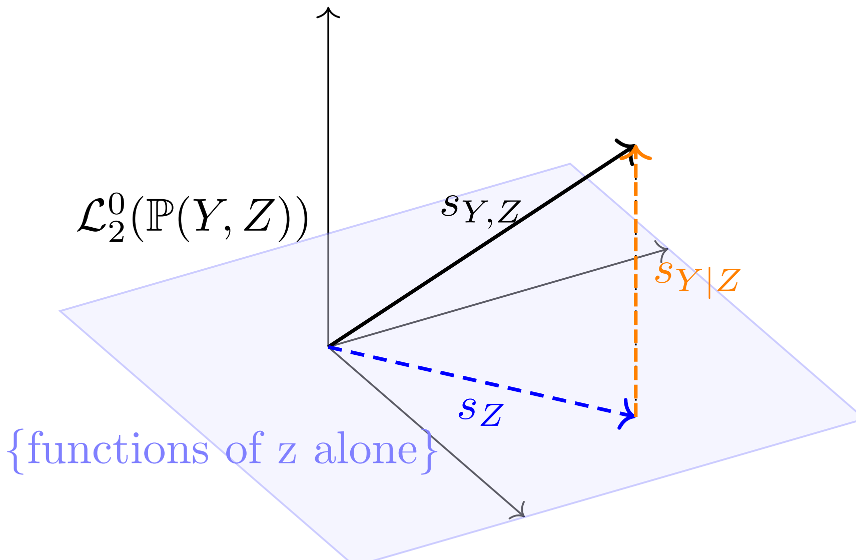

Figure 3.3: The marginal score is an orthogonal projection of the joint score and the conditional score is the residual.

Let's summarize all these results.

Conditional and Marginal Scores

Theorem 3.3. *Let $\mathcal{P} = \mathcal{P}(Y,Z)$ be a model over joint distributions $\mathbb{P}(Y,Z)$ of variables (Y,Z) and let $\mathcal{P}_\epsilon(Y,Z)$ be a differentiable path with score $s(y,z)$. This path induces marginal and conditional paths $\mathcal{P}_\epsilon(Z)$ and $\mathcal{P}_\epsilon(Y|Z)$ with scores*

$$
\begin{aligned}
s(z) &= \mathbb{E}[s(Y,Z)|Z] \\
s(y|z) &= s(y,z) - \mathbb{E}[s(Y,Z)|Z]
\end{aligned}
$$

which sum to $s(y,z)$. The marginal score s_Z is the orthogonal projection of the joint score $s_{Y,Z}$ onto the span of all $\mathcal{L}_2(\mathbb{P})$ functions of z alone, while the conditional score $z_{Y|Z}$ is the residual of this projection. Therefore $s_{Y|Z} \perp s_Z$ in $\mathcal{L}_2(\mathbb{P})$. Moreover

$$
\begin{aligned}
\mathbb{E}[s(Z)] &= 0 \\
\mathbb{E}[s(Y|Z)|Z] &= 0,
\end{aligned}
$$

Proof Sketch for Mean-Square Scores

Let $\mathcal{P}_\epsilon$ be a (mean-square) differentiable path over distributions $\mathbb{P}(Y, Z)$ and let $\epsilon_n \to 0$ and $\mathbb{P}_n = \mathbb{P}_{\epsilon_n}$ be a sequence along the path with $\mathbb{P} = \mathbb{P}_{\epsilon=0}$. Let $\mathbb{Q}(Y, Z)$ be a dominating measure constructed for the sequence $\mathbb{P}_n$ and let $p_n(y, z) = \frac{d\mathbb{P}_n(Y,Z)}{d\mathbb{Q}(Y,Z)}$ be densities. Let $Q(Z)$ and $\mathbb{P}(Z)$ be the marginal distributions of $\mathbb{Q}(Y, Z)$ and $\mathbb{P}(Y, Z)$, respectively, so that $p_n(z) = \frac{d\mathbb{P}_n(Z)}{d\mathbb{Q}(Z)}$ are marginal densities.
Since the path of joint distributions is differentiable, there is a function $g(y, z)$ such that

$$
\left\| \frac{\sqrt{p_n(y,z)} - \sqrt{p_0(y,z)}}{\epsilon_n} - g(y,z) \right\|_{\mathcal{L}_2(\mathbb{Q}(Y,Z))} \to 0
$$

Our goal is to show that the path of marginal distributions is also differentiable, i.e. that there is a function $g(z)$ for which

$$
\left\| \frac{\sqrt{p_n(z)} - \sqrt{p(z)}}{\epsilon_n} - g(z) \right\|_{\mathcal{L}_2(\mathbb{Q}(Z))} \to 0
$$

First decompose

$$
\sqrt{p_n(z)} - \sqrt{p(z)} = \frac{p_n(z) - p(z)}{\sqrt{p_n(z)} + \sqrt{p(z)}}
$$

and notice that, by definition,

$$
\begin{aligned}
p_n(z) - \mathbb{P}(z) &= \int p_n(y,z) - p(y,z)\, d\mathbb{Q}(Y|Z) \\
&= \int \left(\sqrt{p_n(y,z)} - \sqrt{p(y,z)}\right)\left(\sqrt{p_n(y,z)} + \sqrt{p(y,z)}\right) d\mathbb{Q}(Y|Z).
\end{aligned}
$$

This is the equivalent of the more familiar formula $p(z) = \int p(y,z)\, dy$ except that in this case we do not assume that $\mathbb{Q}$ is a product measure. In the familiar case we see $dy = d\mu(Y)$ because $d\mu(Y) = d\mu(Y|Z)$, i.e. Lebesgue measure is a product measure.
Combining these observations,

$$
\frac{\sqrt{p_n(z)} - \sqrt{p(z)}}{\epsilon_n} = \int \underbrace{\frac{\sqrt{p_n(y,z)} - \sqrt{p(y,z)}}{\epsilon_n}}_{g_n(y,z)} \underbrace{\frac{\sqrt{p_n(y,z)} + \sqrt{p(y,z)}}{\sqrt{p_n(z)} + \sqrt{p(z)}}}_{h_n(y,z)} d\mathbb{Q}(Y|Z).
$$

$g_n \overset{\mathcal{L}_2(Q(Y,Z))}{\to} g$ by assumption and since $\sqrt{p_n(y,z)} \overset{\mathcal{L}_2(Q(Y,Z))}{\to} \sqrt{p(y,z)}$, the second factor h_n can be shown to converge in $\mathcal{L}_2\left(Q(Y,Z)\right)$ to $h(y,z) = \sqrt{p(y,z)/p(z)} = \sqrt{p(y|z)}$ with careful handling of the denominator.
Therefore one might conjecture that the $\mathcal{L}_2\left(Q(Z)\right)$ limit of the above is $\int g(y,z)h(y,z)\,\mathrm{d}\mathbb{Q}(Y|Z)$ and indeed this is the case because

$$\begin{aligned}\int \left(\int g_n h_n - gh\,\mathrm{d}\mathbb{Q}(Y|Z)\right)^2 \mathrm{d}\mathbb{Q}(Z) &\le \int\int (g_n h_n - gh)^2\,\mathrm{d}\mathbb{Q}(Y|Z)\,\mathrm{d}\mathbb{Q}(Z)\\ &= \int (g_n h_n - gh)^2\,\mathrm{d}\mathbb{Q}(Y,Z) \to 0\end{aligned}$$

by Jensen's inequality ($0 \le \mathbb{V}[U] = \mathbb{E}[U^2] - \mathbb{E}[U]^2 \implies \mathbb{E}[U]^2 < \mathbb{E}[U^2]$) and the aforementioned convergences.
This shows that $g(z) = \int g(y,z)\sqrt{p(y|z)}\,\mathrm{d}Q(Y|Z)$. Defining $s(z) = 2g(z)/\sqrt{p(z)}$ and $s(y,z) = 2g(y,z)/\sqrt{p(y,z)}$ gives

$$s(z) = \int s(y,z)p(y|z)\,\mathrm{d}Q(Y|Z) = \int s(y,z)\,\mathrm{d}P(Y|Z) = \mathbb{E}[s(Y,Z)|Z=z].$$

Similar arguments may be used to derive the conditional score $s(y|z)$ and then the other facts follow algebraically.

Given this characterization, it's easy to derive the tangent set of a model that has a restricted factor. In a model with a restricted factor, paths can only change the other factors and the scores are the "directions" that change only the non-restricted factors. For example if $p(z)$ is fixed in the model but $p(y|z)$ is free, then the scores are $s(y,z) = s(y|z)+0$, i.e. the z-component $s(z) = 0$ is fixed at 0 so as to not change the fixed marginal density $p(z)$.

Tangent Sets in Factored Models

Theorem 3.4. *Let $\mathcal{P}$ be locally nonparametric around some distribution $\mathbb{P}(Y,Z) = \mathbb{P}(Y|Z)\mathbb{P}(Z)$.*
Let $\mathcal{P}^{(Z)} = \{\mathbb{P}(Y|Z)\tilde{\mathbb{P}}(Z) : \tilde{\mathbb{P}} \in \mathcal{P}\}$, be a restriction of the original model where all the conditional distributions $\mathbb{P}(Y|Z)$ are replaced with the fixed conditional distribution $\mathbb{P}(Y|Z)$, but the marginal distributions of Z are allowed to vary. Then the tangent set at $\mathbb{P}$ in the restricted model is the set of all Z-marginal scores in the original model, i.e.

$$\dot{\mathcal{P}}^{(Z)}_{\mathbb{P}} = \dot{\mathcal{P}}_{\mathbb{P}}(Z),$$

which is all $\mathcal{L}_2^0\left(\mathbb{P}\right)$ functions of z alone.
Similarly, if $\mathcal{P}^{(Y|Z)} = \{\tilde{\mathbb{P}}(Y|Z)\mathbb{P}(Z) : \tilde{\mathbb{P}} \in \mathcal{P}\}$, a mutation of the original model where all the marginal *distributions are fixed, but the* conditional *distributions can still vary, then*

$$\dot{\mathcal{P}}_{\mathbb{P}}^{(Y|Z)} = \dot{\mathcal{P}}_{\mathbb{P}}(Y|Z),$$

which is all $\mathcal{L}_2^0(\mathbb{P})$ functions of (y, z) with z-conditional mean zero.

3.3 Pathwise Differentiability

Now that we understand a bit more about the set of all scores we can return to the central identity above (definition Theorem 3.2). By comparing the asymptotic behavior while moving quickly along differentiable paths implied by regularity to the behavior implied by asymptotic linearity, we deduced that any RAL estimator (with influence function ϕ) satisfies

$$\dot{\Psi}_{\mathbb{P}} s = \mathbb{P} s \phi$$

for any differentiable path through $\mathbb{P}$ with score s. In other words, that equality is satisfied for all scores $s \in \dot{\mathcal{P}}_{\mathbb{P}}$, the tangent set of the model $\mathcal{P}$ at the distribution $\mathbb{P}$.

Let's dig into the left-hand term:

$$\dot{\Psi}_{\mathbb{P}}(s) \overset{\text{def}}{=} \lim_{\epsilon \to 0} \frac{\Psi(\mathbb{P}^{(\epsilon)}) - \Psi(\mathbb{P}^{(0)})}{\epsilon} = \frac{\mathrm{d}}{\mathrm{d}\epsilon}\Big|_{\epsilon=0} \Psi(\mathbb{P}^{(\epsilon)}).$$

For any fixed path this is a standard, vanilla derivative of a one-dimensional function of ϵ. It measures the (instantaneous) rate of change of the estimand as we shift the underlying distribution along the path away from $\epsilon = 0$. This is illustrated in Figure 3.4.

However, that slope is generally different depending on the path one chooses. There may be paths along with the estimand is increasing rapidly, paths along which it is decreasing, and paths along which it is constant. That is why we write $\dot{\Psi}_{\mathbb{P}}(s)$ as a function of s, the "direction" (and "speed") of the path. Since the slope is taken at $\epsilon = 0$, the only thing that matters is the immediate direction of the path at $\mathbb{P}$, i.e. the score s. The instantaneous rate of change of the estimand along two different paths with the same score will be the same. Because it is a derivative along a path, we aptly call the function $s \mapsto \dot{\Psi}_{\mathbb{P}}(s)$ mapping from $\dot{\mathcal{P}}_{\mathbb{P}}$ to $\mathbb{R}$ the *pathwise derivative* of the estimand Ψ at the distribution $\mathbb{P}$ in the model $\mathcal{P}$.

The identity $\dot{\Psi}_{\mathbb{P}}(s) = \mathbb{P} s \phi$ says that the slope $\dot{\Psi}_{\mathbb{P}}(s)$ may be found by taking the inner product of s with *any* influence function ϕ of a RAL estimator. In fact, it doesn't matter *which* RAL estimator, if there are multiple: the result must be the same. We'll return to this point in the next section but we are actually getting ahead of ourselves because we are assuming there *is* a RAL estimator. The argument for Theorem 3.2 only established a condition that must be met by any RAL estimator, not that such an estimator exists. In fact, the result implies something about when RAL estimators *can* exist.

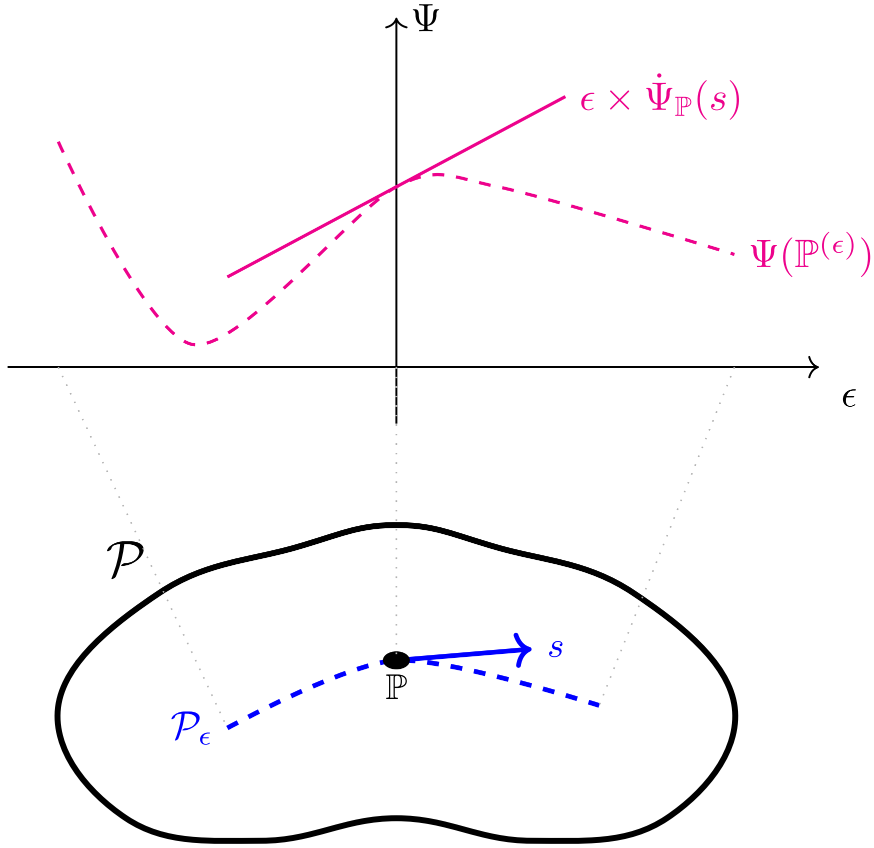

Figure 3.4: The pathwise derivative evaluated at a particular score s gives the rate of change of the estimand if we move the underlying distribution in the direction of the score.

Notice that on the right-hand side, $\mathbb{P}s\phi$ is an inner product. Seen as a function $s \mapsto \mathbb{P}s\phi$ acting on the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$, the inner product is a bounded (i.e. continuous) linear functional (see Definition B.6 and Example B.7). Informally, a small change in the input score must always translate to a small change in the inner product with the fixed function ϕ. Since this inner product is equal to the pathwise derivative, that implies that the pathwise derivative must *also* be linear and continuous in the same sense: if we look at two paths that go in similar directions, the rate of change of the estimand must be similar on those paths.

The interesting thing is that the pathwise derivative has nothing to do with the estimator: it is defined purely by the model $\mathcal{P}$, the distribution $\mathbb{P}$ of interest within it, and the estimand Ψ. The pathwise derivative either exists and is linear and continuous across scores or it isn't. If it isn't a continuous linear functional, then it's not possible for it to be equal to any inner product. But since that equality must hold for any RAL estimator, if the pathwise derivative is not a continuous linear functional, then there cannot be any RAL estimator for that estimand in that model in the first place.

Pathwise Differentiability

Definition 3.3. An estimand Ψ is *pathwise differentiable* at $\mathbb{P} \in \mathcal{P}$ if the pathwise derivative $\dot{\Psi}_{\mathbb{P}}$ exists and is linear and continuous across the tangent set $\mathcal{P}_{\mathbb{P}}$.
If an estimand is not pathwise differentiable, there is no asymptotically linear estimator that is regular in that model.

Warning

Review Definition A.5 for the definition of continuity and Section B.4 for details and definitions relating to Hilbert spaces and Riesz representation. Understanding the details is not nearly as important as understanding the big-picture implications here so it's ok if you skip the mathematical details for now and come back to them on a second pass later.

A pathwise differentiable estimand "varies smoothly" over the model the same way that a ordinary function with a derivative that is continuous has no kinks. At the point of a kink or discontinuity in the estimand, no estimator can be regular.

Exercise 3.1. Let Z be normally distributed with known variance 1 but unknown mean μ, i.e. $\mathcal{P} = \{\mathsf{N}(\mu, 1) : \mu \in \mathbb{R}\}$. Presume we are interested in estimating the quantity $\Psi(\mathbb{P}) = 1(\mathbb{E}_{\mathbb{P}}[Z] > 0)$, i.e. whether the mean is positive or not (we are estimating, not testing).

a. Draw a plot of the function $\mu \mapsto \Psi(\mathsf{N}(\mu, 1))$
b. At $\mathbb{P} = \mathsf{N}(0, 1)$, do you think Ψ is pathwise differentiable? If not, is it because the derivative doesn't exist or is it because the derivative is not continuous?

The example in Exercise 3.1 is contrived. More often the types of “kinks” or discontinuities that cause problems are difficult to visualize in lower-dimensions. It’s not as if the value of most practically interesting nondifferentiable estimands changes suddenly as we move along a single differentiable path. For the most part, the fact that the path is differentiable means that the densities change smoothly as we walk in that fixed direction and that usually implies smooth behavior of the estimand. The kinks are only visible if you rotate around and consider various different directions. I like to imagine standing at a point $\mathbb{P}$ on the “surface” $\Psi(\mathbb{P})$ and looking in some direction: whatever direction I look in, I can see how steep the ascent or descent is. The problem with most practially interesting estimands that are not differentiable isn’t that there is a discontinuity or kink in any given direction, it’s that no matter what direction we look at, there is always another direction where the slope is steeper. In infinite dimensions, kinks can also occur in this “rotational” sense. An estimand is pathwise differentiable at a point in a model only if there is a “direction of maximum slope”.

Example 3.3. Let $\mathcal{P}$ be all distributions of a continuous variable Z. Let’s say that our goal is to estimate the value of the density function of z at the point $z = 0$. There generally isn’t a good motivation for wanting to know this, it’s just a simple example that illustrates ideas that apply identically to more complicated estimands.

We’ll show that there is no RAL estimator of the density at a point by establishing that the pathwise derivative is not bounded over the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$ (at any $\mathbb{P}$). The intuition is that we can find a path that comes very, very close to modifying the density *only* at the point $z = 0$. In other words, we can modify the estimand by arbitrary amounts essentially without changing the overall density.

Fix a distribution $\mathbb{P}$. Since the model is nonparametric, the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$ is all of $\mathcal{L}_2^0(\mathbb{P})$. Since that’s the case, any zero-mean $\mathcal{L}_2(\mathbb{P})$ function can be a score.

Define the sequence of probabilities $t_n = \mathbb{P}\{-1/n \leq Z \leq 1/n\}$. Since these are probabilities of smaller and smaller sets over a continuous random variable they must shrink to zero.

We’ll consider a sequence of scores

$$s_n(z) = \frac{1\left(-\frac{1}{n} \leq z \leq \frac{1}{n}\right) - t_n}{\sqrt{t_n(1-t_n)}}$$

The random variable $1(-1/n \leq Z \leq 1/n)$ is 1 with probability t_n and 0 with probability $1-t_n$, i.e. it is $\mathsf{Bern}(t_n)$. Therefore it has mean t_n and standard deviation $\sqrt{t_n(1-t_n)}$. Consequently, the mean of $s_n(Z)$ is zero and its standard deviation ($\mathcal{L}_2(\mathbb{P})$ norm) is 1. Thus s_n are all $\mathcal{L}_2(\mathbb{P})$ functions with mean zero as desired, i.e. these are scores in the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$. Moreover these are unit-norm scores.

To find the pathwise derivatives $\dot{\Psi}(s_n)$, let’s consider a sequence of paths $\mathcal{P}_\epsilon^{(n)}$ that have the desired scores. For this we can use the convex paths $p_\epsilon^{(n)} = (1 + \epsilon s_n)p$ that we previously used in Example 2.2 and Example 2.5. Now we can evaluate the pathwise derivative directly:

$$
\begin{aligned}
\dot{\Psi}(s_n) &= \frac{\mathrm{d}}{\mathrm{d}\epsilon}\bigg|_{\epsilon=0} \Psi(\mathbb{P}^{(\epsilon)}) \\
&= \frac{\mathrm{d}}{\mathrm{d}\epsilon}\bigg|_{\epsilon=0} (1+\epsilon s_n(0))p(0) \\
&= s_n(0)p(0) \\
&= \frac{1-t_n}{\sqrt{t_n(1-t_n)}} p(0)
\end{aligned}
$$

Since t_n converges to zero, the denominator blows up. Since we only considered unit scores, we proved that there is no pure direction of maximum increase: no matter what direction we look, there is always another direction for which the pathwise derivative is bigger. Therefore the pathwise derivative is unbounded (not continuous). As a consequence, there can be no RAL estimator for the density at a point when the model is all continuous densities.

With more setup and more care taken to define tangent sets, etc., very similar arguments apply if we are estimating the value of a prediction function at a point (i.e. $\mathbb{E}[Y|X=x]$) or a conditional average treatment effect ($\mathbb{E}[Y|X=x, A=1] - \mathbb{E}[Y|X=x, A=0]$) in nonparametric models. These are important examples of practically relevant estimands for which there are no RAL estimators! The theory we develop in the rest of this book does not apply to these estimands which is why approaches to estimate them can be more ad-hoc, context dependent, and difficult to build formal inference around.

Exercise 3.2. Whether an estimand is pathwise differentiable or not depends on the model. Imagine that we know Z is normally distributed with known variance 1 but unknown mean μ, i.e. $\mathcal{P} = \{\mathsf{N}(\mu, 1) : \mu \in \mathbb{R}\}$ and we are interested in estimating the value of the density $p(z)$ at the point $z = 0$. This is again a contrived example but the point is to illustrate ideas in the simplest setting possible. Recall that the density of a $\mathsf{N}(\mu, 1)$ random variable is $p(z) = (2\pi)^{-1/2} e^{-(z-\mu)^2/2}$.

a. What is the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$ at any point $\mathbb{P}$ in this model? What is the dimension of this tangent set?
b. For any score $s \in \dot{\mathcal{P}}_{\mathbb{P}}$, construct the convex path $p_\epsilon = (1+\epsilon s)p$ and use it to compute the pathwise derivative $\dot{\Psi}(s)$.
c. What is $\dot{\Psi}(s)/\|s\|$? According to Definition B.6 and Definition 3.3 what does this imply about the pathwise differentiability of the estimand in this model?
d. Can you guess what a RAL estimator of $p(0)$ might be?

Pathwise differentiability does not necessarily imply that there *is* a RAL estimator. It only provides the possibility of one existing. In the next part of this book we will study how to actually build efficient estimators and we will see that we need certain assumptions even to guarantee asymptotic linearity. However, the nature of these problems depends on the precise estimand in question. From a big picture perspective it is safe to say that pathwise differentiability implies that a RAL estimator exists at least in principle.

3.4 Gradients of the Estimand

Part II

Building Efficient Estimators

4 Plug-In Estimators

4.1 What is a Plug-In?

...

4.2 Plug-In Bias

What's wrong with a plugin $\Psi(\hat{\mathbb{P}}_n)$?

If $\hat{\mathbb{P}}_n$ is fit with ML we always get bias.

Model selection: bigger and smaller models.

Validation error is bias^2 + variance. When we minimize it some bias (eg in μ) remains and it translates to bias in $\hat{\phi}$

Bias is a problem if you want to construct CIs

The plug-in bias does not disappear quickly enough in the asymptotic expansion to be CLT-negligible

Thus CIs etc will be wrong, even though estimate may be ok. — ppl need practice for this (or example)

4.3 Von Mises Expansion (formalizing above)

4.4

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A Convergence

Throughout chapters 3 and 4 we talk a lot about the convergence of random variables so it helps to understand what we mean by these things. In the definition of asymptotic linearity the term $o_{\mathbb{P}}(n^{-1/2})$ already invokes a specific type of convergence (more on that below).

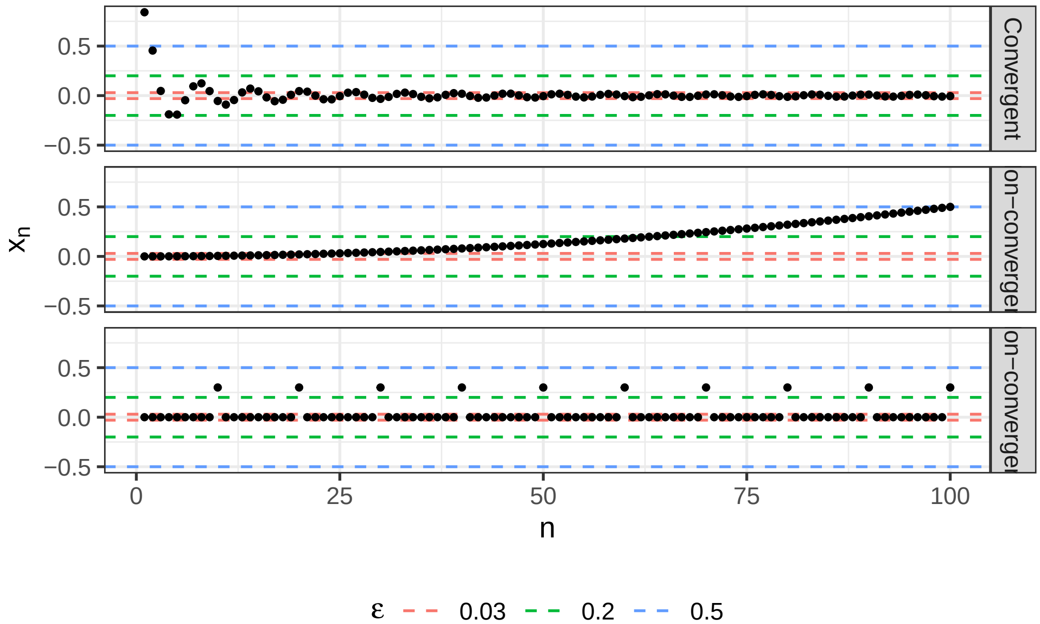
A.1 Sequences

We'll start with the convergence of a sequence of real numbers. A sequence of real numbers is what is sounds like: an infinite ordered list of numbers. We write a sequence like $[x_1, x_2, x_3, \dots x_n \dots]$. Technically we should use something like $\{x_n\}$ to refer to the entire sequence because x_n refers to the nth element, but usually we'll be lazy and just say x_n where it's otherwise clear.

Convergence of Sequences

Definition A.1. A sequence x_n *converges* to the *limit* x if for every $\epsilon > 0$, there is an integer N such that for all $n > N$, we have $|x_n - x| < \epsilon$. To denote this we write either $x_n \to x$ or $\lim_{n\to\infty} x_n = x$.

Intuitively, the values in the sequence "approach" the value x, but this is a little misleading because successive elements of the sequence don't necessarily have to be closer to the limit than preceding values. the sequence can oscillate away and towards the limit as long as eventually it becomes trapped forever in the interval $[x-\epsilon, x+\epsilon]$ around the limit. I think this is a better way to think about it: if you can eventually "trap" the sequence in the interval $[x-\epsilon, x+\epsilon]$ no matter how you choose ϵ, then the sequence converges. Convergence is denoted like $x_n \to x$ or $\lim_{n\to\infty} x_n = x$.



It's important to understand that the convergence of a sequence does not depend on any of the initial values of a sequence. If I chop off the first 100 values of a sequence which converges, that sequence will still converge. If the original didn't converge converge, the truncated sequence will still not converge. If I chop off the first billion entries, the same holds. It doesn't matter how many of the initial values I cut because there are infinitely more after them! The point is that convergence is purely a property of how the "tail" of the sequence behaves. And moreover, it doesn't matter how far away from the beginning of the sequence that behavior "kicks in". For that reason, plots like the ones above are actually totally non-informative about the convergence of the sequence because they can't tell us what's happening in the tails. Nonetheless they are a useful teaching tool to build intuition for what convergence looks like.

There are many, many interesting and useful results relating to limits and convergence of sequences of real numbers. For example,limits obey a kind of "limit algebra" that makes it easy to manipulate them algebraically.

Limit Algebra

Theorem A.1. *If* $x_n \to x$ *and* $y_n \to y$*, then*

$$x_n + y_n \to x + y$$
$$x_n y_n \to xy$$

A.2 Functions

Functions can be said to converge too. A sequence of functions is just what it sounds like: a never-ending list of functions like $\{x^2, 2x^2, 3x^2 \ldots\}$.

Pointwise Convergence of Functions

Definition A.2. The sequence of functions f_n *converges pointwise* to a limit f if $f_n(x) \to f(x)$ as a sequence for every input x.

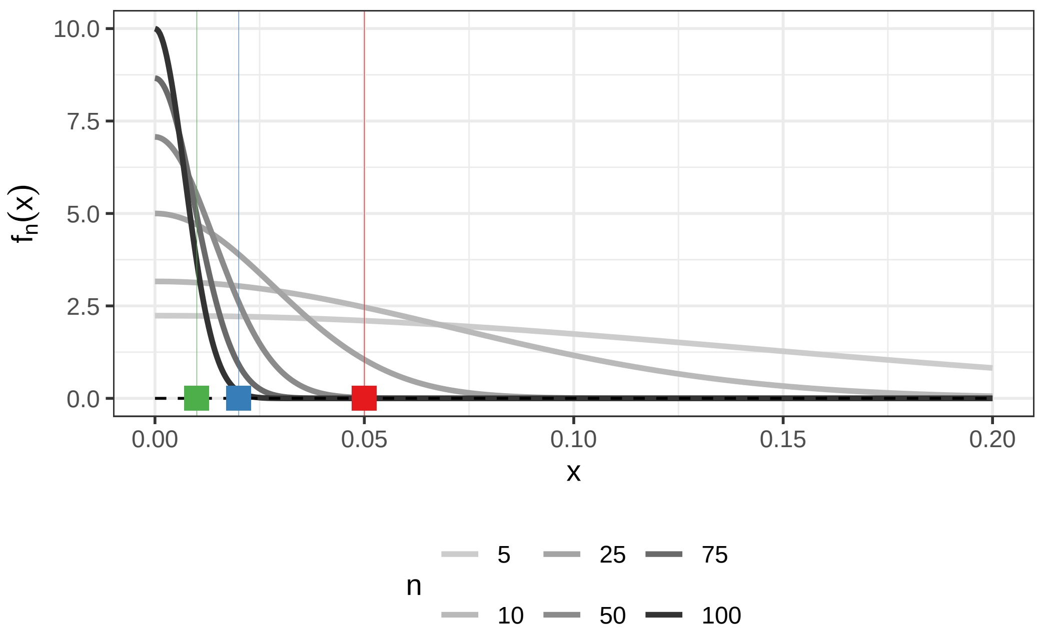
Pointwise convergence is one way to say that the functions f_n are getting closer to f as n increases, but there are actually other ways too! For example, we say that f_n converges to f in mean-squared error if $\int (f_n(x) - f(x))^2 \, \mathrm{d}x \to 0$ as a sequence. For reasons we'll see later this is sometimes denoted as $f_n \overset{\mathcal{L}_2}{\to} f$.

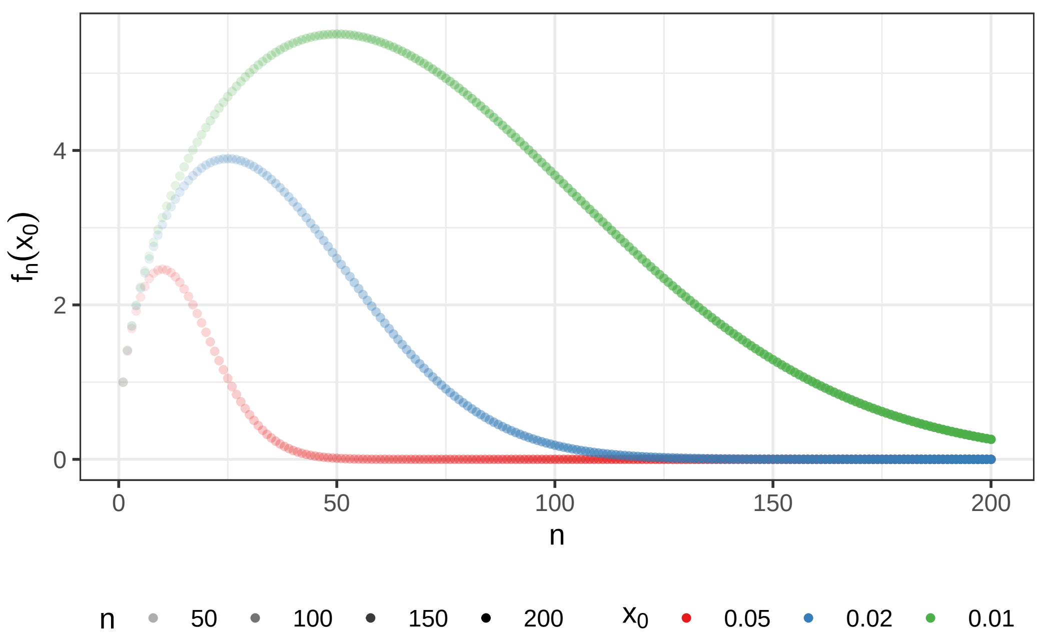
The reason I bring this up is to point out the fact that there are many different senses of "convergence" when you start talking about objects that are more complicated than real numbers. A function can converge to a limit in a mean-squared sense but not in a pointwise sense, and vice versa! Sometimes one type of convergence implies another and sometimes not (without additional assumptions).

Example A.1. In the plots shown here we investigate the convergence of

$$f_n(x) = \sqrt{n} e^{-(nx)^2}$$

on the domain $x \in (0, \infty)$.





As you can see visually, at any point $x_0 \in (0, \infty)$ the sequence of points $a_n = f_n(x_0)$ converges to zero. Therefore we say $f_n \to 0$ on the domain $(0, \infty)$. However, if you calculate the mean-squared error $\int_0^\infty (f_n - 0)^2 \, \mathrm{d}x$ for this sequence of functions (your favorite LLM can do this for you), you'll see that it stays at a constant, nonzero value of $\sqrt{\pi/8}$. Therefore, while this sequence converges pointwise, it does not converge in a mean-squared sense.

There is also a more general notion of convergence of which the convergence of real number sequences and mean-squared convergence of functions are special cases.

Metric Space Convergence

Definition A.3. Let $\mathcal{X}$ be some set and let $d : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a *metric* (also called a *distance function*) satifying the following natural properties of "distance between two points":

a. $d(x, x) = 0$ for all $x \in \mathcal{X}$
b. $d(x, y) > 0$ as long as $x \neq y$
c. $d(x, y) = d(y, x)$
d. $d(x, y) \leq d(x, z) + d(z, y)$, i.e. taking a detour to z on the way from x to y increases the total distance traveled

The set $\mathcal{X}$ and metric d are together called a *metric space* $(\mathcal{X}, d)$.
In a metric space, we say that a sequence of objects $x_n \in \mathcal{X}$ $(\mathcal{X}, d)$-converges to a limit x if for every $\epsilon > 0$, there is an integer N such that for all $n > N$, we have $d(x_n, x) < \epsilon$. Equivalently, we can say that $x_n \overset{(\mathcal{X},d)}{\to} x$ if and only if $d(x_n, x) \to 0$ in the conventional sense as a sequence of real numbers.

This definition generalizes the definition used for sequences real numbers by generalizing the usual "distance" $d(x, y) = |x - y|$ on the reals to other possible choices that make sense on other sets of objects. Usually we just say x_n "converges" to x instead of saying that the sequence $(\mathcal{X}, d)$-converges, leaving the set and metric implicit when clear from context. It's also common to see something like $x_n \overset{\mathcal{X}}{\to} x$, slightly abusing notation and letting $\mathcal{X}$ represent both the set $\mathcal{X}$ and the metric space $(\mathcal{X}, d)$. In this convention, the metric comes "bundled" with the set. In practice certain sets almost always use the same metric (e.g. absolute value of differences for $\mathbb{R}$, Euclidian norm for $\mathbb{R}^d$: see Example A.2) but really there is no single privileged metric that goes with each set. When multiple spaces or metrics are in play you may see the more explicit $(\mathcal{X}, d)$ notation because sequences may converge under one metric but not another.

Example A.2. For finite-dimensional vectors $v \in \mathbb{R}^d$, the Euclidian norm is defined as $\|v\| = \sqrt{\sum v_j^2}$. From this, the Euclidean distance is defined as $d(u, v) = \|u - v\|$. It is easy to verify that this distance satisfies the properties of a metric. Then we can say that $v_n \to v$ if $\|v_n - v\| \to 0$.

Exercise A.1. Let $d(x, y) = 1(x = y)$.

a. Show that this satisfies the properties of a metric on $\mathbb{R}$ (an alternative to the usual $d(x, y) = |x - y|$).
b. Show that a sequence x_n converges to a limit x under this metric only if there is an N after which the elements of the sequence are all x, i.e. $x_n = x$ for $n > N$.

There is actually an even more general notion of convergence (convergence in topological space) that also generalizes metric convergence as well as pointwise convergence of functions (which can't be seen as convergence in any metric space) but we don't need to bother with it in this book since we'll always work in metric spaces.

Convergence of sequences and functions is a central topic in real analysis. You can consult an analysis book or lecture series to learn more. I recommend starting with Abbott (2015) and then proceeding to the first three chapters of Kreyszig (1991).

A.2.1 Uniform Convergence

Uniform Convergence of Functions

Definition A.4. f_n converges to f *uniformly* if the largest difference between f and f_n across all values of x at each n converges to 0:

$$\sup_x |f_n(x) - f(x)| \to 0.$$

Notice that as as n changes, the point x at which the difference $|f_n(x) - f(x)|$ is maximized might change!

Uniform convergence is another example of convergence in a metric space (Definition A.3): you can check that $d(f, g) = \sup_x |g(x) - f(x)|$ is a valid metric on the set of all functions.

Example A.3. let $f = 0$ and consider a sequence $f_n(x) = 1(n < x \leq n + 1)$. You can think of the sequence f_n as a bump that moves further and further down the x-axis as n increases. If you stand still at any value x, the bump comes and passes over you and then departs. Therefore, for n large enough (specifically $n > x + 1$), $f_n(x) = 0$ exactly forevermore and thus $f_n \to f$ pointwise.

That might seem a little weird to you considering that obviously there is still always a bump in f_n that doesn't "go to zero" no matter how big n is. That's part of the intuiton that uniform convergence tries to capture. If you look at the largest value of $f_n - f = f_n$, you will find that it is always 1, attained at any value $x \in [n, n + 1]$. Therefore $\sup_x |f_n(x) - f(x)| = 1 \not\to 0$, meaning this sequence of functions does not converge *uniformly*.

Exercise A.2. In Example A.1 we considered the pointwise convergence of $f_n(x) = \sqrt{n}e^{-(nx)^2}$. Just by looking at the plots, explain why this sequence does not converge uniformly.

Similar concepts pop up in different places across mathematics. in general, when we're looking at some property of the worst-case behavior of a mapping across a variety of inputs simultaneously we usually use the word "uniform" to describe it.

A.2.2 Continuity

Convergence is the foundation for many other useful concepts. One of these is *continuity*.

Continuity

Definition A.5. A function $f : \mathcal{X} \to \mathcal{Y}$ is *continuous* at a point in its domain $x \in \mathcal{X}$ if $f(x_n) \xrightarrow{\mathcal{Y}} f(x)$ for all sequences $x_n \xrightarrow{\mathcal{X}} x$.
A second way to write this is that $d_{\mathcal{X}}(x, x_n) \to 0$ implies $d_{\mathcal{Y}}(f(x), (x_n)) \to 0$. A third way is that $\lim_{n\to\infty} f(x_n) = f(\lim_{n\to\infty} x_n)$ for sequences $x_n \to x$.

Informally, a function is continuous if small perturbations in its input can only create small perturbations in the output. In this definition I've abused notation a bit to let $\mathcal{X}$ and $\mathcal{Y}$ represent metric spaces so that I don't have to explicitly write the metrics.

A.3 Random Variables

Finally we get to random variables. When rigorously defined, random variables are actually functions and can converge in many different ways, but most of them will not be important for the rest of this book. For our purposes, we will be able to think about random variables pretty much exclusively as distributions. The type of convergence we will mostly be concerned with is *convergence in distribution*. You can think of a sequence of random variables Z_n as a sequence of distributions. Since all distributions have cumulative distribution functions (CDFs) you can also think of a sequence of CDFs $F_n(z) = F_{Z_n}(z)$. You may lastly also imagine a sequence of densities or mass functions if you like. Not all distributions actually have densities in the way you may be used to thinking about them, but that shouldn't stop you from using densities to build your intuition. I usually picture something like this Joy Division album cover when I think of a sequence of random variables:

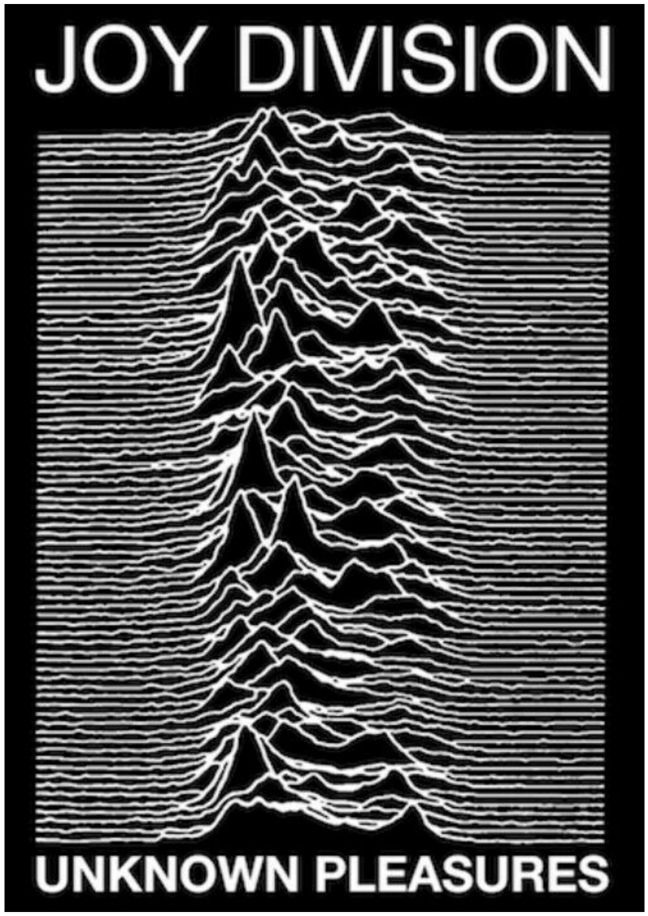


Figure A.1: Joy Division's famous Unknown Pleasures album cover

Convergence in Distribution

Definition A.6. The sequence Z_n converges in distribution (or "converges in law" or "weakly converges") to a limit Z if and only if the CDFs F_n converge pointwise (as functions) to the CDF F of Z (at all points where F is continuous). We write this as $Z_n \rightsquigarrow Z$.

The reason convergence in distribution is so often used is because this is the kind of convergence implied by the central limit theorem (CLT). The (simplest version) of the CLT says that if Z_i are IID random variables with finite variance σ^2, then

$$\sqrt{n}\left(\frac{1}{n}\sum Z_i - \mathbb{E}[Z]\right) \rightsquigarrow \mathsf{N}(0, \sigma^2).$$

This might look a little complicated but if you rename $X_n = \sqrt{n}\left(\frac{1}{n}\sum Z_i - \mathbb{E}[Z]\right)$ you see that this is just a particular sequence of random variables (because the Z_i are random variables) and we can read the CLT as saying $X_n \rightsquigarrow \mathsf{N}(0, \sigma^2)$.

As with convergence of real number sequences, you should keep in mind what this definition does *not* imply. It does not imply that the sequence converges "nicely" in the sense that all subsequent elements are always "closer" to the limit than previous elements. It does not tell us anything about *when* the sequence gets "close enough" to its limit: we could get "close" by $n = 10$ or it might take up until $n = 10^{100}$! Convergence says nothing about the behavior at the beginning of the sequence, and the "beginning" could be very, very long.

The other type of convergence of random variables we'll use is convergence in probability. I'll give you the full definition for completion, but it's a bit tricky and honestly for the purposes of this book you can get away without really understanding it.

Convergence in Probability

Definition A.7. A sequence of random variables Z_n converges in probability to a random variable Z if $\mathbb{P}\{|Z_n - Z| > \epsilon\} \to 0$ as a sequence of real numbers for any $\epsilon > 0$. We denote this as $Z \xrightarrow{\mathbb{P}} Z$.

The reason this definition can be tricky is because in general you have to consider the joint distribution of Z_n and Z to calculate the probability invoked in the definition. For this reason, a sequence that converges in distribution might not converge in probability (but the converse is true: convergence in probability is stronger than convergence in distribution).

Thankfully, in this book, the only place we'll see convergence in probability is when we're talking about convergence to a fixed constant z (i.e. a degenerate distribution where all the mass is placed at a single value). In this case, the definition simplifies: $Z \xrightarrow{\mathbb{P}} z$ if and only if $Z_n \rightsquigarrow z$. So the only kind of convergence you really need to recognize to read this book is convergence in distribution.

Exercise A.3. Let $Z_n \sim \text{Unif}(0, 1{+}1/n)$. What distribution does this converge to? Draw a picture to sharpen your intuition, then prove the convergence rigorously by taking the pointwise limit of the CDFs.

Limits of random variables obey some "limit algebra" rules that are the equivalents to the limit algebra rules for sequences of real numbers (Theorem A.1).

Distributional Limit Algebra (Slutsky's Lemma)

Theorem A.2. *If $X_n \rightsquigarrow X$ and $Y_n \xrightarrow{\mathbb{P}} y \in \mathbb{R}$, then*

$$\begin{aligned} X_n + Y_n &\rightsquigarrow X + y \\ X_n Y_n &\rightsquigarrow Xy \end{aligned}$$

See Lemma 2.8 in Van der Vaart (2000) for a proof.

A.3.1 Little $o_{\mathbb{P}}$

Often in asymptotic statistics we want to say something like

$$Y_n = X_n + U_n \text{ where } U_n \xrightarrow{\mathbb{P}} 0.$$

This is kind of annoying to write! Probabilists and statisticians have come up with a more convenient way to say the same thing:

$$Y_n = X_n + o_{\mathbb{P}}(1).$$

You can read the term $o_{\mathbb{P}}(1)$ as "some sequence of random variables that converges in probability to 0". The benefit of this is that we don't even have to give this sequence a name (like U_n, for example) if we don't care to. Moreover, we've hidden away any limit arrows and are left with an equation that we can manipulate algebraically.

This notation is extended to handle cases where we want to say

$$Y_n = X_n + U_n \text{ where } r_n U_n \xrightarrow{\mathbb{P}} 0.$$

The abbreviation for this is $Y_n = X_n + o_{\mathbb{P}}(1/r_n)$. The way I "see" this from the notation is that $r_n U_n = o_{\mathbb{P}}(1)$, so we "divide" by r_n to get

$$U_n = o_{\mathbb{P}}(1)/r_n \equiv o_{\mathbb{P}}(1/r_n).$$

The last equality here defines what we mean when we say some variable U_n is $o_{\mathbb{P}}(1/r_n)$: namely that U_n times r_n converges in probability to 0. When this is the case we say that "U_n converges at the *rate* $1/r_n$". Sometimes the sequence r_n can also itself be a random variable R_n, but the meaning is the same: if you see $U_n = o_{\mathbb{P}}(1/R_n)$ it means that $U_n R_n \xrightarrow{\mathbb{P}} 0$.

The intuition for "rates" is that often we're interested in random variables that go to 0 so quickly that even if they get blown up by some increasing sequence, the product still goes to 0. For example, if some variable U_n is $o_{\mathbb{P}}(n^{-1/2})$, that means that $\sqrt{n}U_n$ still goes to 0, even though $\sqrt{n}$ is sequence that gets bigger and bigger. The rate r_n ($r_n = n^{1/2}$ in this example) is the factor by which we can afford to "blow up" the sequence of variables U_n and still have it converge to 0. Another way to think of it is that $U_n = o_{\mathbb{P}}(1/r_n)$ means that $U_n < C/r_n$ with high probability (for some constant C and large enough n). Notice that this is a strict inequality: more on that later.

Example A.4. Let's say that $Z_n \sim \mathsf{N}(0, \sigma_n^2)$ where $\sigma_n = n^{-1/2}$. This is what we'd expect the large-sample sampling distribution of $\widehat{\psi} - \psi$ to look like for any asymptotically linear estimator (see Definition 1.1).

Below, you can see some density plots for this sequence and how they change if they are blown up by different sequences. In the top two rows, the scaling factor is 1 (not increasing) or $n^{1/4}$ (increasing very slowly) and you can see that the densities are still narrowing as n increases, albiet a little more slowly when blown up by $n^{1/4}$. However, in the third row the sequence is blown up by the moderately increasing factor $n^{1/2}$. You can see this is enough to perfectly "stabilize" the distribution: it neither narrows nor widens as n increases. Lastly, in final row we've blown up Z_n by the massive factor n and you can see the distribution widens very quickly as n increases.

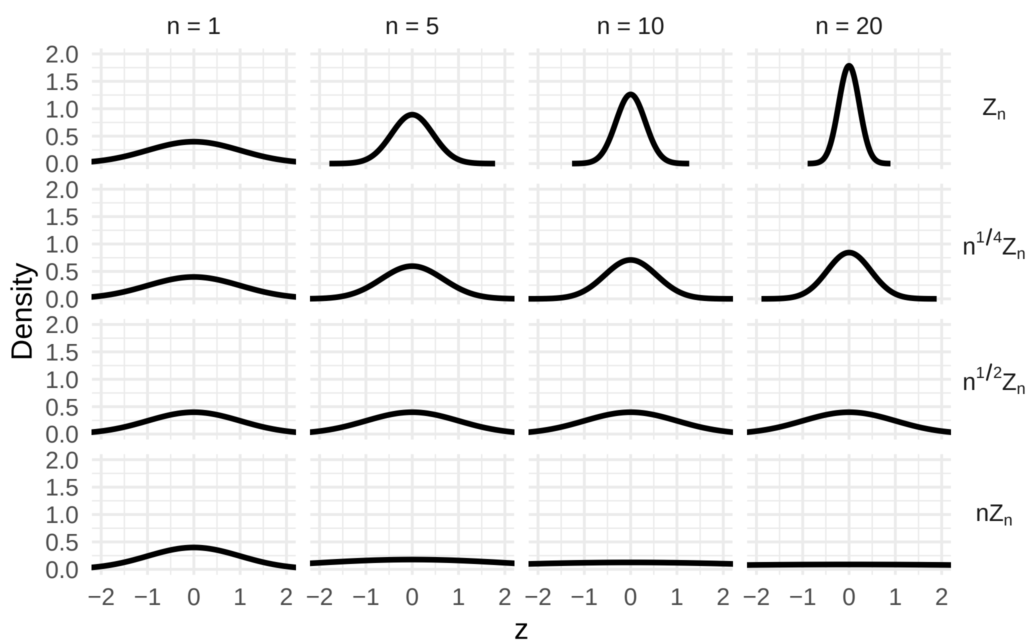


Figure A.2: Scaled normal densities.

Since we can see $Z_n \xrightarrow{\mathbb{P}} 0$ we could write $Z_n = o_{\mathbb{P}}(1)$. Similarly, since we see $n^{1/2} Z_n \xrightarrow{\mathbb{P}} 0$ we could also write $Z_n = o_{\mathbb{P}}(n^{-1/4})$, which is a stronger statement. However, we *cannot* write $Z_n = o_{\mathbb{P}}(n^{-1/2})$ because $n^{1/2}Z_n$ is not collapsing to zero. Similarly we cannot write $Z_n = o_{\mathbb{P}}(n^{-1})$.

Exercise A.4. Let $U_n \sim \text{Unif}(0, 1/n)$. Is $U_n = o_{\mathbb{P}}(1)$? Is $U_n = o_{\mathbb{P}}(1/n)$?

Writing $U_n = o_{\mathbb{P}}(1/r_n)$ tells us something about the sequence U_n without entirely specifying what that sequence is. The use of the equals sign is a big abuse of notation because obviously two sequences can be "equal" to the same $o_{\mathbb{P}}(1/r_n)$ without being equal to each other. Moreover anything that is $o_P(n^{-1/2})$ is also $o_P(1)$ (something that converges to zero when blown up still converges to zero if it is not), but not the other way around. The utility of the $o_{\mathbb{P}}$ notation is that it lets us show some important property of a sequence of random variables within the context of an equation. It just saves a line or two of exposition. In that sense it's not that different from other conventions like consistently using capital letters for random variables: even without extra exposition you already know an important property of the objects you're looking at.

The most common use of this notation is to tell the reader that some term is "negligible" (in large samples) relative to some other term of greater analytic importance. For example, let's say that $X_n \sim \mathsf{N}(0, \sigma^2/n)$ and I know that $Y_n = X_n + U_n$. By the definition of X_n, we know that

$$\sqrt{n}X_n \sim \mathsf{N}(0, \sigma^2).$$

This does *not* converge to zero: it stays stable at this normal distribution with "spread" σ even as n increases. If we can show that $\sqrt{n}U_n \xrightarrow{\mathbb{P}} 0$ (also written $U_n = o_{\mathbb{P}}(n^{-1/2})$), then we would know that U_n is "negligible" compared to X_n in large samples: U_n converges to 0 even when blown up by $\sqrt{n}$ whereas X_n doesn't. Therefore, asymptotically, $Y_n \approx X_n$. To make this a little more precise we would write $Y_n = X_n + o_{\mathbb{P}}(n^{-1/2})$ to indicate "how quickly" this approximation kicks in.

Alternatively, if you saw $Y_n = X_n + o_{\mathbb{P}}(n^{-1/2})$ without knowledge of the distribution of X_n then you would know that $Y_n \approx X_n$ asymptotically only if X_n converges more slowly (if at all) to zero than $n^{-1/2}$. Alternatively, you would know that Y_n converges to zero at an $n^{-1/2}$ rate if and only if $X_n = o_{\mathbb{P}}(n^{-1/2})$.

Usually the most important part of a rate will be some polynomial factor like $n^{1/2}$. You should remember that the larger the exponent, the faster a polynomial grows. If you have a negative exponent, the larger in magnitude it is the faster it decays. A term that is $o_{\mathbb{P}}(n^{-1})$ decays very quickly, while $o_{\mathbb{P}}(n^{-1/2})$ decays more slowly and $o_{\mathbb{P}}(n^{-1/3})$ more slowly still.

A final note is that a similar notation is often used to describe *deterministic* sequences. If $r_n a_n \to 0$, you will sometimes see that written as $a_n = o(r_n^{-1})$.

Example A.5. If a function $f(z)$ is differentiable at $z = 0$ with derivative $\dot{f}$ it means that

$$\lim_{t \to 0} \frac{f(t) - f(0)}{t} = \dot{f}$$

Another way to write this is that

$$\frac{f(t) - f(0)}{t} - \dot{f} \overset{t \to 0}{\to} 0,$$

or, using the little o notation,

$$\frac{f(t) - f(0)}{t} - \dot{f} = o(1).$$

Multiplying by t we get another way to write differentiability:

$$(f(t) - f(0)) - \dot{f}t = o(t).$$

This way of writing the definition more clearly illustrates that $\dot{f}$ is the slope of a linear function $g(t) = \dot{f}t$ which approximates the difference $f(t) - f(0)$ well-enough that the remainder of the approximation disappears at a better-than-linear rate $o(t)$ (recall $t \to 0$). This is really the key to differentiability: you can construct a linear approximation to any function at a point, but for the function to be *differentiable*, that approximation error has to improve linearly or better as you zoom in at a steady rate to the approximation point.

A.3.2 Big $O_{\mathbb{P}}$

The last thing to know about is the "big oh-p" notation. If $U_n = O_{\mathbb{P}}(1/r_n)$, then it means that $r_n U_n$ is *stochastically bounded* (also called *asymptotically tight*). Essentially, a sequence that is stochastically bounded does not diverge. For a sequence of random variables to "not diverge" we mean that its probability mass mostly concentrates in some region $[-b, b]$ (but doesn't necessarily accumulate at a single point). We'll formalize this in a minute.

For example, let's say that the random variables Z_n converge to a normal distribution: since the limit is stable, we'd like to say that this sequence does not diverge. But because the normal is unbounded, there is no region $[-b, b]$ that contains all of the mass of the limit distribution. However, for a given tolerance ϵ, we can always find a region that contains $1 - \epsilon$ percent of the mass.

To rigorously define stochastic bounding we'll just generalize this insight.

Stochastic Bounding

Definition A.8. A sequence of random variables Z_n is stochastically bounded if for any given tolerance ϵ we can find a region $[-b, b]$ so that $\lim_n \mathbb{P}(|Z_n| < b) > 1 - \epsilon$.

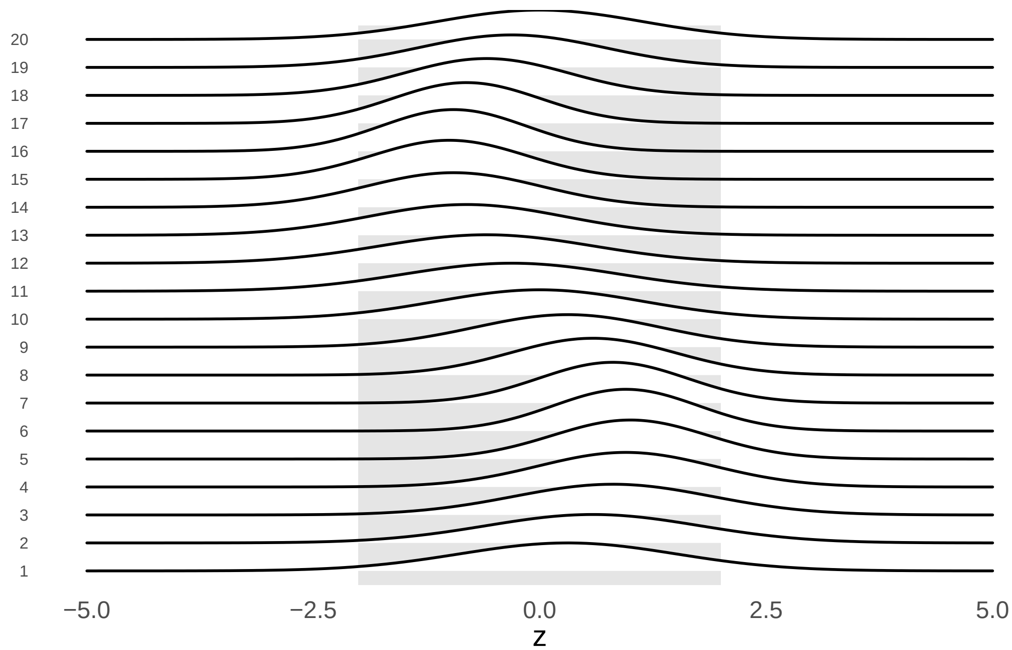


Figure A.3: The sequence of random variables $\mathsf{N}(\sin(n), \sin(n) + 2)$ does not converge in distribution yet is still stochastically bounded.

Saying a sequence is stochastically bounded is only *slightly* weaker than saying it converges to zero. If we multiply a stochastically bounded sequence by any sequence that converges to zero, however slowly, the resulting sequence will converge to zero. On the other hand, if we multiply it by any sequence that diverges, however slowly, then the resulting sequence also diverges. A stochastically bounded sequence that does not converge to zero is therefore on the knife's edge between convergence and divergence. Another way to think of it is that $U_n = O_{\mathbb{P}}(1/r_n)$ means that $U_n \leq C/r_n$ with high probability (for a constant C and large enough n). Notice that for big O it's less than *or equal to* whereas for little o it's a strict inequality.

Any sequence of random variables that converges in distribution is stochastically bounded. This makes sense if we're thinking of stochastic boundedness as "almost" convergence to zero: if the sequence converges to a (nonzero) stable limit distribution, any additional shrinking will collapse it to zero.

Example A.6. The CLT says that under a variety of conditions

$$\sqrt{n}\, \underbrace{\left(\frac{1}{n}\sum Z_i - \mathbb{E}[Z]\right)}_{X_n} \rightsquigarrow \mathsf{N}(0, \sigma^2)$$

Since the demeaned sample average X_n converges to a stable distribution when scaled by $\sqrt{n}$, it does not diverge. Therefore we can say $X_n = O_{\mathbb{P}}(n^{-1/2})$. However, the limit distribution is not deterministically zero. Therefore $X_n \neq o_{\mathbb{P}}(n^{-1/2})$.

Often the big O is used to indicate that r_n is *exactly* the rate that “stabilizes” the asymptotic distribution of U_n so that is neither converging to zero nor diverging. Any rate that is slightly slower (like $n^{1/2-\epsilon}$) and $r_n U_n$ would collapse to a point. Anything slightly faster (like $n^{1/2+\epsilon}$) and it would blow up.

Example A.7. Recall the example above where $Z_n = \mathsf{N}(0, \sigma_n^2)$ where $\sigma_n = n^{-1/2}$. In this case we saw that $n^{1/2} Z_n$ does *not* converge to zero in probability. However, $n^{1/2} Z_n$ *is* stochastically bounded because it converges to a stable limit distribution (in this case $N(0,1)$). Therefore while we cannot write $Z_n = o_{\mathbb{P}}(n^{-1/2})$, we *can* write $Z_n = O_{\mathbb{P}}(n^{-1/2})$ with the big O.

Technically something that is $O_{\mathbb{P}}(1/r_n)$ might also be $o_{\mathbb{P}}(1/r_n)$ (big O means less than or equal to, which *could* be strictly less). But usually for given rates $1/a_n < 1/b_n$ (e.g. $n^{-1/2} < n^{-1/3}$) you should roughly understand that $o_{\mathbb{P}}(1/a_n) < O_{\mathbb{P}}(1/a_n) < o_{\mathbb{P}}(1/b_n)$.

A.3.3 Convergence Algebra

Once you’ve understood the definitions of the symbols, the O notation lets you do some quick “algebraic” manipulations of asymptotic statements. Some of the following are copied from Chapter 2 of Van der Vaart (2000):

$o_{\mathbb{P}}$ Algebra

Theorem A.3.

statement	*intuition*
$o_{\mathbb{P}}(1) + o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1)$	*the sum of two convergent sequences converges*
$O_{\mathbb{P}}(1) + o_{\mathbb{P}}(1) = O_{\mathbb{P}}(1)$	*a stochastically bounded sequence plus a convergent seqeunce remains stochastically bounded*
$O_{\mathbb{P}}(1) o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1)$	*multiplying a stochastically bounded sequence by a convergent sequence is enough to tip it over the edge and make it converge*
$(c + o_{\mathbb{P}}(1))^{-1} = O_{\mathbb{P}}(1)$	*the reciprocal of a nonzero constant plus a convergent term will be stochastically bounded (because the denominator will eventually be close to $c \neq 0$)*

if $r_n \to \infty$ *then* $O_{\mathbb{P}}(r_n^{-1}) = o_{\mathbb{P}}(1)$	*any sequence that is still stochastically bounded when "blown up" by* r_n *must by itself converge*

As per the abuse of notation, these are really implications rather than equalities. For example, the first statment is an abbreviation for $X_n = o_{\mathbb{P}}(1) + o_{\mathbb{P}}(1) \implies X_n = o_{\mathbb{P}}(1)$ which itself is an abbreviation for $X_n \to 0, Y_n \to 0 \implies X_n + Y_n \to 0$.

B $\mathcal{L}_2$ Space

It will help you a lot to learn efficiency theory if you have a solid understanding of $\mathcal{L}_2$ and a few basic ideas from functional analysis. Many students of statistics are missing this background so I'll try and give you the necessary intuition here.

B.1 Vector Spaces

The first thing to understand is that functions are a kind of vector. This might make no sense to you if you're only familiar with the "vectors" you learned about in your first linear algebra class, which are finite-dimensional objects like $x = [x_1, x_2, x_3]$. But the formal definition of "vector" is actually much broader than this.

Real Vector Space

Definition B.1. A (real) *vector space* is a set $\mathcal{V}$ that satisfies the following properties:

1) **Closure under scalar multiplication**: Multiplication between elements of $\mathcal{V}$ and real numbers is defined. For all $v \in \mathcal{V}$ and real numbers c, the product cv is also in $\mathcal{V}$.
2) **Closure under addition**: Addition between elements of $\mathcal{V}$ are defined. For any two elements $u, v \in \mathcal{V}$, their sum $u + v$ is again in $\mathcal{V}$.

The elements $v \in \mathcal{V}$ are what we call vectors.

Technically this is a special case of vector space called a *real* vector space (one built on the real scalar field) but for our purposes that will suffice.

Exercise B.1. Why is the interval $[0, 1]$ not a vector space? Why is the set of all rational numbers not a (real) vector space?

Our usual (finite-dimensional) vectors are a special case of this. The vectors in the space $\mathbb{R}^d$ are all the ordered tuples of d real numbers. If $x = [x_1, \dots x_d]$ and $y = [y_1, \dots y_d]$, we define scalar multiplication as $cx = [cx_1 \dots cx_d]$ and $x + y = [x_1 + y_1, \dots x_d + y_d]$. Clearly if I add two of these objects together or scale one of them using these definitions I get another vector in the same set, which means that we can call this set a vector space and its elements vectors.

Technically we should also include the definitions of scaling and addition as part of the vector space since the set by itself doesn't naturally come "equipped" with these definitions.

If you were to "code" a vector space, you might do it something like this:

```
class Vector:

    def __init__(self::Vector):
        ... # define what kind of object the "vector" is

    def __add__(self::Vector, other::Vector):
        ... # define how to add two vectors so you can evaluate v + u

    def __mult__(self::Vector, c::Number):
       ... # define how to scale a vector by a number so you can evaluate c*v

VectorSpace = Set(  # a set containing various Vectors (usually infinite)
    Vector(...),
    Vector(...),
    ...
)
```

This has no practical value, I'm just trying to help you wrap your head around the ideas in case you are more familiar with programming than you are with abstract mathematics.

The definitions I gave for addition and scaling may seem so natural to you that you don't even think about them but in fact they are just one specific way we could define these operations for finite dimensional ordered tuples. You can check that the given definitions satisfy the usual properties of "addition" and "multiplication" (e.g. $a + (b + c) = (a + b + c)$) so it is sensible to call them by those names.

You can make a vector space out of pretty much anything upon which you can define some kind of scaling and addition. For example, consider the space of all functions that map real numbers to real numbers. If you give me a function f, let me define multiplication of a function by a scalar c to produce another function in the following way: cf is a new function that maps reals to reals where the output $(cf)(x)$ is given by $c \cdot f(x)$. Similarly, define addition of two functions as $(f + g)(x) = f(x) + g(x)$, i.e. the function sum $f + g$ is the unique function which when evaluated at any x gives the value $f(x) + g(x)$. Again this might seem so natural as to not need explanation (or to be made confusing by explanation) but you should be aware that these are man-made definitions of "addition" and "scaling" for functions rather than numbers. Equipped with these operations, the set of all functions from reals to reals becomes a vector space and the functions within it are the vectors.

B.1.1 Functions as Vectors

In linear algebra or vector calculus you may have built some intuition about finite-dimensional vectors by imagining them as "arrows" from the origin to a particular point in 3D (or higher dimensional) space. This helps us reason about vectors geometrically using our physical sense of space. For example, you have probably seen the usual definition of the "length" of a vector (the *norm*) for d-dimensional vectors: $\|x\| = \sqrt{\sum^d x_i^2}$). You probably also know the *inner product* (dot product) $x \cdot y = \sum^d x_i y_i$ (more generally written $\langle x, y \rangle$) and may recall that the inner product measures the *angle* between two vectors according to the law of cosines: $\cos(\theta_{x,y}) = \frac{x \cdot y}{\|x\|\|y\|}$. If the inner product is zero, the two vectors are *orthogonal*. You may also recall that the norm is actually defined from the inner product, i.e. $\|x\|^2 = x \cdot x$.

Seeing functions as vectors ("arrows") helps us apply similar geometrical reasoning. We can talk about the "length" of a function or the "angle" between two functions. We just need to come up with definitions for those things that satisfy our expectations for how geometry should work. But before doing that I want to give you some more intuition to help you "see" functions as vectors outside of an abstract definition of a vector space.

I like to think of a function as a (continuously) infinite list of numbers like this:

$$f = [\dots f(-1), \dots f(0), \dots f(1.1), \dots f(\pi), \dots].$$

Each "element" of this function/vector is the *value* that the function takes at a particular argument x. Instead of using function evaluation notation with the parentheses I could instead write it with the arguments as *indices*:

$$f = [\dots f_{-1}, \dots f_0, \dots f_{1.1}, \dots f_\pi, \dots].$$

This looks the same as a d-dimensional vector $x = [x_1, \dots x_d]$ except that now the set of indices is e.g. all real numbers $\mathbb{R}$ instead of the finite set $\{1, \dots d\}$. To me, this perspective makes functions "feel" like very long, densely packed vectors. Since the coordinate axes of a standard d-dimensional vector space correspond to the vector indices $\{1, 2, \dots d\}$, the "axes" of a vector space of functions might be thought of as corresponding to all the possible arguments x of the function(s). Different functions evaluate to different values along each of these (continuously infinite) different axes. Thinking this way shows that definitions of scaling and addition now work exactly the same "elementwise" way for finite-dimensional vectors and functions.

To complete the picture it's also helpful to notice that finite-dimensional vectors can also be written as functions. Instead of writing e.g. $x = [x_1, x_2, x_3]$ I could say that x is a function on the domain $\{1, 2, 3\}$ such that $x(1) = x_1$, $x(2) = x_2$, and $x(3) = x_3$. Indices and arguments are essentially two different ways to view or construct the same thing. The difference between a function (e.g. on the reals) and a finite-dimensional vector is just their domains if you want to see them both as functions (or their index sets if you want to see them both as "vectors").

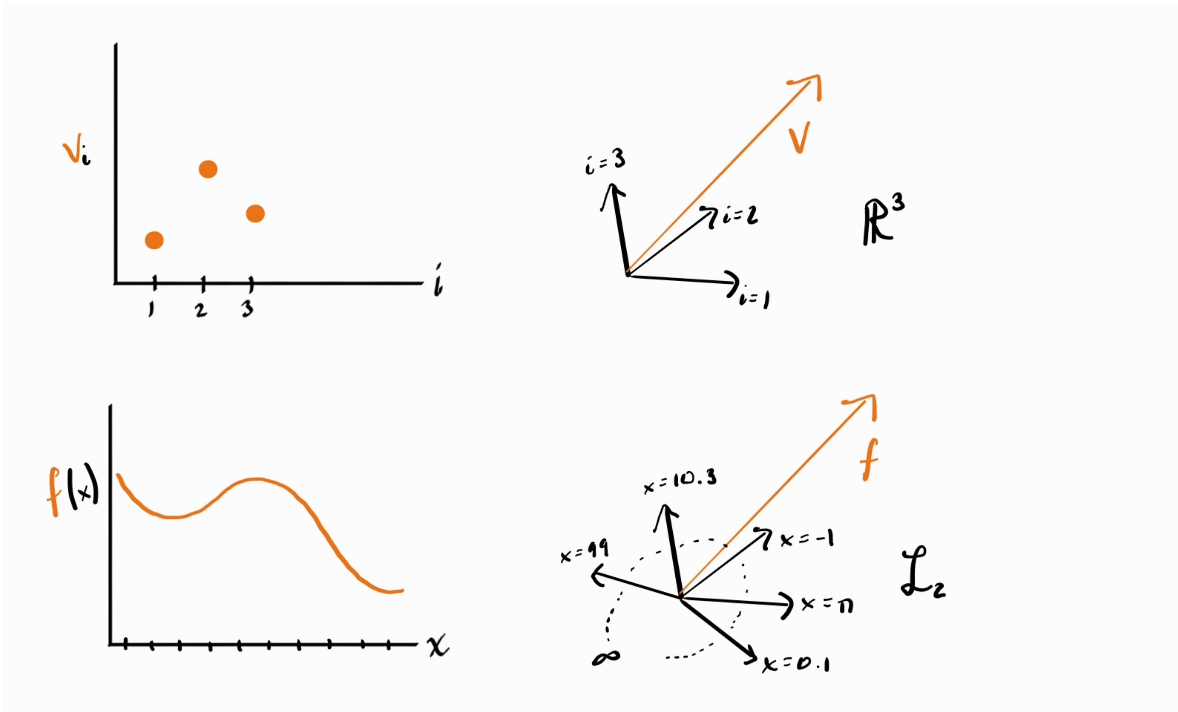


Figure B.1: Functions as vectors and vice-versa

Hopefully at this point the visual intuition doesn't seem so crazy to you, even when you're working with infinite dimensional spaces. But if you're having trouble, follow the advice of computer scientist Geoffrey Hinton: "to deal with a 14-dimensional space, visualize a 3-D space and say 'fourteen' to yourself very loudly. Everyone does it."

It's funny, but this kind of thinking is a staple of mathematics. We all use stupid pictures or imaginations grounded in our physical experience which we know are "wrong" in some way (e.g. 3 dimensions instead of 14). It's the only way to build intuition! The important thing is knowing that the pictures are wrong in some ways. But the more wrong pictures you have (e.g. from different "perspectives"), the better, since you can switch to a picture that "correctly" illustrates the phenomenon of interest.

B.2 $\mathcal{L}_2$ Space

Now that we've built that intuition we can start to formalize some of our geometric ideas. First up: what is the "length" of a function? If the usual length of a finite vector is determined by the (square root of the) sum of the squares of its elements, then we could extend that definition to a function. The only problem is that $\sum_{x \in \mathbb{R}} f(x)^2$ is not well-defined: the finite sum is not appropriate to deal with an index set that is continuously infinite. But we do already have a "continuously infinite sum": the integral. Thus we might define

$$\|f\|^2 = \int_{-\infty}^{\infty} f(x)^2 \, \mathrm{d}x$$

to be our norm. We call this the $\mathcal{L}_2$ norm (there are other choices; the "2" is for the square inside the integral). Although there are other possible norms, the $\mathcal{L}_2$ norm is nice because it is compatible with an inner product, namely

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(x) g(x) \, \mathrm{d}x.$$

You should again think of this as the continuously-summed version of the finite-dimensional dot product. Hopefully the intuition from thinking of functions as long, dense "vectors" is helpful to you here. If you replace the integral with a sum and the function call notation with an index you will see how this norm and inner product are analogous to the finite-dimensional versions you're more familiar with.

There are some functions which have infinite or undefined $\mathcal{L}_2$ norm. For example, $f(x) = x^2$ would have infinite norm since (for now) we're integrating over the entire number line and the area under the curve of $(x^2)^2$ goes off to infinity. To preserve the geometric intuition of our vector space, we cannot include these functions (the same way there is no finite-dimensional vector that has infinite norm). The vector space $\mathcal{L}_2$ (read: "el-two") is therefore defined as the set of all functions with finite $\mathcal{L}_2$ norm (not quite but see below). Sometimes this is also called the space of "square-integrable" functions. You can check that this is indeed a vector space, i.e. that two functions with finite square integrals sum to a function with a finite square integral and that a function with a finite square integral scales to another function with finite square integral.

Example B.1. Consider the functions $f = 1_{[0,1]}$ and $g = 1_{[2,3]}$. According to our inner product, these two functions are orthogonal (compute the integral and see). If this is not obvious to you at first, consider the finite-dimensional vectors $[0, 1]$ and $[1, 0]$: hopefully these are "obviously" orthogonal to you: you can visualize them as unit vectors along the two axes of $\mathbb{R}^2$. Our two functions are no different: f is nonzero along the "axes" corresponding to the indices in $[0, 1]$ while g is nonzero only along "axes" $[2, 3]$. If you imagine them as long dense "vectors" you'll see that f and g have ones in different places, the same way that $[0, 1]$ and $[1, 0]$ do.

A further wrinkle is that until now I've used the standard Riemman integral (with the differential $\mathrm{d}x$) to define $\mathcal{L}_2$ but that is not quite right. In reality it's slightly more complicated. We formally define $\mathcal{L}_2(\mathbb{P})$ to be the space of $\mathbb{P}$-measureable, square-Lebesgue-integrable functions. The measurability is a technical condition that rules out some extremely esoteric corner cases that you don't need to worry about: read up on measure theory if you want to learn more.

Although the $\mathcal{L}$ in $\mathcal{L}_2$ stands for "Lebesgue", the "Lebesgue" part is also not terribly important. The Lebesgue integral is a formalization of the integral that can integrate strictly more functions than the Riemann integral, but any function that is Riemann integrable Lebesgue-integrates to the same value so it doesn't make much difference in terms of intuition.

The important part is the $\mathbb{P}$: one of the strengths of the Lebesgue integral is that it can be defined with respect to any "measure" (probability distributions are formally mathematical objects called measures). When you integrate "with respect to a distribution $\mathbb{P}$" you write $\int f \, \mathrm{d}\mathbb{P}$ or $\int f(z) \, \mathrm{d}\mathbb{P}(z)$ to indicate more explicitly that $\mathbb{P}$ is a distribution over Z and that f takes arguments in the range of Z. If you're not familiar with measure theory (I don't expect you to be), when you see an integral with respect to a continuous distribution $\mathbb{P}(Z)$ with density function $p(z)$ you should read it as

$$\int f(z) \, \mathrm{d}\mathbb{P}(z) = \int f(z) \; p(z) \mathrm{d}z = \mathbb{E}_{\mathbb{P}}[f(Z)]$$

In other words, you can think of the differential $\mathrm{d}\mathbb{P}$ as weighting different parts of the domain of z in the integral by the probability of z falling into them. Integrating a function with respect to a distribution over Z is the same as taking its expectation over the distribution of Z.

However, $\mathcal{L}_2(\mathbb{P})$ spaces can also be defined when $\mathbb{P}(Z)$ is a discrete distribution (with probability mass function $p(z)$). In that case, $\int f(z) \, \mathrm{d}\mathbb{P}(z) = \sum f(z) \; p(z) = \mathbb{E}_{\mathbb{P}}[f(Z)]$. The integral is also defined for distributions that are neither strictly continuous or discrete (e.g. Z is 0 half the time and a standard normal the other half; think of a normal density with an infinite "spike" of mass 1/2 at 0). You don't have to worry about this to get all the intuition you need for this book.

You'll notice that some functions will be square-integrable under some distributions but not under others. There isn't really one single $\mathcal{L}_2$ space when we're working with probability distributions. In the wild, if you see "$\mathcal{L}_2$", that's short for the $\mathcal{L}_2$ space based on the Lebesgue measure: roughly speaking, the integrals used to define the norm and inner product don't use any weights on the real number line. Nonetheless I may sometimes refer to "$\mathcal{L}_2$" as a shorthand for $\mathcal{L}_2(\mathbb{P})$ for a specific distribution that is clear from context or for any generic distribution. Similarly, if we write a norm or inner product we should technically always specify the relevant space, i.e. $\|f\|_{\mathcal{L}_2(\mathbb{P})}$ or $\langle f, g \rangle_{\mathcal{L}_2(\mathbb{P})}$. However, to minimize notational burden we often leave off the details of the space and/or base measure when they are clear from context.

Example B.2. $f(z) = z^{-1/2}$ is square-integrable if $Z \sim \text{Unif}[2, 3]$ but not if $Z \sim \text{Unif}[0, 1]$.

The very last detail we need to attend to is that two different (pointwise) functions need to be considered "equivalent" in $\mathcal{L}_2$. This is a very minor detail that you can quickly forget about for most purposes but I'll go over it here for completeness.

In finite-dimensional vector spaces, if you have two vectors v and u and $\|v - u\| = 0$, then $v = u$. We'd like this to work in $\mathcal{L}_2$ but it doesn't without a little extra massaging.

Exercise B.2. Take $f(z) = 1(z = 0)$ and $g(z) = 0$. Argue that under any distribution $\mathbb{P}$ which has a traditional density function we get $\|f - g\| = 0$ for the $\mathcal{L}_2(\mathbb{P})$ norm.

The problem highlighted by Exercise B.2 is that the $\mathcal{L}_2$ distance "ignores" pointwise differences (in fact any countable number of them). Therefore in order to get our vector space to work the same way as $\mathbb{R}^d$ we need to say that f and g are effectively the same function as far as $\mathcal{L}_2(\mathbb{P})$ is concerned. Formally, we say that the elements (vectors) of $\mathcal{L}_2(\mathbb{P})$ are not functions themselves, but mutually exclusive groups of functions that are equivalent to each other for purposes of computing norms or inner products. We call these groups *equivalence classes*. Again, this is not really that critical to understand and doesn't have many implications as far as this book is concerned. Practically speaking, everyone considers *functions* to be the vector elements of $\mathcal{L}_2$ spaces, not equivalence classes of functions.

Now we can finally state our "official" definition of $\mathcal{L}_2(\mathbb{P})$:

$\mathcal{L}_2(\mathbb{P})$

Definition B.2. Given a distribution $\mathbb{P}$, The vector space $\mathcal{L}_2(\mathbb{P})$ is the set of equivalence classes of functions $[f] = \{g : \mathbb{P}(g - f)^2 = 0\}$ for which the norm $\|f\|^2 = \mathbb{P}f^2$ is defined and finite. We equip this set with the inner product $\langle f, g\rangle = \mathbb{P}fg$ which induces the norm.
We define $\mathcal{L}_2$ to be the *unweighted* equivalent where the inner product is given by $\int fg\,\mathrm{d}z$

Exercise B.3. Let $\mathbb{P}(Z)$ be such that Z Unif$(0, 2\pi)$. Calculate the $\mathcal{L}_2(\mathbb{P})$ inner product between $\cos(z)$ and $\sin(z)$. Based on the law of cosines, what does that mean the "angle" is between these functions?

The (squared) $\mathcal{L}_2(\mathbb{P})$ norm has an important probabilistic interpretation: it represents the (uncentered) second moment of $f(Z)$:

$$\|f\|^2 = \mathbb{E}[f(Z)^2].$$

For functions that have zero mean ($\mathbb{E}[f(Z)] = 0$), this is the variance $\|f\|^2 = \mathbb{V}[f(Z)]$. Therefore the "length" of a (zero mean) random variable is nothing other than its standard deviation.

Exercise B.4. Let a random variable Z and functions f, g be such that $\mathbb{E}[f(Z)] = \mathbb{E}[g(Z)] = 0$. Use the definitions of expectation and covariance to show that $E[f(Z)^2] = \|f\|^2$ and $\mathbb{C}[f(Z), g(Z)] = \langle f, g\rangle$ where the norm and inner product are with respect to $\mathcal{L}_2(\mathbb{P})$.

$\mathcal{L}_2$ is the canonical example of a *Hilbert space*, which is just an inner product space that contains its limits. So if you see that term anywhere and aren't familiar with it, just think of

$\mathcal{L}_2$ or $\mathbb{R}^d$. We won't look at Hilbert spaces besides these in this book but nonetheless I'll use the term "Hilbert space" just so I don't have to write out "$\mathcal{L}_2$ or $\mathbb{R}^d$" every time.

These topics are part of the foundations of measure theory and functional analysis. If you're curious about function spaces and integrals you should check out a book or take a course on one of these subfields (after some introductory real analysis). My recommendations for self-study are in the introduction to this book.

B.3 Triangles in Inner Product Spaces

In this section we'll investigate some properties of Hilbert spaces that come from simple geometric arguments about triangles. These properties are generic to all Hilbert spaces so when I say "vector" in the following I mean it in the general sense that could indicate a function, not just a finite-dimensional vector. To prove these properties I'll use the general definition of an inner product.

Inner Product

Definition B.3. A (real) inner product is any function $f(u, v)$ of two vectors that satisfies:

1. **Symmetry**: $f(u, v) = f(v, u)$
2. **Linearity**: $f(cv + w, u) = cf(v, u) + f(w, u)$ for vectors u, v, w and scalar c
3. **Positive-Definiteness**: $f(v, v) > 0$ for all $v \neq 0$

We use the special notation $\langle u, v\rangle$ to denote an inner product. If there are multiple inner products in play we use a subscript on the right bracket to distinguish them.

You can check that the usual dot product and the $\mathcal{L}_2(\mathbb{P})$ inner product we defined above satisfy these conditions. In what follows below I'll also use the fact that the norm is induced by the inner product: $\|v\|^2 = \langle v, v\rangle$. As a consequence of this and the definition above we have the simple corollaries $\|cv\| = c\|v\|$ and $\|v\| = 0 \implies v = 0$.

B.3.1 Pythagorean Theorem

You may recall the Pythagorean theorem from geometry. For a right triangle with two right-angle adjacent sides of length a and b, the length of the hypotenuse c is given by

$$c^2 = a^2 + b^2$$

We say two vectors are *orthogonal* (which we write $u \perp v$) if their inner product is zero: $\langle u, v\rangle = 0$. This is another way to formalize the notion of there being a right angle between vectors. Recall that you can think of $v - u$ as (the translation to the origin of) the "arrow" going from the point u to the point v. Therefore when $u \perp v$, you can think of v and u are the "sides" of a right triangle whose hypotenuse is $v - u$. In that case, the Pythagorean theorem tell us that $\|v - u\|^2 = \|v\|^2 + \|u\|^2$. In $\mathcal{L}_2$ and general Hilbert spaces things work the same way.

General Pythagorean Theorem

Theorem B.1. *In any Hilbert space, if f and g are orthogonal, then $\|f - g\|^2 = \|f\|^2 + \|g\|^2$.*

Proof of Theorem B.1

Recall that $\|f\|^2 = \langle f, f, \rangle$ and that $f \perp g$ means $\langle f, g\rangle = 0$. Then,

$$\begin{aligned}\|f - g\|^2 &= \langle f - g, f - g\rangle \\ &= \langle f, f\rangle - 2\langle f, g\rangle + \langle g, g\rangle \\ &= \|f\|^2 - 0 + \|g\|^2.\end{aligned}$$

Exercise B.5. Let $\mathbb{P}(Z)$ be such that Z Unif$(0, 10)$. Let $f(z) = 1(0 < z < 1)$ and let $g(z) = 1(5 < z < 6)$. Check that $f \perp g$ in $\mathcal{L}_2(\mathbb{P})$ and verify that $\|f - g\|^2 = \|f\|^2 + \|g\|^2$.

B.3.2 Projection

The Pythagorean theorem often comes up in the context of projection. Recall that in finite-dimensional vector spaces a *subspace* is a smaller vector space that is contained in the original vector space. For example, in 3-dimensional space, the x-y plane is a 2-dimensional subspace because any linear combination of vectors in that plane remain in the plane: thus it is a vector space in its own right. Subspaces are often constructed as a span (linear combination) of some set of vectors, e.g. the span of $\{[-2, 0, 0], [0, 5, 0]\}$ gives the x-y plane in 3D space. In infinite dimensions subspaces are defined in exactly the same way.

Although they sound similar, you should not confuse a sub*space* with a sub*set*. Any set of vectors can be a subset of the original vector space. A subspace, however, has to still itself be a vector space in its own right.

Example B.3. Consider the subset of $\mathcal{L}_2(P)$ comprised of all functions that have zero mean under $\mathbb{P}$, i.e. functions f where $\mathbb{E}[f(Z)] = 0$. This subset $\mathcal{V}$ is actually a sub*space* because it is a vector space. To verify that we just need to check that $\mathcal{V}$ is a vector space: if we add and scale elements with it, the results are still in $\mathcal{V}$.

To that end, let f and g be any functions in $\mathcal{V}$ and let c be a scalar constant. Since

$$\mathbb{E}[cf(Z) + g(Z)] = c\mathbb{E}[f(Z)] + \mathbb{E}[g(Z)] = 0,$$

$cf + g$ satisfies the condition required for membership in $\mathcal{V}$, proving it is a vector space.

Exercise B.6. Let $\mathbb{P}$ be the standard uniform distribution. Convince yourself that in $\mathcal{L}_2(\mathbb{P})$, the set of linear functions $\{f : f(z) = mz + b\}$ is a subspace and is spanned by $\{f(z) = 1, f(z) = z\}$ (thus making it a two-dimensional subspace!). Can you think of a subspace of this subspace? Can you think of a subset of linear functions that is *not* a subspace?

Now that we have our definition of subspaces we can define orthogonal projection in the context of inner product spaces.

Orthogonal Projection

Definition B.4. The orthogonal *projection* of a vector v onto a (closed) subspace $\mathcal{T}$ is defined to be the vector in $\mathcal{T}$ that is closest to v. In this book we'll use the notation

$$\mathcal{T} v = \operatorname*{argmin}_{t \in \mathcal{T}} \|v - t\|$$

to denote orthogonal projections.

Does Definition B.4 make sense?

For the definition above to make sense, we have to prove that there *is* a minimum-distance point to v in $\mathcal{T}$: it could be that there is none (what is the one rational number that is closest to π?) or that it is not unique (what is the integer that's closest to $1/2$?). For the purposes of this book you can take it as fact that it is well-defined. The proof is not complicated, but it requires more fluency with convergence and infima/suprema than I'd like to assume you have (as well as the parallelogram law). A key required assumption which is satisfied in all cases within this book is that the subspace is *closed*, i.e. that it contains its limits. If you are curious, you can take a screenshot of this definition and ask a large language model to supply the proof.

In terms of intuition, you can think of projection as "dropping down" the vector v onto the "plane" $\mathcal{T}$. The idea is that the projection $\mathcal{T} v$ is the "best approximation" of v that we can manage in $\mathcal{T}$.

In the case that $\mathcal{T}$ is a one-dimensional subspace of the larger vector space $\mathcal{V}$ we have a handy formula for computing the projection.

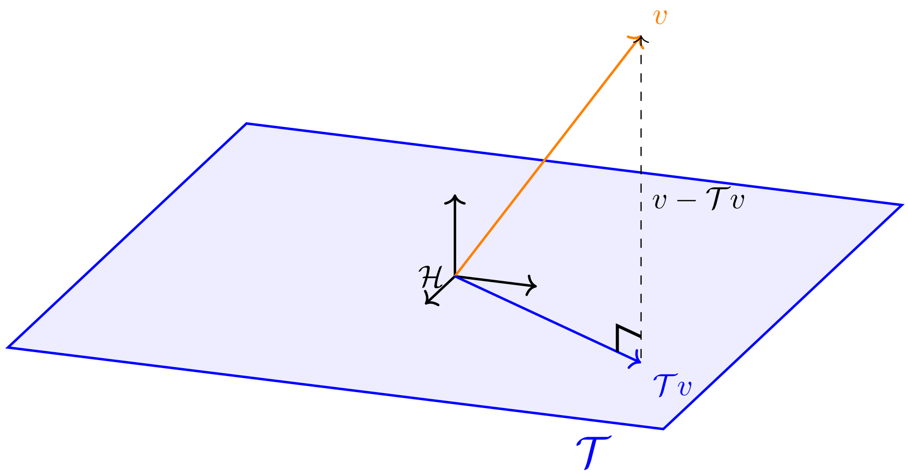


Figure B.2: A visual representation of orthogonal projection. Notice that the residual $v - \mathcal{T}v$ is orthogonal to all vectors in $\mathcal{T}$.

1D projection

Theorem B.2. *If* $\mathcal{T} = \{c\tilde{t} : c \in \mathbb{R}\}$ *for some fixed, nonzero vector* $\tilde{t}$*, then*

$$\mathcal{T}v = \left\langle v, \frac{t}{\|t\|} \right\rangle \frac{t}{\|t\|}$$

for any $t \in \mathcal{T}$*.*

Proof of Theorem B.2

The projection $\mathcal{T}v$ is defined to be the vector t^* in $\mathcal{T}$ that minimizes the distance $\|t - v\|$. Pick any nonzero vector $t \in \mathcal{T}$. Since every vector in $\mathcal{T}$ can be written as ct for some scalar c, we can write the minimizer $t^* = c^*t$. Using this and the fact that minimizing the distance is the same as minimizing the squared distance, the optimization problem is to find c that minimizes

$$\begin{aligned} \|ct - v\|^2 &= \langle ct - v, ct - v \rangle \\ &= c^2\langle t, t \rangle - 2c\langle t, v \rangle + \langle v, v \rangle \end{aligned}$$

To find the minimizer we can take the derivative of this with respect to c and set it equal to zero, giving $2\|t\|^2c^* - 2\langle t, v \rangle = 0$. Solving for c^* and plugging into $t^* = c^*t$ gives the result.

This formula gives us a nice geometric interpretation to the inner product if you consider a unit vector $\tilde{t}$. In this case, the projection formula simplifies to $\mathcal{T}v = \langle v, \tilde{t} \rangle \tilde{t}$. The inner product of v with the unit vector $\tilde{t}$ is therefore the length of the projection of f onto the span of $\tilde{t}$. The inner product with vectors t of different norm will give the same number, just scaled by that norm. Formally, if we let $\mathcal{T}$ be the span of t,

$$\langle v, t \rangle = \|\mathcal{T}v\| \|t\|.$$

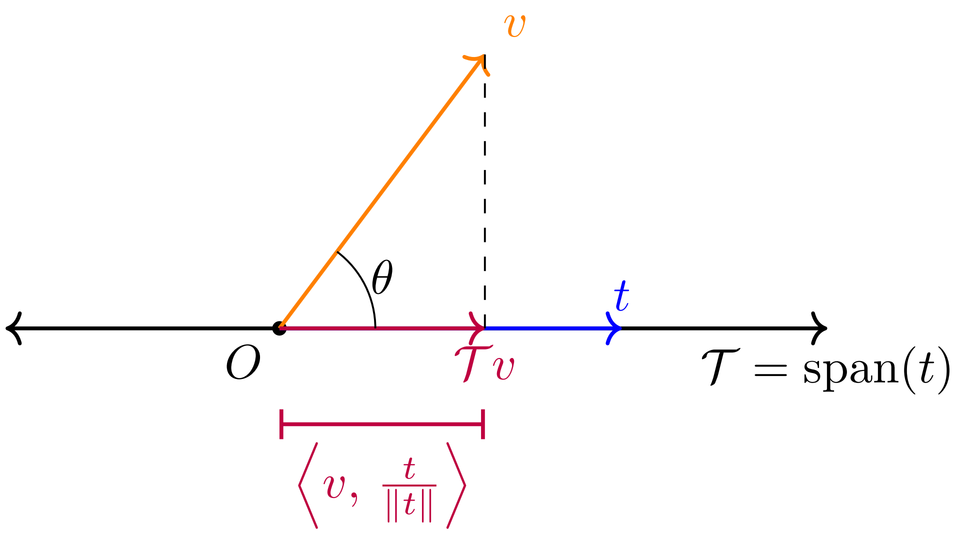


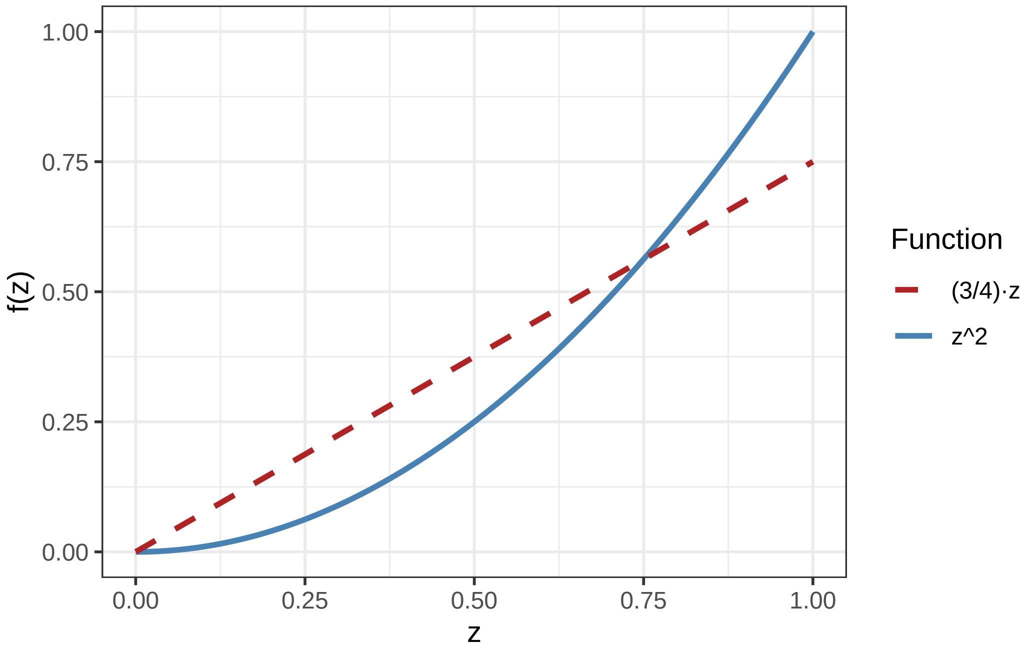
Figure B.3: Visual representation of the inner product giving the length of a projection.

From this picture you can see that the angle between any $t \in \mathcal{T}$ (not necessarily norm 1) and v can be extracted by taking the inverse cosine of the lengths of the "adjacent" over the "hypotenuse" (the classical definition of cosine). Those lengths are $\langle v, t/\|t\| \rangle$ and $\|v\|$. Dividing and pulling out the norm gives the law of cosines $\cos(\theta) = \frac{\langle v, t \rangle}{\|v\| \|t\|}$.

Example B.4. Let's presume that $\mathbb{P}(Z)$ is the uniform distribution over $[0, 1]$ and let $\mathcal{T}$ be the subspace of $\mathcal{L}_2(\mathbb{P})$ comprising of of all linear functions with no intercept, i.e. functions that can be written as $f(z) = cz$ for some slope c. You can verify that this is a subspace.

Let's let $v(z) = z^2$ and figure out what the projection of this function is onto $\mathcal{T}$. In other words, what is the linear function that best approximates z^2 in a mean-squared sense over the domain $[0, 1]$? To do this we'll use our formula: $\mathcal{T}v = \frac{\langle t, v \rangle}{\|t\|^2} t$

Pick $t(z) = z$ as the representative from $\mathcal{T}$. Then $\langle t, v \rangle = \int_0^1 zz^2 dz = 1/4$ and $\|t\|^2 = \int_0^1 z^2 dz = 1/3$. Thus $\mathcal{T}v = \frac{3}{4}x$. Visually your eye might agree that this is the best (no-intercept) linear approximation:



We call what's left over when you subtract the projection from the original vector the *residual*. An important geometric property of projections that you may recall from linear algebra (and which I have already been showing in the figures) is that residuals are always orthogonal to the entire subspace that gets projected on (see Figure B.2).

Orthogonality of Residuals

Theorem B.3. *For any subspace $\mathcal{T} \subseteq \mathcal{V}$ and vector $v \in \mathcal{V}$ in a Hilbert space for which $\mathcal{T}v$ exists and is unique, we have that $t \perp (v - \mathcal{T}v)$ for all $t \in \mathcal{T}$.*

Proof of Theorem B.3

For any $t \in \mathcal{T}$ we compute the (squared) distance between v and a perturbed version of its projection $\mathcal{T}(v) + \epsilon t$ for an arbitrary nonzero perturbation direction $t \in \mathcal{T}$:

$$
\begin{aligned}
d_t(\epsilon) &= \| v - (\mathcal{T}v + \epsilon t)\|^2 \\
&= \langle v - \mathcal{T}v + \epsilon t, v - \mathcal{T}v + \epsilon t\rangle \\
&= \|v - \mathcal{T}v\|^2 + 2\epsilon \langle v - \mathcal{T}v, t\rangle + \epsilon^2\|t\|^2
\end{aligned}
$$

The fact that $\mathcal{T}v$ is the minimizer of the distance to v means that if we perturb it by ϵ in any direction t, we must be increasing the distance. Therefore for all t, $d_t(\epsilon)$ is minimized at $\epsilon = 0$. Since d_t is a smooth function in ϵ for any t (a quadratic, in fact), if we take the derivative at $\epsilon = 0$ it must be zero:

$$0 = d_t'(0) = \langle v - \mathcal{T}v, t\rangle + 0 \cdot \|t\|^2.$$

which establishes the orthogonality $\langle v - \mathcal{T}v, t\rangle = 0$ for all $t \in \mathcal{T}$.

A special case of this is when $t = \mathcal{T}v$, which by definition is a vector in $\mathcal{T}$. The residual and the projection form two sides of a right triangle with the original vector as the hypotenuse. Therefore we often encounter the Pythagorean theorem in the context of projection:

$$\|v\|^2 = \|\mathcal{T}v\|^2 + \|v - \mathcal{T}v\|^2.$$

Exercise B.7. Let $\mathbb{P}$ be the standard uniform distribution. Your friend claims that the $\mathcal{L}_2(\mathbb{P})$ projection of the function $f(z) = 1$ onto the subspace of all polynomials has norm equal to 1/2. Without any calculation, explain why this is impossible. Again without any calculation, explain what the norm of the desired projection actually is.

At this point you should recognize we're abusing notation by using $\mathcal{T}$ as both a subspace and as a projection operator (function) that takes a given vector v and returns another vector which is the projection onto $\mathcal{T}$. Besides convenience, one reason to write it like this is that $\mathcal{T}v$ evokes the matrix-vector multiplication notation Ax. In fact, orthogonal projection *is* synonymous with a special kind of matrix multiplication in $\mathbb{R}^d$. More generally, you should bring that intuition with you: projection is a special kind of linear mapping.

Linearity of Orthogonal Projection

Theorem B.4. *For any subspace $\mathcal{T}$, projection is a linear mapping in the sense that $\mathcal{T}(cv + u) = c\mathcal{T}v + \mathcal{T}u$.*

Proof of Theorem B.4

Fix $v, u \in V$ and $c \in \mathbb{R}$. For any vector v we know we can project it onto some space $\mathcal{T}$ to obtain the projection $\mathcal{T}v \in \mathcal{T}$ and a residual $v - \mathcal{T}v$ orthogonal to $\mathcal{T}$. Therefore let's decompose

$$\begin{aligned} cv + u &= \mathcal{T}(cv + u) + [cv + u - \mathcal{T}(cv + u)] \\ cv &= c(\mathcal{T}v + [v - \mathcal{T}]) \\ u &= \mathcal{T}v + [u - \mathcal{T}] \end{aligned}$$

The two equations on the bottom sum to the one on the top. The first term on the right-hand side of each equation is a vector in $\mathcal{T}$ and the second is a vector orthogonal to it. The part of $cv + u$ that is in $\mathcal{T}$ can be composed only of the parts of v and u that are in $\mathcal{T}$ because no vectors that are orthogonal to a subspace can ever sum to a nonzero vector within the subspace. Thus $\mathcal{T}(cv+u) = c\mathcal{T}v + \mathcal{T}u$ which proves the linearity.

The proof of Theorem B.4 is based on the common theme that many interesting vector spaces can be built up as a kind of "sum" of subspaces.

Orthogonal Sum and Complement

Definition B.5. Two subspaces $\mathcal{V}_1$ and $\mathcal{V}_2$ are orthogonal if every vector $v_1 \in \mathcal{V}_1$ is orthogonal to every vector $v_2 \in \mathcal{V}_2$. A vector space $\mathcal{V}$ is the orthogonal sum of subspaces $\mathcal{V}_1$ and $\mathcal{V}_2$ if every vector $v \in \mathcal{V}$ can be written as $v = v_1 + v_2$ for some (unique) vectors $v_1 \in \mathcal{V}_1$ and $v_2 \in \mathcal{V}_2$. If this is the case we write $\mathcal{V} = \mathcal{V}_1 \oplus \mathcal{V}_2$. When two orthogonal subspaces sum to the total vector space $\mathcal{V}$, we say that they are orthogonal complements of each other and usually write them as $\mathcal{T}$ and $\mathcal{T}^\perp$ (which is which does not matter).

When I think of orthogonal sums, I usually think of 3D space being the "sum" of the x-y plane and the z-axis. The x-y plane is itself the sum of the x- and y-axes. Aleternatively you could formulate 3D space as the sum of the z-y plane and the x-axis. This shows that there are many ways to "break up" a space into orthogonal subspaces. The "named" axes don't matter either, I could also break up 2D space into the sum $\mathbb{R}^2 = \{c[1,1] : c \in \mathbb{R}\} \oplus \{c[1,-1] : c \in \mathbb{R}\}$.

In the proof of the linearity of orthogonal projection, we used the fact that by the orthogonality of projection, $\mathcal{V} = \mathcal{T} \oplus \mathcal{T}^\top$. We broke the original vectors and their sum into their projections and residuals and then noted that by orthogonality the only way to make the sums match up was to match up the parts corresponding to each orthogonal complement.

This also gives a general method to decompose a vector space into an orthogonal sum. First pick an arbitrary nonzero vector t. Then project every other vector onto its span $\mathcal{T}$ and take all the residuals. The set of projections and set of residuals are then orthogonal subspaces that sum to the full vector space.

Exercise B.8. Prove the above claim by following these steps:

1. Show that the set of projections $\{\mathcal{T}v : v \in \mathcal{V}\}$ and the set of residuals $\{v - \mathcal{T}v : v \in \mathcal{V}\}$ are indeed vector spaces (check that the sum of two vectors that satisfy the definition again satisfies the definition…)
2. Prove that any vector from the projection subspace is orthogonal to any vector from the residual subspace (already done)
3. Find the unique way to write every $v \in \mathcal{V}$ as a sum of two vectors from these subspaces (practically already done)

Example B.5. Let $\mathbb{P}$ be the standard uniform distribution and let $\mathcal{V} = \{f : f(z) = mz + b\}$ be the subspace of linear functions in $\mathcal{L}_2(\mathbb{P})$. Let's find a way to decompose this subspace into an orthogonal sum.

Pick $f(z) = 1$. The span of f is clearly $\mathcal{T} = \{z \mapsto c : c \in \mathbb{R}\}$. Using our 1D projection formula and taking the representative $f(z) = 1$ (which has norm 1), the residual of any function $g(z) = mz + b$ projected onto $\mathcal{T}$ is given by

$$\begin{aligned}(mz - b) - \mathcal{T}(mz - b) &= (mz - b) - \int_0^1 mz + b \,\mathrm{d}z \\ &= m(z - 1/2)\end{aligned}$$

Therefore

$$\{z \mapsto mz + b : m, b \in \mathbb{R}\} = \{z \mapsto c : c \in \mathbb{R}\} \oplus \{z \mapsto m(z - 1/2) : m \in \mathbb{R}\}$$

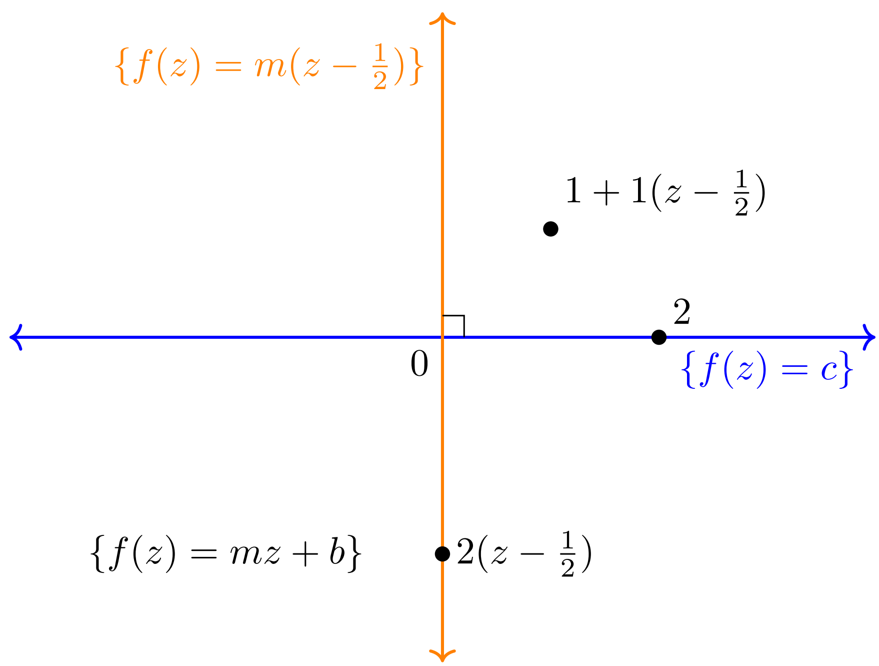


Figure B.4: A visual representation of the orthogonal sum. Each "point" in this vector space is a linear function defined over $z \in [0, 1]$. The two orthogonal "axes" are the constant functions and the functions that have the form $m(z - \frac{1}{2})$. If you picked a different initial axis, you could construct a different orthogonal "coordinate system" for these functions.

B.3.3 Cauchy-Schwarz

One important consequence of the general Pythagorean theorem is that we can easily bound inner products with a product of norms

Cauchy-Schwarz Inequality

Theorem B.5.

$$|\langle f, g\rangle| \leq \|f\|\|g\|$$

for any two elements f, g in the space. The equality is obtained if and only if $f = cg$ for some scalar c.

Proof of Theorem B.5

Let $\mathcal{G}$ be the span of g and consider the projection $\mathcal{G}f = \frac{\langle g,f\rangle}{\|g\|^2}g$. Recall that we previously used our one-dimensional projection formula to show

$$\langle f, g\rangle = \|\mathcal{G}f\|\|g\|,$$

a geometric interpretation to the inner product as the scaled length of the "adjacent" side of a right triangle. The Cauchy-Schwarz inequality is immediate from this because $\|\mathcal{G}f\| \leq \|f\|$ since these are the side and hypotenuse of a right triangle, respectively. Equality holds if and only if $f \in \mathcal{G}$ already, i.e. if $f = cg$ for some scalar c.

One way to remember the Cauchy-Schwarz inequality is that it makes it sensible to define an "angle" as $\cos(\theta) = \frac{\langle f,g\rangle}{\|f\|\|g\|}$ since the right-hand side is guaranteed to be between 1 and -1. The probabilistic version of this in $\mathcal{L}_2(\mathbb{P})$ is that for two zero-mean functions $\mathbb{E}[f(Z)] = \mathbb{E}[g(Z)] = 0$,

$$\frac{\langle f, g\rangle}{\|f\|\|g\|} = \frac{\mathbb{C}[f(Z), g(Z)]}{\sqrt{\mathbb{V}[f]\mathbb{V}[g]}},$$

which is the correlation between the random variables $f(Z)$ and $g(Z)$. This must also be between -1 and 1. This also gives a geometric interpretation to the correlation coefficient: it's a measure of the "angle" between two random variables viewed as vectors in $\mathcal{L}_2(\mathbb{P})$.

Example B.6 (Bounding an Integral)**.** Suppose I know $f \in \mathcal{L}_2(\mathbb{P})$. In many problems we might end up with an expression like $\mathbb{E}[f(Z)]$ which we might not know how to compute without knowing exactly what f is. However, Cauchy-Schwarz gives us one useful way to approximate such a term in the sense that we can at least say it's no bigger than a certain number. To wit,

$$|\mathbb{E}[f(Z)]| = \left|\int f(Z) \cdot 1 \, \mathrm{d}\mathbb{P}\right| = |\langle f, 1\rangle| \leq \|f\|.$$

We even know that the bound is *sharp* (we can't do any better without knowing more). That's because if $f = c$ (a constant), then $\mathbb{E}[c] = \|c\|$.

Example B.7 (Continuity of Inner Product)**.** Let's say we want to show that the inner product is continuous in either argument. Definition A.5 says that for the inner product to be continuous it must be that $\langle v_n, u\rangle \to \langle v, u\rangle$ for any sequence $v_n \xrightarrow{\mathcal{V}} v$ where we use the $\mathcal{V}$-norm to define the required distance metric $d_{\mathcal{V}}(v, u) = \|v - u\|_{\mathcal{V}}$. The inner products converge if $|\langle v_n, u\rangle - \langle v, u\rangle| = |\langle v_n - v, u\rangle| \to 0$ for all sequences such that $\|v_n - v\| \to 0$. By Cauchy-Schwarz,

$$|\langle v_n - v, u\rangle| \le \|v_n - v\|\|u\|.$$

Since the right-hand side converges to zero for any convergent sequence v_n, the left hand side does too and thus the inner product is continuous in the first argument. Symmetry of the inner product means this is also true of the second argument.

B.3.4 Triangle Inequality

Lastly, we have the aptly-named triangle-inequality, which essentially says that the fastest way from point A to point B is a straight line. If you take a detour to point C, the total distance that you travel is larger than if you had gone directly to B.

Triangle Inequality

Theorem B.6. *For any norm $\|\cdot\|$ and vectors a, b, c,*

$$\|a - b\| \le \|a - c\| + \|c - b\|.$$

Proof of Theorem B.6

Let $u = a - c$ and $v = c - b$ so that $a - b = u + v$.

$$\begin{aligned}\|u + v\|^2 &= \langle u + v, u + v\rangle \\ &= \|u\|^2 + 2\langle u, v\rangle + \|v\|^2 \\ &\le \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 \quad \text{(Cauchy-Schwarz)} \\ &= (\|u\| + \|v\|)^2\end{aligned}$$

All norms that are induced by inner products satisfy the triangle inequality by design, as we have proven. However, the triangle inequality is actually part of the definition of a norm: any function that someone wants to call a norm must satisfy the triangle inequality in order for them to call it that.

Sometimes we are interested in proving that one side of a triangle is small, but what we know about are the other two sides. That's when the triangle inequality comes in handy.

Example B.8. If I have a sequence of real numbers $x_n \to x$ and another $y_n \to y$, is it true that $x_n y_n \to xy$?

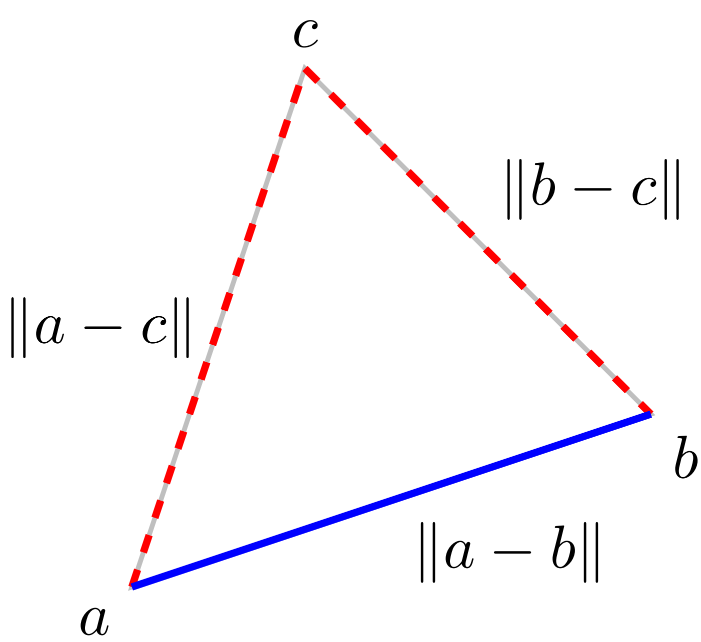


Figure B.5: Illustration of the triangle inequality.

> Warning
>
> If you don't understand (in a precise way) what I mean by convergence of a sequence you should review Section A.1.

Let's consider the difference $|x_n y_n - xy|$. To prove the claim we have to show that this is eventually always smaller than any given ϵ. We can do this by taking a "detour" to the point(s) $x_n y$:

$$|x_n y_n - xy| \leq |x_n y_n - x_n y| + |x_n y - xy|.$$

Perhaps it goes without saying, but the absolute value is a norm on $\mathbb{R}$ induced by the inner product $\langle x, y \rangle = xy$ so we are justified in applying the triangle inequality. Our purpose in taking this detour is that we're trying to somehow reduce the distance $|x_n y_n - xy|$ into some combination of $|x_n - x|$ and $|y_n - y|$ because we know we can make those distances as small as we like by taking n large enough (that's the definition of convergence).

For each of the terms above, we can now use Cauchy-Schwarz, e.g. $|x_n y_n - x_n y| = |x_n(y_n - y)| \leq |x_n||y_n - y|$, therefore

$$|x_n y_n - xy| \leq |x_n||y_n - y| + |y||x_n - x|.$$

We're almost there but we have the term $|x_n|$ which is neither a constant like $|y|$ nor converging to zero. Here again we can use the triangle inequality on $|x_n|$ and take a detour to x to expand $|x_n| = |x_n - 0| = |x_n - x| + |x - 0|$, leaving us with

$$|x_n y_n - xy| \leq |x||y_n - y| + |x_n - x||y_n - y| + |y||x_n - x|.$$

Now everything is in terms of constants and the two distances which we know go to zero. Therefore we have the desired convergence.

When you first see proofs like the one in the example above it can feel overwhelming: how did I know where to use the triangle inequality? How did I know when to use Cauchy-Schwarz? It all seems like black magic and cheap tricks that you can understand in retrospect, but could never have constructed yourself. But there actually are certain “tells” that you can use to build your intuition.

For example, when you see a norm of a sequence, you should immediately want to approximate it by the norm of the limit $|x_n| \approx |x|$ for convenience since the latter is just a constant that you can deal with easily. The triangle inequality then comes to mind as a way to do this since we can “detour” x_n to x and then get the approximation $|x_n| \leq |x_n - x| + |x|$. Cauchy-Schwarz should come to mind any time you see pretty much any inner product. $|yx_n - yx|$ is a mess but $|y||x_n - x|$ is something we can understand. With continued practice you will get very good at just “seeing” what to do. If you’re interested in that I recommend picking up a real analysis textbook and working through the problems.

B.4 Riesz Representation

In this section we’ll talk about *Riesz representation*. The basic idea here is that any linear function whose inputs are elements of a vector space can be entirely characterized by its direction of steepest ascent and by its slope in that direction. We’ll first walk through the ideas in finite-dimensional inner product spaces where things are easier to visualize and we don’t need any definitions from real analysis before proceeding to the general situation in possibly infinite-dimensional Hilbert spaces.

B.4.1 Warm-Up in $\mathbb{R}^d$

You may recall from linear algebra or vector calculus that many useful functions map vectors to other vectors or vectors to numbers. In $\mathbb{R}^d$, given a $p \times d$ matrix A, the matrix-vector “multiplication” Av can actually be seen as a function that maps $v \in \mathbb{R}^d$ to $Av \in \mathbb{R}^p$. Similarly, the vector dot product $v \mapsto a \cdot v$ for some vectors $a, v \in \mathbb{R}^d$ maps $v \in \mathbb{R}^d$ to $a \cdot v \in \mathbb{R}$. Both of these example functions on vectors are *linear* in the sense that $f(cv + u) = cf(v) + f(u)$ for a scalar c and vectors v, u.

It's clear that the dot product between an argument v and some fixed vector a is linear in the argument v, but actually the relationship goes both ways: *all* linear functions can be written as an inner (dot) product between the argument and some other fixed vector in the vector space. Formally, for any function f which is linear and maps $\mathbb{R}^d$ to $\mathbb{R}$, there is some fixed $t_f \in \mathbb{R}^d$ that "represents" the function in the sense that for all $f(v) = \langle t_f, v \rangle$ for all $v \in \mathbb{R}^d$. We call the vector t_f the "Riesz representer" of the function f.

Mechanically this is pretty easy to show. To start, we can always write v as a sum of its components along the orthonormal basis vectors e_j that span our vector space: $v = \sum_j v_j e_j$. Now we apply the function f to both sides and use the linearity:

$$f(v) = f\left(\sum_j v_j e_j\right) = \sum_j v_j f(e_j) = \langle v, t_f \rangle$$

where we've defined the vector $t_f = [f(e_1), f(e_2), \ldots f(e_d)]$. Since f maps vectors to numbers, each element $f(e_j)$ is indeed a scalar value. It's important to recognize that t_f is a fixed vector associated with the fixed function f: it doesn't depend on the argument v.

While the proof is simple, it doesn't give much intuition for what the result *means* or give any sense of *why* it's true. To do that it's helpful to look at some pictures. In the figure below I've drawn the graph of some linear function of vectors in $\mathbb{R}^2$: it looks like a plane that passes through the origin. For each vector $v = [v_1, v_2]$ on the $\mathbb{R}^2$ plane, you can find the scalar value of $f(v)$ by drawing a vertical up to the graph that represents f.

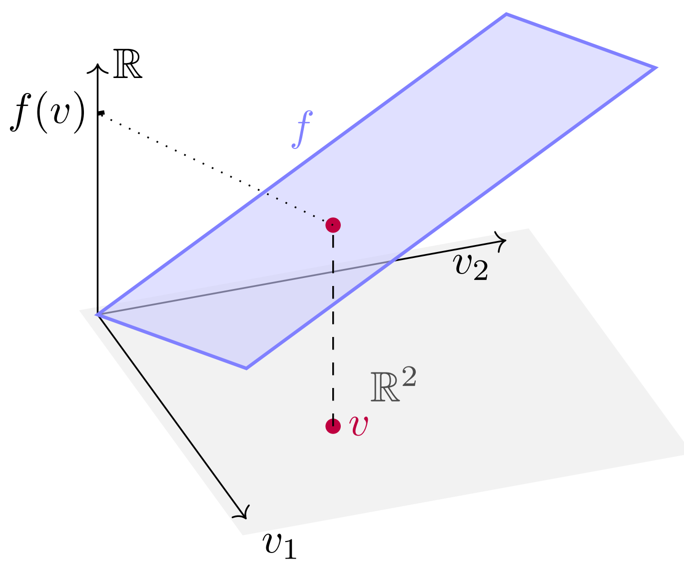


Figure B.6: A linear function mapping $v \in \mathbb{R}^2$ to $f(v) \in \mathbb{R}$.

It should be somewhat intuitive to you that a linear function has a linear (planar) graph but nonetheless let's think about why that is. The graph must pass through the origin because by linearity $f(0) = 0 \times f(v) = 0$. Now, for any fixed vector v, the graph of ϵ vs $f(\epsilon v) = \epsilon f(v)$ is a straight line between with intercept 0 and slope $f(v)$. Thus the graph of f is linear in all directions and therefore planar.

Let's look at this graph from the top down. I've drawn this in the figure and you can see that the level sets of f (the "elevation lines" on a map) are straight lines that all go in the same direction. The direction orthogonal to these level sets is the direction in which f is increasing the fastest. If you stood on the surface, that would be the "uphill" direction. I've also drawn a similar picture for another linear function that I call g. For g you can see that the uphill direction is different and that the level sets are closer together, meaning that the climb is "steeper" in the uphill direction than it is for f.

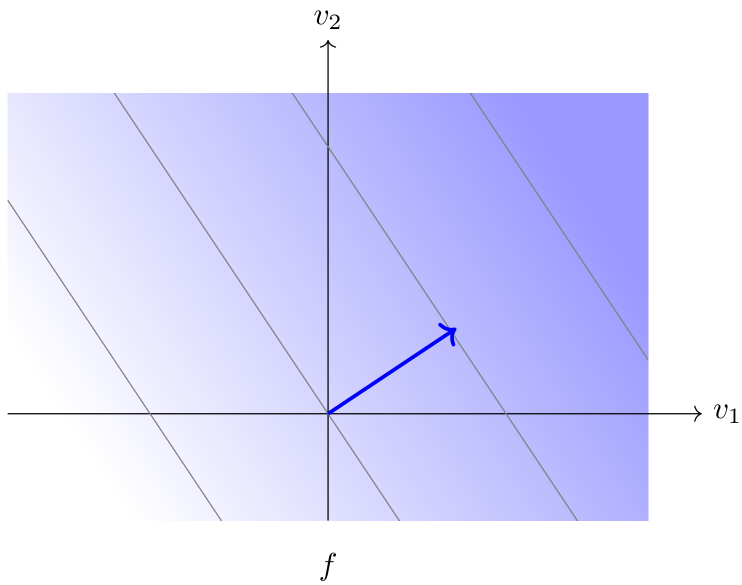


Figure B.7: The same linear function as seen in Figure B.6, but seen from the top down. Darker blue indicates larger values of $f(v)$. Grey lines are the level sets $f(v) = c$, spaced at equal intervals in c so that the density of those lines indicates the slope.

Looking at the two examples, you may realize that the uphill direction and the steepness in that direction are *everything* you need to know in order to characterize any linear function f. The uphill direction can be represented by a unit vector and the steepness is a factor that it is scaled by. We call that scaled unit vector t_f, our vector representer of f. This vector contains all the information there is to know about f, and vice versa. Technically they are not the same object, but we can always reconstruct one from the other.

It remains to be shown that this vector is the same representer t_f that we derived above. To find the "most uphill" direction, we see how much we climb if we take a step v of unit length $\|v\| = 1$ in any direction away from some starting point. It doesn't matter from where we start so for convenience we'll start from zero. Formally, the "most uphill" direction is therefore

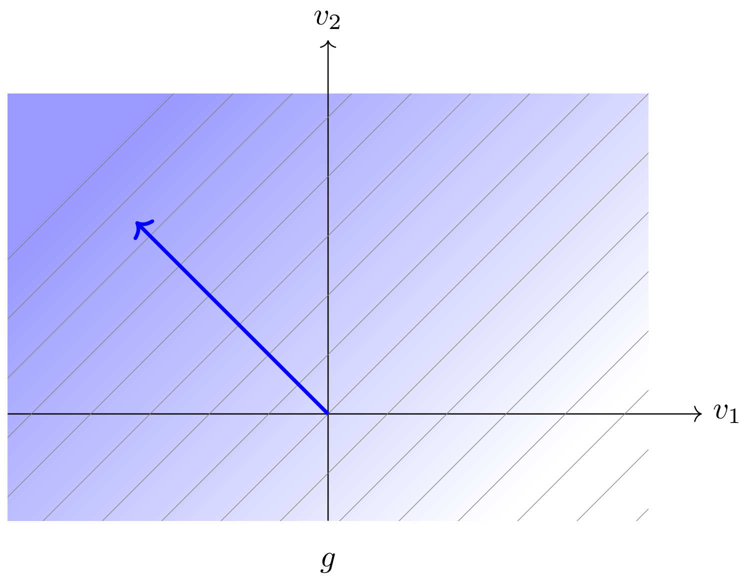


Figure B.8: A different linear function from the top down.

$$
\begin{aligned}
v^* &= \underset{v \in \mathbb{R}^d : \|v\|=1}{\operatorname{argmax}} \; f(v) \\
&= \underset{v \in \mathbb{R}^d : \|v\|=1}{\operatorname{argmax}} \sum v_i f(e_i) \quad \text{(linearity)}
\end{aligned}
$$

i.e. we want to maximize the inner product $\langle v, t_f \rangle$ over all v with norm 1. By Cauchy-Schwarz, this is attained by letting $v = t_f / \|t_f\|$. If we look at the amount by which f climbs over one unit step in this direction, it is $f(t_f/\|t_f\|) = \langle t_f, t_f \rangle / \|t_f\| = \|t_f\|$. Therefore the representer t_f is nothing more than the direction in which f is increasing the most, with its norm indicating how quickly f increases in that direction.

Exercise B.9. How is the Riesz representer of a linear function from $\mathbb{R}^d$ to $\mathbb{R}$ related to its gradient?

I've explained intuitively why this representer vector uniquely identifies any linear function, but I haven't explained why taking the inner product of the argument with the representer is equivalent to evaluating the function. To understand that again I think it helps to look at the pictures above that show the level sets of the function. If you imagine walking along one of those level sets (orthogonal to the representer), the value of the function does not change. In other words, the function f is completely insensitive to any motion at any point that goes in a direction that is orthogonal to the direction of steepest increase. Let's decompose $\mathcal{V}$ orthogonally into $\mathcal{T} = \operatorname{span}(t_f)$ and $\mathcal{T}^\perp$. What I've said is formally that $f(v + t^\perp) = f(v)$ for any motion $t^\perp$ in $\mathcal{T}^\perp$.

Now imagine moving in an arbitrary direction u. Whatever direction you go, you can decompose it into a part that is in the direction of steepest ascent and the part that is in the orthogonal complement of that subspace, i.e. write $u = t + t^{\perp}$. By the linearity of f, $f(u) = f(t) + f(t^{\perp}) = f(t)$. In other words, f doesn't even care about any *component* of motion that isn't in the direction of steepest ascent. Finally, let's think about what the vector t is: it's the part of u that lies in $\mathcal{T}$, i.e. the projection $\mathcal{T}u = \left\langle u, \frac{t_f}{\|t_f\|} \right\rangle \frac{t_f}{\|t_f\|}$. The inner product is just a constant so when we evaluate $f(u) = f(\mathcal{T}u)$ it gets passed right through and we're left with

$$f(u) = f(\mathcal{T}u) = \left\langle u, \frac{t_f}{\|t_f\|} \right\rangle f\left(\frac{t_f}{\|t_f\|}\right) = \left\langle u, \frac{t_f}{\|t_f\|} \right\rangle \|t_f\|$$

where the last equality obtains because the increase in f under one unit step in the direction of maximal increase is $\|t_f\|$, as we just showed. Therefore $f(u) = \langle u, t_f \rangle$ as desired.

The overall lesson is that linear functions completely ignore motion in any direction that isn't the direction of steepest increase. Because of this, the only part of the input that matters is the part in the direction of steepest increase, which is given by projecting the input onto that direction. Since the projection is related to the inner product, we can replace function evaluation with taking that inner product as long as we choose a representative in the subspace of maximum increase that has a norm equivalent to the increase in f when moving one unit in that direction.

To be clear, none of this reasoning proves the fact that every linear function on $\mathbb{R}^d$ has a Riesz representer any better than the 2-line argument I gave at the beginning of this section. But I hope that walking through this has helped you "feel it" in your bones rather than "know it" as an abstract fact.

B.4.2 Infinite-Dimensional Riesz Representation

Now we'll talk about the more general case when the Hilbert space of interest could be infinite-dimensional. For the most part we'll be interested in $\mathcal{L}_2$ or subspaces thereof. If we talk about functions defined *on* $\mathcal{L}_2$, we're talking about functions *of functions*. In functional analysis, people sometimes call these *operators*, but there's really nothing special about them: they are still functions. When an operator has real-valued output (i.e. maps functions to numbers), it's often called a *functional*.

Example B.9 (Expectation Functional)**.** The function $f \mapsto \mathbb{E}_{\mathbb{P}}[f(Z)]$ is a functional (a linear functional, in fact). The expectation functional is defined on all of $\mathcal{L}_2(\mathbb{P})$ since every square-integrable function is integrable (see Example B.6).

In general Hilbert spaces the story of Riesz representation is essentially the same except there is one little wrinkle: not all linear functions actually have a direction of steepest ascent. The problem is that there are too many "directions" to go in within infinite-dimensional space. No matter what direction you "look" (e.g. imagine standing at the origin), there may be some other direction you could look in where the slope (the unit rise in the function over a unit change in the input) is steeper! In contrast, in 3D (or d-dimensional) space, the fact that there are only 3 (d) dimensions means that there is *always* some direction of steepest ascent: you quickly run out of directions to look for steeper slopes.

Example B.10 (Evaluation at a Point)**.** Let $Z \sim \text{Unif}(-1, 1)$ and let $T : f \mapsto f(0)$ be the functional that takes $f \in \mathcal{L}_2(\mathbb{P})$, evaluates it at the point $z = 0$, and returns the result. Clearly T is linear under the definition of function addition and scaling in $\mathcal{L}_2$ because

$$T(cf + g) = (cf + g)(0) = cf(0) + g(0) = cTf + Tg.$$

Interestingly, however, T does not have a "direction of steepest ascent". To understand this, let's first define the slope of T as the rise in T over a unit change in the input f. Equivalently, we may consider the rise in T divided by the amount of change in the input:

$$\frac{T(f) - T(f + h)}{\|h\|}.$$

Since T is linear, the slope in a given direction is the same at all points so we can take $f = 0$ to simplify things without loss of generality. Now let's pick some directions h (these are functions; recall functions are the vectors in $\mathcal{L}_2$) and see what the magnitude of the slope $\frac{|Th|}{\|h\|}$ is. Consider the functions (directions, vectors)

$$h_n(z) = 2\sqrt{n}\mathbb{1}\left(-\frac{1}{n} \leq z \leq \frac{1}{n}\right).$$

These functions have norms

$$\|h_n\|^2 = \frac{1}{2}\int_{-n^{-1}}^{n^{-1}} 2^2\sqrt{n}^2 \,\mathrm{d}z = 1.$$

On the other hand,

$$Th_n = h_n(0) = 2\sqrt{n} \to \infty.$$

By increasing $n \to \infty$ I can find a direction $h = h_n$ where the slope $Th/\|h\| = 2\sqrt{n}$ is bigger than any given number. It is therefore not possible to find a direction of steepest ascent: there is always a steeper slope in another direction!

In the field of functional analysis, functionals that don't have a direction of steepest ascent are called *unbounded*.

Bounded Linear Functional

Definition B.6. A linear functional $T : \mathcal{V} \to \mathbb{R}$ for a Hilbert space $\mathcal{V}$ is called bounded if

$$\sup_{v \in \mathcal{V}} \frac{|Tv|}{\|v\|} < \infty.$$

The precise value of the supremum is called the *operator norm* of the functional, sometimes written $\|T\|_{\mathrm{op}}$.

"Boundedness" in this sense is not the same as usual sense of the word, which would be $\sup_v Tv < \infty$. No linear function can be bounded in the traditional sense because no matter what Tv is for some vector v, I can scale v by a large constant c and make $T(cv) = cTv$ as large as I want. But that's not what boundedness means in the sense of functional analysis. In a functional analysis context, boundedness means that there is a maximum value to the *slope* of T, not that the *output* of T has a maximum value.

Exercise B.10. Every functional of the form $v \mapsto \langle t, v \rangle$ is linear by the definition of any inner product. Use the Cauchy-Schwarz inequality to prove that any such functional is also bounded.

Now we can finally state the general-purpose Riesz representation theorem:

Riesz Representation

Theorem B.7. *Every bounded linear functional $T : \mathcal{V} \to \mathbb{R}$ on a Hilbert space can be written as*

$$Tv = \langle t, v \rangle$$

for some fixed $t \in \mathcal{V}$ which we call the Riesz representer *of T.*

Proof of Theorem B.7

First we'll quickly prove that T is continuous. It's counter-intuitive, but linear functionals in infinite dimensions can be discontinuous. In fact, continuity is the same thing as boundedness, but we'll just prove the implication in one direction. Consider a sequence of vectors $v_n \to v$ in $\mathcal{V}$. Since T is bounded, $|Tv_n - Tv| = |T(v_n - v)| \leq \|T\|_{\mathrm{op}} \|v_n - v\| \to 0$, proving that T is continuous (a sequence of inputs converging to a limit means the sequence of outputs converges to the function evaluated at that limit).
Based on our intuition from the finite dimensional case, let's take $\tilde{t}$ to be the maximizer of Tv over all unit-norm vectors v. Since T is bounded, $|Tv|$ has a maximum value over the set of unit-norm vectors. Moreover, since the set of unit-norm vectors is closed and bounded and T is continuous, T attains its supremum somewhere: there is a unit-norm $\tilde{t}$ such that $T\tilde{t} = \sup_{\|v\|=1} Tv$ (this is called the extreme value theorem; you can look it

up but it's not important to get the gist of things).
Now we'll show that the function T only responds to changes in its input that are aligned with $\tilde{t}$, just like in the finite-dimensional case. Let $\mathcal{T}$ be the span of $\tilde{t}$. Pick an arbitrary input v and consider the decomposition

$$v = \underbrace{(v - \mathcal{T}v)}_{u} + \mathcal{T}v.$$

Now let's calculate the slope of T in the direction of $\tilde{t} + cu$ as a function of c: how much we perturb away from the direction of the steepest slope $\tilde{t}$. That is

$$s(c) = \frac{T(\tilde{t} + cu)}{\|\tilde{t} + cu\|} = \frac{T\tilde{t} + cTu}{\sqrt{1 + c^2\|u\|}}$$

where we've expanded the numerator using linearity and calculated the denominator with the Pythagorean theorem ($u \perp \tilde{t}$ by construction). Since the slope is maximized in the direction $\tilde{t}$, we know that $s(c)$ has its maximum at $c = 0$, meaning that the derivative is zero there: $s'(0) = 0$. However, from the above, a tedious application of the product and chain rules and evaluating at $c = 0$ gives $s'(0) = Tu$. But since we already know that must be zero, we conclude $Tu = 0$. In other words, T does not care about any component of its input that is not aligned with the direction of maximum increase $\tilde{t}$.
As a consequence of this,

$$\begin{aligned} T(v) &= T(u + \mathcal{T}v) \\ &= T(\mathcal{T}v) \\ &= T(\langle \tilde{t}, v\rangle \tilde{t}) \\ &= \langle T(\tilde{t})\, \tilde{t}, v\rangle \end{aligned}$$

Where we've used the 1D projection formula $\mathcal{T}v = \langle \tilde{t}, v\rangle \tilde{t}$ and linearity to move terms around. This establishes that $T(\tilde{t})\,\tilde{t}$ is the Riesz representer of T. The result once again aligns with the intuition from the finite-dimensional case: the representer is the (unit) direction of steepest ascent, scaled by the slope of the function in that direction.

Exercise B.11. Is $f \mapsto E[f(Z)]$ a bounded linear functional over all of $\mathcal{L}_2(\mathbb{P})$? If so, what is its Riesz representer?

Exercise B.12. For a bounded functional T, prove that $\|T\|_{\text{op}} = \|t\|$ where t is the Riesz representer of T.

If you are paying very, very close attention, you may have noticed that the pointwise evaluation functional is actually *not* well-defined on $\mathcal{L}_2(\mathbb{P})$. The reason is that, as you may recall, the elements of $\mathcal{L}_2(\mathbb{P})$ are not really functions, but *equivalence classes* of functions. For an operator that actually works on individual functions to be well-defined on $\mathcal{L}_2(\mathbb{P})$, it must have the same output for each function in any given equivalence class.

There *are* unbounded linear functionals that are well-defined on $\mathcal{L}_2$ but it's actually not possible to construct one explicitly. Proving what they look like *implicitly* requires more functional analysis than I'd like to get into in this book. Moreover, the examples of "unbounded linear functionals" that come up in statistics are more often like my technically ill-defined example: functionals that are actually directly defined on a space of individual functions, but we pretend are defined on $\mathcal{L}_2$. Ultimately it doesn't make much of a difference because the conclusion is the same either way: Riesz representation doesn't work for these functionals.